

# E2366793637\_2

The equation  $2x^5 + xy + 2y^3 - 1 = 0$  has no integer solutions

Size  $h = 85$ , length  $l = 12$

**Theorem 1.** *The equation*

$$2x^5 + xy + 2y^3 - 1 = 0 \tag{1}$$

*has no integer solutions.*

The proof uses 48 auxiliary polynomials  $H_0, \dots, H_{47}$ , 6 of which are constants, and a graph on these vertices with 167 edges; to each edge  $\{i, j\}$  we attach the Hilbert symbol  $(H_i, H_j)_v$  of the values at a hypothetical solution. By the product formula, the product of these symbols over all edges and all places is 1. We show that the product over the edges is 1 at every odd prime outside a finite set  $S$ , compute it at the primes of  $S$  and at the real place, and obtain a contradiction. Apart from polynomial identities, which can be checked by expanding both sides, and finite checks of congruences, the proof uses only basic properties of the Legendre symbol and of the Hilbert symbol.

## 1 Legendre symbols and Hilbert symbols

For a prime  $p$  and a non-zero integer  $N$ , we write  $v_p(N)$  for the exponent of  $p$  in the prime factorization of  $N$ . For an integer  $a$  and an odd prime  $p$ , the Legendre symbol  $(a/p)$  is 0 if  $p \mid a$ , 1 if  $a$  is a non-zero square modulo  $p$ , and  $-1$  otherwise. It is multiplicative in  $a$ , and it depends only on  $a$  modulo  $p$  [1, Chapter 5].

For a prime  $p$ , let  $\mathbb{Q}_p$  be the field of  $p$ -adic numbers, and put  $\mathbb{Q}_\infty = \mathbb{R}$ . For  $v$  a prime or  $v = \infty$ , and non-zero  $a, b \in \mathbb{Q}$ , the Hilbert symbol  $(a, b)_v$  is 1 if the equation  $z^2 = ar^2 + bs^2$  has a solution  $(r, s, z) \neq (0, 0, 0)$  in  $\mathbb{Q}_v$ , and  $-1$  otherwise. In particular,  $(a, b)_v = 1$  for every  $v$  if this equation has a non-zero solution in  $\mathbb{Q}$ . For an odd integer  $t$ , we put  $\varepsilon(t) = (t - 1)/2$  and  $\omega(t) = (t^2 - 1)/8$ . We use the following facts [2, Chapter III, Proposition 2 and Theorems 1 and 3].

**Theorem 2.** *Let  $a, a'$  and  $b$  be non-zero integers, and let  $v$  be a prime or  $v = \infty$ .*

- (i)  $(a, b)_\infty = -1$  if and only if  $a < 0$  and  $b < 0$ .
- (ii) Let  $p$  be an odd prime, and write  $a = p^\alpha u$  and  $b = p^\beta w$  with  $p \nmid uw$ . Then  $(a, b)_p = (-1)^{\alpha\beta\varepsilon(p)}(u/p)^\beta(w/p)^\alpha$ .
- (iii) Write  $a = 2^\alpha u$  and  $b = 2^\beta w$  with  $u$  and  $w$  odd. Then  $(a, b)_2 = (-1)^{\varepsilon(u)\varepsilon(w) + \alpha\omega(w) + \beta\omega(u)}$ .
- (iv)  $(a, b)_v = 1$  for all but finitely many  $v$ , and  $\prod_v (a, b)_v = 1$ , the product over all primes  $v$  and  $v = \infty$ .
- (v)  $(a, b)_v = (b, a)_v$  and  $(aa', b)_v = (a, b)_v(a', b)_v$ .

**Lemma 3.** *Let  $a$  and  $b$  be non-zero integers. If  $p$  is an odd prime with  $p \nmid a$  and  $(a/p) = 1$ , then  $(a, b)_p = 1$ . If  $a \equiv 1 \pmod{8}$ , then  $(a, b)_2 = 1$ .*

*Proof.* The first claim is Theorem 2(ii) with  $\alpha = 0$ . For the second,  $\varepsilon(a)$  and  $\omega(a)$  are even, and Theorem 2(iii) with  $\alpha = 0$  gives  $(a, b)_2 = 1$ .  $\square$

**Lemma 4.** *Let  $p$  be a prime. For a non-zero integer  $X$ , write  $X = p^j u$  with  $p \nmid u$ . If  $p$  is odd, put  $c_p(X) = (j, e) \in \mathbb{F}_2^2$ , where  $(u/p) = (-1)^e$ ; if  $p = 2$ , put  $c_2(X) = (j, e_1, e_2) \in \mathbb{F}_2^3$ , where  $u \equiv (-1)^{e_1} 5^{e_2} \pmod{8}$ ; all entries are taken modulo 2. Then  $c_p(XX') = c_p(X) + c_p(X')$  and  $(X, X')_p = (-1)^{\beta_p(c_p(X), c_p(X'))}$  for non-zero integers  $X, X'$ , where  $\beta_p$  is the symmetric bilinear form*

$$\begin{aligned} \beta_p((j, e), (j', e')) &= jj' \varepsilon(p) + je' + j'e && \text{if } p \text{ is odd,} \\ \beta_2((j, e_1, e_2), (j', e'_1, e'_2)) &= e_1 e'_1 + je'_2 + j'e_2. \end{aligned}$$

*Proof.* For odd  $p$ , this is Theorem 2(ii), as the Legendre symbol is multiplicative. For  $p = 2$ , the odd residues 1, 3, 5, 7 modulo 8 are  $(-1)^{e_1} 5^{e_2}$  with  $(e_1, e_2) = (0, 0), (1, 1), (0, 1), (1, 0)$ , respectively; in each case  $\varepsilon(u) \equiv e_1$  and  $\omega(u) \equiv e_2 \pmod{2}$ , so Theorem 2(iii) gives the formula, and  $c_2$  is additive because  $(e_1, e_2) \mapsto (-1)^{e_1} 5^{e_2}$  is an isomorphism from  $\mathbb{F}_2^2$  to the group of odd residues modulo 8.  $\square$

**Lemma 5.** *Let  $p$  be a prime,  $k \geq 1$ , and let  $X$  be a non-zero integer with  $X \equiv r \pmod{p^k}$ , where  $0 \leq r < p^k$ . Then  $c_p(X) \in b + D$  for the vector  $b$  and the subspace  $D$  given as follows. If  $r = 0$ , then  $b = 0$  and  $D$  is the whole space. Otherwise write  $r = p^j r'$  with  $p \nmid r'$ , so that  $j < k$ . If  $p$  is odd, or if  $p = 2$  and  $k - j \geq 3$ , then  $b = c_p(r)$  and  $D = 0$ . If  $p = 2$  and  $k - j = 2$ , then  $b = (j, e_1, 0)$  with  $r' \equiv (-1)^{e_1} \pmod{4}$ , and  $D$  is spanned by  $(0, 0, 1)$ . If  $p = 2$  and  $k - j = 1$ , then  $b = (j, 0, 0)$ , and  $D$  is spanned by  $(0, 1, 0)$  and  $(0, 0, 1)$ .*

*Proof.* If  $r \neq 0$ , then  $v_p(X) = j$  and  $X/p^j \equiv r' \pmod{p^{k-j}}$ . For odd  $p$ , the Legendre symbol of  $X/p^j$  is therefore that of  $r'$ . For  $p = 2$ , the residue of  $X/2^j$  modulo 8 is that of  $r'$  if  $k - j \geq 3$ , and its residue modulo 4, which determines  $e_1$ , is that of  $r'$  if  $k - j = 2$ .  $\square$

**Lemma 6.** *Let  $p$  be a prime, and let  $E$  be a set of edges on a finite set of vertices. For every vertex  $i$ , let  $b_i$  be a vector and  $D_i$  a subspace of  $\mathbb{F}_2^2$  (of  $\mathbb{F}_2^3$  if  $p = 2$ ), and put  $\sigma_i = \sum_{j \sim i} b_j$ , the sum over the edges  $\{i, j\} \in E$ . Suppose that  $\beta_p(d, \sigma_i) = 0$  for every vertex  $i$  and every  $d \in D_i$ , and that  $\beta_p(d, d') = 0$  for every edge  $\{i, j\} \in E$  and all  $d \in D_i$  and  $d' \in D_j$ . Then*

$$\sum_{\{i, j\} \in E} \beta_p(c_i, c_j) = \sum_{\{i, j\} \in E} \beta_p(b_i, b_j) \quad \text{whenever } c_i \in b_i + D_i \text{ for every } i.$$

*By bilinearity, it suffices to check the hypotheses for  $d$  and  $d'$  in spanning sets of  $D_i$  and  $D_j$ .*

*Proof.* Write  $c_i = b_i + d_i$  with  $d_i \in D_i$ . As  $\beta_p$  is bilinear and symmetric, the left side equals the right side plus  $\sum_i \beta_p(d_i, \sigma_i) + \sum_{\{i, j\} \in E} \beta_p(d_i, d_j)$ , and both sums vanish.  $\square$

**Lemma 7.** *Let  $f(z) = az^3 + bz^2 + cz + d$  be a real polynomial with  $a \neq 0$  and discriminant  $\Delta < 0$ . Then  $f$  has exactly one real root  $\eta$ , and  $f(z)$  has the sign of  $a(z - \eta)$  for every real  $z \neq \eta$ .*

*Proof.* The discriminant is  $\Delta = a^4 \prod_{i < j} (\rho_i - \rho_j)^2$ , where  $\rho_1, \rho_2, \rho_3$  are the complex roots of  $f$ . If all roots were real, then  $\Delta \geq 0$ . Hence  $f$  has a non-real root  $\rho$ ; then  $\bar{\rho} \neq \rho$  is also a root, and the third root  $\eta$  is real. Thus  $f(z) = a(z - \eta)(z - \rho)(z - \bar{\rho}) = a(z - \eta)|z - \rho|^2$  for real  $z$ , and  $|z - \rho|^2 > 0$ .  $\square$

## 2 The graph

Put  $F(x, y) = 2x^5 + xy + 2y^3 - 1$ , so that (1) is the equation  $F(x, y) = 0$ . The vertices are the polynomials

$$\begin{aligned} H_0 &= 2y, \\ H_1 &= 2y - 1, \\ H_2 &= 2x + 2y, \\ H_3 &= x + 2y - 1, \\ H_4 &= 2y - 2, \\ H_5 &= 2y + 2, \\ H_6 &= -x + 2y - 1, \\ H_7 &= -x + 2y + 1, \\ H_8 &= -2x + 2y, \\ H_9 &= 2x + 2y - 1, \\ H_{10} &= 2x + 2y + 1, \\ H_{11} &= x + 2y + 2, \\ H_{12} &= 2x + 2y - 2, \\ H_{13} &= 2x + 2y + 2, \\ H_{14} &= -2x + 2y - 2, \\ H_{15} &= x^2 + 2y, \\ H_{16} &= x^2 + x - 2y, \\ H_{17} &= x^2 + x + 2y, \\ H_{18} &= x^2 + x - 2y + 1, \\ H_{19} &= x^2 - x - 2y + 1, \\ H_{20} &= x^2 - 2x - 2y, \\ H_{21} &= x^2 + x + 2y + 1, \\ H_{22} &= x^2 + x - 2y + 2, \\ H_{23} &= x^2 - x - 2y + 2, \\ H_{24} &= x^2 - 2x - 2y - 1, \\ H_{25} &= 2x^2 + 2y, \\ H_{26} &= 2x^2 + 2y - 1, \\ H_{27} &= 2x^2 + 2x + 2y, \\ H_{28} &= 2x^2 + 2y - 2, \\ H_{29} &= 2x^2 + 2y + 2, \\ H_{30} &= 2x^2 - x + 2y - 1, \\ H_{31} &= 2x^2 - x + 2y + 1, \\ H_{32} &= 2x^2 - 2x + 2y, \\ H_{33} &= 2x^2 + 2x + 2y - 1, \\ H_{34} &= 2x^2 + x + 2y - 2, \\ H_{35} &= 2x^2 + 2x + 2y + 2, \\ H_{36} &= 2x^2 - 2x + 2y + 2, \end{aligned}$$

$$\begin{aligned}
H_{37} &= 2x^2 - 2y, \\
H_{38} &= 2x^2 - 2y + 1, \\
H_{39} &= 2x^2 + x - 2y + 1, \\
H_{40} &= 2x^2 - x - 2y + 1, \\
H_{41} &= 2x^2 + x - 2y + 2, \\
H_{42} &= 1, \\
H_{43} &= -1, \\
H_{44} &= -2, \\
H_{45} &= 2, \\
H_{46} &= 6, \\
H_{47} &= -3.
\end{aligned}$$

The edges are  $\{0, 3\}, \{0, 6\}, \{0, 7\}, \{0, 8\}, \{0, 20\}, \{0, 26\}, \{0, 27\}, \{0, 31\}, \{0, 35\}, \{1, 5\}, \{1, 9\}, \{1, 10\}, \{1, 12\}, \{1, 18\}, \{1, 26\}, \{1, 27\}, \{1, 30\}, \{1, 32\}, \{2, 4\}, \{2, 12\}, \{2, 25\}, \{2, 35\}, \{2, 36\}, \{3, 5\}, \{3, 18\}, \{3, 25\}, \{3, 31\}, \{3, 34\}, \{3, 37\}, \{4, 8\}, \{4, 14\}, \{4, 17\}, \{4, 19\}, \{4, 25\}, \{4, 32\}, \{4, 34\}, \{4, 38\}, \{4, 39\}, \{4, 40\}, \{5, 17\}, \{5, 18\}, \{5, 20\}, \{5, 26\}, \{5, 32\}, \{5, 40\}, \{6, 9\}, \{6, 11\}, \{6, 13\}, \{6, 16\}, \{6, 28\}, \{6, 33\}, \{6, 38\}, \{6, 39\}, \{6, 40\}, \{7, 11\}, \{7, 17\}, \{7, 21\}, \{7, 24\}, \{7, 25\}, \{7, 33\}, \{7, 35\}, \{7, 39\}, \{8, 11\}, \{8, 16\}, \{8, 24\}, \{8, 28\}, \{8, 29\}, \{8, 30\}, \{8, 31\}, \{8, 34\}, \{9, 13\}, \{9, 17\}, \{9, 20\}, \{9, 25\}, \{9, 28\}, \{9, 33\}, \{9, 34\}, \{9, 39\}, \{10, 17\}, \{10, 18\}, \{10, 35\}, \{10, 37\}, \{10, 39\}, \{11, 13\}, \{11, 29\}, \{11, 34\}, \{11, 36\}, \{11, 40\}, \{12, 18\}, \{12, 25\}, \{12, 26\}, \{12, 34\}, \{12, 39\}, \{13, 14\}, \{13, 38\}, \{13, 40\}, \{14, 20\}, \{14, 23\}, \{14, 25\}, \{14, 27\}, \{14, 41\}, \{15, 17\}, \{15, 20\}, \{15, 21\}, \{15, 26\}, \{16, 21\}, \{16, 27\}, \{16, 29\}, \{16, 39\}, \{17, 25\}, \{17, 29\}, \{17, 30\}, \{17, 32\}, \{17, 40\}, \{17, 41\}, \{18, 19\}, \{18, 21\}, \{18, 26\}, \{18, 37\}, \{19, 24\}, \{19, 33\}, \{19, 37\}, \{19, 38\}, \{19, 39\}, \{20, 21\}, \{20, 37\}, \{21, 25\}, \{21, 26\}, \{21, 27\}, \{21, 29\}, \{21, 31\}, \{22, 23\}, \{22, 30\}, \{23, 37\}, \{23, 41\}, \{24, 31\}, \{25, 28\}, \{25, 39\}, \{26, 31\}, \{26, 33\}, \{26, 34\}, \{26, 36\}, \{28, 30\}, \{29, 36\}, \{30, 40\}, \{31, 34\}, \{31, 35\}, \{33, 41\}, \{34, 39\}, \{34, 40\}, \{35, 37\}, \{36, 38\}, \{37, 39\}, \{38, 40\}, \{39, 40\}, \{9, 43\}, \{10, 44\}, \{12, 43\}, \{15, 45\}, \{17, 45\}, \{18, 45\}, \{22, 46\}, \{23, 45\}, \{24, 45\}, \{30, 43\}, \{31, 47\}, \{41, 45\}. For a vertex  $i$ , we write  $j \sim i$  if  $\{i, j\}$  is an edge. For every non-constant vertex  $i$ , Table 1 gives a positive integer  $m_i$  and polynomials  $W_i, A_i$  and  $B_i$  with integer coefficients such that$

$$m_i^2 \prod_{j \sim i} H_j - W_i^2 = A_i F + B_i H_i. \quad (2)$$

For every edge  $\{i, j\}$  between two non-constant vertices, Table 2 gives a positive integer  $k_{ij}$  and polynomials  $\alpha_{ij}, \beta_{ij}$  and  $\gamma_{ij}$  with integer coefficients such that

$$\alpha_{ij} F + \beta_{ij} H_i + \gamma_{ij} H_j = k_{ij}. \quad (3)$$

These identities can be checked by expanding both sides. For the signs of the vertices, we regard  $F$  as a polynomial in  $y$ ; it has degree 3 and leading coefficient 2, and its discriminant is

$$\Delta(x) = -432x^{10} + 432x^5 - 8x^3 - 108.$$

For  $i \in I = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41\}$ , the vertex  $H_i$  has the form  $k_i y + r_i(x)$  with a non-zero integer  $k_i$ , and Table 3 lists the polynomial  $\varphi_i(x) = k_i^3 F(x, -r_i(x)/k_i)$ , which has integer coefficients.

### 3 Proof of Theorem 1

Suppose that  $(x, y) \in \mathbb{Z}^2$  is a solution of (1), so that  $F(x, y) = 0$ ; from now on the polynomials denote their values at  $(x, y)$ .

**Step 1: the values are non-zero.** The non-constant  $H_i$  are non-zero by Step 2.

**Step 2: signs.** *The signs of  $H_i$  for  $i \in I$ .* If  $\Delta(x) < 0$ , then by Lemma 7, applied to  $F(x, z)$  as a polynomial in  $z$ ,  $y$  is its only real root, and  $F(x, z)$  has the sign of  $2(z - y)$  for  $z \neq y$ . For  $i \in I$  and  $z = -r_i(x)/k_i$ , we have  $z - y = -H_i/k_i$ , so if moreover  $\varphi_i(x) = k_i^3 F(x, z) \neq 0$ , then  $H_i \neq 0$ , and  $H_i$  has the sign of  $-\varphi_i(x)$ . Expanding  $\Delta(8 + t)$  and the  $\varphi_i(8 + t)$  as polynomials in  $t$  gives in each case coefficients of one sign and a non-zero constant term, and likewise for  $\Delta(-11 - t)$  and the  $\varphi_i(-11 - t)$ . Hence, for  $x \geq 8$  and for  $x \leq -11$ , we have  $\Delta(x) < 0$ , and  $\varphi_i(x)$  is non-zero and has the sign of the constant term of the corresponding expansion; the resulting signs of the  $H_i$  are listed in Table 3. It remains to consider  $-10 \leq x \leq 7$ . Checking the residues of  $y$ , we find that for  $x \in \{-10, -8, -6, -4, -2, 0, 2, 4, 6\}$ , we have  $F(x, y) \not\equiv 0 \pmod{2}$  for every  $y$ , for  $x \in \{-5, 1, 7\}$ , we have  $F(x, y) \not\equiv 0 \pmod{3}$  for every  $y$ , for  $x \in \{-9, -3, -1\}$ , we have  $F(x, y) \not\equiv 0 \pmod{5}$  for every  $y$ , for  $x = -7$ , we have  $F(x, y) \not\equiv 0 \pmod{7}$  for every  $y$ , for  $x = 5$ , we have  $F(x, y) \not\equiv 0 \pmod{11}$  for every  $y$ , and for  $x = 3$ , we have  $F(x, y) \not\equiv 0 \pmod{19}$  for every  $y$ . Hence (1) has no solution with  $-10 \leq x \leq 7$ .

For a place  $v$ , put  $B_v = \prod_{\{i,j\}} (H_i, H_j)_v$ , the product over the edges. By Theorem 2(i),  $(H_i, H_j)_\infty = -1$  exactly when  $H_i < 0$  and  $H_j < 0$ . Here  $H_i > 0$  for  $i \in \{42, 45, 46\}$  and  $H_i < 0$  for  $i \in \{43, 44, 47\}$ . The other signs are given in Table 3 for  $x \geq 8$  and for  $x \leq -11$ . In both cases, counting the edges whose two values are negative gives  $B_\infty = 1$ .

**Step 3: odd primes outside  $S$ .** Let  $S = \{2, 3, 5, 7, 11, 13, 17\}$  be the set of primes dividing 2, the  $m_i$ , the  $k_{ij}$  and the constant vertices. Let  $p \notin S$  be an odd prime; we show that  $B_p = 1$ . For an edge  $\{i, j\}$  such that  $p$  divides neither  $H_i$  nor  $H_j$ ,  $(H_i, H_j)_p = 1$  by Theorem 2(ii). Now let  $i$  be a vertex with  $p \mid H_i$ . Then  $H_i$  is not constant, as  $p \notin S$ . For every edge  $\{i, j\}$ ,  $p \nmid H_j$ : if  $H_j$  is constant, because  $p \notin S$ , and otherwise by (3), because  $p \nmid k_{ij}$ . In particular, no edge has both values divisible by  $p$ . By Theorem 2(v), the edges at  $i$  contribute  $\prod_{j \sim i} (H_i, H_j)_p = (H_i, \Pi_i)_p$ , where  $\Pi_i = \prod_{j \sim i} H_j$  is not divisible by  $p$ . By (2),  $m_i^2 \Pi_i \equiv W_i^2 \pmod{p}$ , and  $p \nmid m_i$ , so  $\Pi_i$  is a non-zero square modulo  $p$ , and  $(H_i, \Pi_i)_p = (\Pi_i, H_i)_p = 1$  by Lemma 3. Hence  $B_p = 1$ .

**Step 4: the primes in  $S$ .** Let  $p \in S$ . By Lemma 4,  $B_p = (-1)^N$ , where  $N = \sum_{\{i,j\}} \beta_p(c_p(H_i), c_p(H_j))$ , the sum over the edges. Suppose that  $x \equiv a$  and  $y \equiv b \pmod{p^k}$ . Then  $H_i \equiv H_i(a, b) \pmod{p^k}$ , and Lemma 5 gives a vector  $b_i$  and a subspace  $D_i$  with  $c_p(H_i) \in b_i + D_i$ ; for a constant vertex, we take  $b_i = c_p(H_i)$  and  $D_i = 0$ . If the hypotheses of Lemma 6 hold for the edges, then  $N = \sum_{\{i,j\}} \beta_p(b_i, b_j)$ , so  $B_p$  is determined by  $a, b$  and  $k$ . Starting from the solutions of (1) modulo  $p$ , we refine a residue class  $x \equiv a, y \equiv b \pmod{p^k}$  to the classes modulo  $p^{k+1}$  that contain solutions of (1) modulo  $p^{k+1}$ , until these hypotheses hold, or until no solution remains. Table 4 lists the resulting terminal classes together with the value of  $B_p$ ; every integer solution lies in one of them. The result is  $B_2 = 1, B_3 = -1, B_5 = 1, B_7 = 1, B_{11} = 1, B_{13} = 1, B_{17} = 1$ .

**Step 5: contradiction.** By Theorem 2(iv), applied to each edge,  $\prod_v B_v = 1$ , the product over all primes and  $v = \infty$ . By Step 3, the factors with  $p \notin S$  are 1, so  $\prod_{p \in S} B_p = B_\infty = 1$  by Step 2. But by Step 4,  $\prod_{p \in S} B_p = -1$ . This contradiction proves the theorem.  $\square$

## References

- [1] K. Ireland and M. Rosen, *A Classical Introduction to Modern Number Theory*, 2nd ed., Graduate Texts in Mathematics 84, Springer, New York, 1990.
- [2] J.-P. Serre, *A Course in Arithmetic*, Graduate Texts in Mathematics 7, Springer, New York, 1973.

Table 1: The data of the identities (2).

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|                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $i = 0, m = 1, \text{ neighbours } 3, 6, 7, 8, 20, 26, 27, 31, 35$           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| $W$                                                                          | $8x^4 + 2x^3 - 4x^2 - 2$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| $A$                                                                          | $-16x^9 + 48x^8y + 24x^8 + 32x^7y^2 - 136x^7y + 40x^7 - 192x^6y^3$<br>$+208x^6y^2 + 96x^6y - 28x^6 + 128x^5y^4 + 96x^5y^3 - 600x^5y^2$<br>$+148x^5y - 40x^5 - 448x^4y^4 + 384x^4y^3 + 80x^4y^2 - 124x^4y - 24x^4$<br>$+256x^3y^5 + 752x^3y^4 - 1232x^3y^3 + 84x^3y^2 - 52x^3y + 8x^3$<br>$-480x^2y^5 + 792x^2y^4 - 48x^2y^3 - 252x^2y^2 - 150x^2y + 16x^2$<br>$-64xy^6 + 1744xy^5 - 1096xy^4 - 260xy^3 + 58xy^2 + 152xy - 128y^7 - 672y^6$<br>$+1624y^5 - 20y^4 - 96y^3 - 374y^2 + 134y + 4$                                                                                                                                                                                                                                                                                                                                              |
| $B$                                                                          | $-576x^4y^6 + 1216x^4y^5 - 184x^4y^4 - 164x^4y^3 + 182x^4y^2$<br>$+202x^4y - 90x^4 - 256x^3y^7 - 1392x^3y^6 + 1920x^3y^5 - 128x^3y^4$<br>$+388x^3y^3 - 610x^3y^2 + 153x^3y - 18x^3 - 416x^2y^7 - 440x^2y^6$<br>$-56x^2y^5 + 272x^2y^4 + 492x^2y^3 + 31x^2y^2 - 198x^2y - 83x^2$<br>$+64xy^8 - 1936xy^7 + 1432xy^6 - 456xy^5 + 888xy^4 - 700xy^3 + 41xy^2 - 30xy + 74x$<br>$+128y^9 + 416y^8 - 1752y^7 + 212y^6 - 176y^5 + 1106y^4 - 152y^3 - 44y^2 - 187y$<br>$+67$                                                                                                                                                                                                                                                                                                                                                                       |
| $i = 1, m = 1, \text{ neighbours } 5, 9, 10, 12, 18, 26, 27, 30, 32$         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| $W$                                                                          | $12x^4 + 6x^3 - 6x^2$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| $A$                                                                          | $128x^8y + 128x^8 + 384x^7y^2 + 320x^7y - 64x^7 + 384x^6y^3 + 576x^6y^2$<br>$-96x^6y - 288x^6 + 128x^5y^4 + 960x^5y^3 + 288x^5y^2 - 560x^5y - 16x^5$<br>$+832x^4y^4 + 768x^4y^3 - 752x^4y^2 - 400x^4y + 288x^4 + 256x^3y^5$<br>$+256x^3y^4 + 160x^3y^3 - 448x^3y^2 - 312x^3y + 224x^3 - 704x^2y^5$<br>$+1440x^2y^4 - 464x^2y^3 - 1224x^2y^2 + 1168x^2y - 288x^2 - 640xy^6$<br>$-640xy^5 + 1024xy^4 - 592xy^3 - 376xy^2 + 880xy - 306x - 128y^7 - 3072y^6$<br>$+3008y^5 + 320y^4 - 2152y^3 + 2432y^2 - 992y + 92$                                                                                                                                                                                                                                                                                                                          |
| $B$                                                                          | $-1856x^4y^6 - 672x^4y^5 + 1472x^4y^4 - 672x^4y^3 + 168x^4y^2$<br>$+696x^4y - 396x^4 - 256x^3y^7 - 2240x^3y^6 + 1792x^3y^5 + 832x^3y^4$<br>$-1064x^3y^3 + 1584x^3y^2 - 600x^3y - 208x^3 - 192x^2y^7 + 128x^2y^6$<br>$+1264x^2y^5 - 624x^2y^4 + 224x^2y^3 + 548x^2y^2 - 994x^2y + 304x^2$<br>$+640xy^8 + 512xy^7 + 1408xy^6 - 400xy^5 - 880xy^4 + 620xy^3 - 984xy^2 - 176xy$<br>$+306x + 128y^9 + 2880y^8 - 1184y^7 - 848y^6 - 256y^5 - 880y^4 + 768y^3 - 832y^2$<br>$+808y - 92$                                                                                                                                                                                                                                                                                                                                                          |
| $i = 2, m = 1, \text{ neighbours } 4, 12, 25, 35, 36$                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| $W$                                                                          | $8x^4 - 12x^2 - 4$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| $A$                                                                          | $-32x^3 + 16x^2y - 16x^2 + 16xy^2 - 32xy + 112x + 80y + 16$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| $B$                                                                          | $48x^4y^2 - 112x^4y - 32x^4 + 32x^3y^2 + 64x^3y - 24x^3 + 48x^2y^3$<br>$-72x^2y^2 - 32x^2 - 16xy^4 + 32xy^3 - 16xy^2 - 40x + 16y^4 - 80y^3 - 48y^2 + 56$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| $i = 3, m = 8, \text{ neighbours } 0, 5, 18, 25, 31, 34, 37$                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| $W$                                                                          | $-60x^4 + 50x^3 - 50x^2 + 34x + 18$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| $A$                                                                          | $2048x^5y^2 + 2048x^5y + 2048x^4y^2 + 2048x^4y + 512x^3y^2 + 512x^3y$<br>$-1800x^3 + 4096x^2y^3 + 4096x^2y^2 + 3000x^2 - 8192xy^4 - 3072xy^3$<br>$+4096xy^2 - 1024xy - 4250x - 2048y^5 + 1024y^4 - 3264y^3 - 6656y^2 - 964y + 2986$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| $B$                                                                          | $-6144x^4y^4 - 3712x^4y^3 + 6144x^4y^2 + 5000x^4y + 3108x^4$<br>$-3840x^3y^4 - 7296x^3y^3 - 2320x^3y^2 + 584x^3y - 632x^3 - 1536x^2y^5$<br>$-2560x^2y^4 - 3152x^2y^3 - 2976x^2y^2 - 640x^2y - 832x^2 + 7168xy^6$<br>$+3584xy^5 - 3424xy^4 - 1152xy^3 + 3424xy^2 + 5274xy + 2812x + 2048y^7 + 4096y^6$<br>$+5312y^5 + 3168y^4 + 3060y^3 - 2064y^2 - 4360y - 2662$                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| $i = 4, m = 1, \text{ neighbours } 2, 8, 14, 17, 19, 25, 32, 34, 38, 39, 40$ |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| $W$                                                                          | $-32x^4 - 20x^2 - 50x - 18$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| $A$                                                                          | $256x^{14} - 256x^{13}y + 128x^{13} - 256x^{12}y^2 + 128x^{12}y - 192x^{12}$<br>$+256x^{11}y^3 - 128x^{11}y^2 - 832x^{11}y + 288x^{11} - 128x^{10}y^3$<br>$-704x^{10}y^2 + 416x^{10}y - 96x^{10} + 2368x^9y^3 - 2976x^9y^2 + 1696x^9y$<br>$-144x^9 + 1152x^8y^4 - 1312x^8y^3 + 768x^8y^2 - 528x^8y + 128x^8$<br>$-1536x^7y^5 + 2560x^7y^4 + 1312x^7y^3 - 3728x^7y^2 + 1984x^7y - 384x^7$<br>$-256x^6y^6 + 896x^6y^5 + 1696x^6y^4 - 432x^6y^3 + 16x^6y^2 - 496x^6y$<br>$+160x^6 + 128x^5y^6 - 5952x^5y^5 + 12416x^5y^4 - 9408x^5y^3$<br>$+2384x^5y^2 - 216x^5y - 32x^5 - 3904x^4y^6 + 4672x^4y^5 - 4112x^4y^4$<br>$+4544x^4y^3 - 3112x^4y^2 + 1232x^4y - 104x^4 + 2688x^3y^7 - 7648x^3y^6$<br>$+2528x^3y^5 + 8576x^3y^4 - 9432x^3y^3 + 4528x^3y^2 - 1240x^3y - 352x^3$<br>$+1536x^2y^8 - 3584x^2y^7 - 448x^2y^6 + 1008x^2y^5 - 1592x^2y^4$ |

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$$\begin{aligned}
& +3064x^2y^3 - 2752x^2y^2 + 1168x^2y - 208x^2 + 256xy^9 - 896xy^8 \\
& +5856xy^7 - 19280xy^6 + 18608xy^5 - 5952xy^4 + 64xy^3 + 964xy^2 - 568xy - 528x \\
& -128y^9 + 7360y^8 - 13344y^7 + 12288y^6 - 8792y^5 + 7240y^4 - 3884y^3 + 592y^2 \\
& -40y - 1616 \\
B \quad & 576x^4y^8 + 5696x^4y^7 + 2560x^4y^6 - 6160x^4y^5 + 5288x^4y^4 \\
& -1748x^4y^3 + 1468x^4y^2 + 4x^4y + 828x^4 - 2688x^3y^9 + 4704x^3y^8 \\
& +2304x^3y^7 - 3488x^3y^6 + 2384x^3y^5 - 1812x^3y^4 + 3272x^3y^3 \\
& -20x^3y^2 + 1692x^3y + 1176x^3 - 1536x^2y^{10} + 1920x^2y^9 - 512x^2y^8 \\
& +8224x^2y^7 + 256x^2y^6 + 816x^2y^5 + 1656x^2y^4 + 796x^2y^3 \\
& +1958x^2y^2 + 866x^2y + 1714x^2 - 256xy^{11} + 640xy^{10} - 5664xy^9 + 11856xy^8 \\
& -3216xy^7 + 1760xy^6 - 4636xy^5 + 372xy^4 - 126xy^3 + 138xy^2 + 640xy + 1164x \\
& +128y^{11} - 6208y^{10} + 3040y^9 - 2656y^8 + 4184y^7 - 7104y^6 + 2284y^5 \\
& -2640y^4 + 1020y^3 + 694y^2 + 990y + 970 \\
i = 5, m = 2, \text{ neighbours } 1, 3, 17, 18, 20, 26, 32, 40 \\
W \quad & -78x^3 + 84x^2 - 60x + 90 \\
A \quad & 32x^8y - 16x^8 + 64x^7y^2 - 112x^7y + 40x^7 - 128x^6y^2 + 64x^6y \\
& -64x^5y^3 - 96x^5y^2 + 120x^5y - 28x^5 - 352x^4y^3 + 320x^4y^2 \\
& -48x^4y - 12x^4 - 480x^3y^4 + 992x^3y^3 - 584x^3y^2 + 128x^3y - 12x^3 \\
& -64x^2y^5 + 1072x^2y^4 - 1832x^2y^3 + 1184x^2y^2 - 368x^2y + 52x^2 \\
& +448xy^5 - 1280xy^4 + 1552xy^3 - 932xy^2 + 250xy - 3062x + 64y^6 + 736y^5 - 648y^4 \\
& -20y^3 + 72y^2 + 22y + 6542 \\
B \quad & 1248x^4y^5 - 3568x^4y^4 + 5712x^4y^3 - 7296x^4y^2 + 8048x^4y - 8202x^4 \\
& +480x^3y^6 - 2464x^3y^5 + 5648x^3y^4 - 8844x^3y^3 + 11032x^3y^2 \\
& -11884x^3y + 12046x^3 + 64x^2y^7 - 1392x^2y^6 + 4280x^2y^5 \\
& -6968x^2y^4 + 8792x^2y^3 - 10030x^2y^2 + 10665x^2y - 9334x^2 - 448xy^7 \\
& +1184xy^6 - 1952xy^5 + 2408xy^4 - 2840xy^3 + 6546xy^2 - 7015xy + 3869x - 64y^8 \\
& -1184y^7 + 3112y^6 - 4340y^5 + 5276y^4 - 5782y^3 - 754y^2 + 790y - 779 \\
i = 6, m = 16, \text{ neighbours } 0, 9, 11, 13, 16, 28, 33, 38, 39, 40 \\
W \quad & 774x^4 - 336x^3 - 1368x^2 - 744x + 522 \\
A \quad & 32768x^{10}y + 131072x^9y^2 + 147456x^8y^3 + 163840x^8y^3 + 344064x^8y^2 \\
& +204800x^8y + 65536x^7y^4 + 32768x^7y^3 + 49152x^7y^2 + 61440x^7y \\
& -360448x^6y^4 - 876544x^6y^3 - 299008x^6y^2 - 69632x^6y - 196608x^5y^5 \\
& -1097728x^5y^4 - 663552x^5y^3 + 102400x^5y^2 - 94208x^5y \\
& -458752x^4y^5 - 57344x^4y^4 + 356352x^4y^3 + 323584x^4y^2 - 73728x^4y \\
& -163840x^3y^6 + 819200x^3y^5 + 385024x^3y^4 + 45056x^3y^3 \\
& +323584x^3y^2 + 7168x^3y - 299538x^3 - 65536x^2y^7 + 1081344x^2y^6 \\
& +729088x^2y^5 - 831488x^2y^4 + 176128x^2y^3 + 257024x^2y^2 - 6144x^2y \\
& +260064x^2 + 753664xy^7 + 1286144xy^6 - 356352xy^5 - 405504xy^4 + 106496xy^3 \\
& -131072xy^2 - 50176xy + 1002384x + 196608y^8 + 1048576y^7 - 679936y^6 - 303616y^5 \\
& -1988096y^4 - 1646704y^3 - 553304y^2 + 122828y + 781470 \\
B \quad & 360448x^4y^7 - 1851392x^4y^6 - 1188864x^4y^5 - 2313216x^4y^4 \\
& -2840800x^4y^3 - 1036976x^4y^2 + 311192x^4y + 1330524x^4 + 196608x^3y^8 \\
& -2277376x^3y^7 - 2172928x^3y^6 - 2307072x^3y^5 - 1742272x^3y^4 \\
& +56192x^3y^3 + 1474016x^3y^2 + 2105614x^3y + 1848924x^3 + 65536x^2y^9 \\
& -1671168x^2y^8 - 2281472x^2y^7 - 2310144x^2y^6 - 591744x^2y^5 \\
& +1639616x^2y^4 + 2088988x^2y^3 + 2532412x^2y^2 + 1836938x^2y + 135414x^2 \\
& -851968xy^9 - 1925120xy^8 - 1798144xy^7 + 319744xy^6 + 2939136xy^5 \\
& +3406712xy^4 + 3239964xy^3 + 846552xy^2 - 563726xy - 1270134x - 196608y^{10} \\
& -1015808y^9 - 24576y^8 + 258560y^7 + 3002112y^6 + 2684912y^5 + 1616976y^4 \\
& -189604y^3 - 1736488y^2 - 1140800y - 508986 \\
i = 7, m = 64, \text{ neighbours } 0, 11, 17, 21, 24, 25, 33, 35, 39 \\
W \quad & -2056x^4 + 794x^3 + 998x^2 - 118x - 730 \\
A \quad & 65536x^{10}y + 131072x^9y^2 + 294912x^8y^3 + 589824x^8y^2 + 327680x^8y \\
& +524288x^7y^3 + 557056x^7y^2 - 671744x^7y + 65536x^6y^3 \\
& -3276800x^6y^2 - 2981888x^6y - 65536x^5y^4 - 6324224x^5y^3 \\
& -12730368x^5y^2 - 5390336x^5y - 131072x^4y^5 - 6127616x^4y^4 \\
& -22708224x^4y^3 - 23150592x^4y^2 - 5947392x^4y - 2949120x^3y^5 \\
& -21463040x^3y^4 - 41451520x^3y^3 - 26566656x^3y^2 - 4390912x^3y \\
& -2113568x^3 - 524288x^2y^6 - 10911744x^2y^5 - 37142528x^2y^4 \\
& -47169536x^2y^3 - 20152320x^2y^2 - 2465792x^2y + 1632464x^2 - 2818048xy^6 \\
& -13402112xy^5 - 37339136xy^4 - 35266560xy^3 - 11116544xy^2 - 1900544xy + 1736670x \\
& +131072y^7 + 9306112y^6 + 22867968y^5 - 2674688y^4 - 19164032y^3 - 11744832y^2
\end{aligned}$$


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$-8393084y - 2492110$   
 $B \quad 131072x^4y^7 + 8454144x^4y^6 + 43638784x^4y^5 + 34824192x^4y^4$   
 $-4232960x^4y^3 - 8481920x^4y^2 - 11723512x^4y - 2914180x^4$   
 $+3211264x^3y^7 + 35307520x^3y^6 + 63610880x^3y^5 + 27963904x^3y^4$   
 $-3198976x^3y^3 - 12513904x^3y^2 - 14012960x^3y + 896200x^3$   
 $+524288x^2y^8 + 14647296x^2y^7 + 47742976x^2y^6 + 47809536x^2y^5$   
 $+11850240x^2y^4 - 6403360x^2y^3 - 17356464x^2y^2 - 6262720x^2y$   
 $+1615000x^2 + 2752512xy^8 + 12058624xy^7 + 24610816xy^6 + 15919104xy^5$   
 $+3524544xy^4 - 2224352xy^3 - 12056112xy^2 + 1169742xy - 1460620x - 131072y^9$   
 $-8192000y^8 - 16674816y^7 + 11601920y^6 + 16967552y^5 + 14236288y^4 + 68668y^3$   
 $-7058704y^2 - 2343064y - 3025010$

$i = 8, m = 1, \text{ neighbours } 0, 4, 11, 16, 24, 28, 29, 30, 31, 34$   
 $W \quad 186x^4 + 98x^3 - 50x^2 - 74x + 16$   
 $A \quad 64x^{10}y^2 - 64x^{10}y + 128x^9y^3 - 96x^9y^2 - 32x^9y - 128x^8y^3$   
 $-304x^8y^2 + 432x^8y + 128x^7y^4 - 672x^7y^3 + 248x^7y^2 + 296x^7y$   
 $-768x^6y^4 - 32x^6y^3 + 1272x^6y^2 - 472x^6y - 832x^5y^5$   
 $-1312x^5y^4 + 896x^5y^3 + 1472x^5y^2 - 224x^5y - 128x^4y^6$   
 $-160x^4y^5 - 16x^4y^4 + 1608x^4y^3 - 1032x^4y^2 - 272x^4y$   
 $-1152x^3y^6 - 752x^3y^5 + 1432x^3y^4 + 2844x^3y^3 - 2156x^3y^2$   
 $-216x^3y - 17298x^3 - 128x^2y^7 + 2528x^2y^6 - 2520x^2y^5 + 160x^2y^4$   
 $-916x^2y^3 + 232x^2y^2 + 644x^2y - 18228x^2 + 1408xy^7 - 192xy^6 + 24xy^5$   
 $+2392xy^4 - 3728xy^3 + 12xy^2 + 84xy + 4498x + 960y^8 + 8192y^7 - 6096y^6$   
 $-18408y^5 - 2160y^4 + 25194y^3 + 22212y^2 - 6866y + 256$   
 $B \quad 128x^4y^8 + 4128x^4y^7 - 1056x^4y^6 - 15840x^4y^5 - 3612x^4y^4$   
 $+24246x^4y^3 + 21644x^4y^2 - 6610x^4y - 18408x^4 + 1280x^3y^8$   
 $-624x^3y^7 - 11792x^3y^6 - 5336x^3y^5 + 26724x^3y^4 + 20202x^3y^3$   
 $-7426x^3y^2 - 27057x^3y - 3026x^3 + 128x^2y^9 - 2848x^2y^8$   
 $-6792x^2y^7 - 2632x^2y^6 + 22160x^2y^5 + 19396x^2y^4 - 25046x^2y^3$   
 $-25761x^2y^2 - 12072x^2y + 13917x^2 - 2368xy^9 - 7712xy^8 + 1336xy^7$   
 $+18776xy^6 + 20340xy^5 - 26298xy^4 - 43669xy^3 - 11874xy^2 + 15940xy + 11052x$   
 $-960y^{10} - 7680y^9 + 5840y^8 + 17480y^7 + 6832y^6 - 26898y^5 - 31800y^4$   
 $+5274y^3 + 12405y^2 + 11170y - 3433$

$i = 9, m = 4, \text{ neighbours } 43, 1, 6, 13, 17, 20, 25, 28, 33, 34, 39$   
 $W \quad -96x^4 + 84x^3 - 54x^2 + 36x + 27$   
 $A \quad 1024x^{11}y - 512x^{11} - 1024x^{10}y^2 + 3584x^{10}y - 1536x^{10} - 2048x^9y^3$   
 $+2048x^9y^2 - 2304x^9y + 896x^9 - 5120x^8y^3 + 8448x^8y^2 - 18048x^8y$   
 $+7552x^8 - 6144x^7y^4 + 11776x^7y^3 - 22528x^7y^2 - 4864x^7y + 6976x^7$   
 $+9216x^6y^4 + 6912x^6y^3 - 65536x^6y^2 + 39104x^6y - 4608x^6$   
 $+5120x^5y^5 + 56320x^5y^4 - 87552x^5y^3 - 5504x^5y^2 + 39936x^5y$   
 $-11328x^5 + 2048x^4y^6 + 63488x^4y^5 - 14336x^4y^4 - 118912x^4y^3$   
 $+97600x^4y^2 - 12224x^4y - 4544x^4 + 33792x^3y^6 + 75008x^3y^5$   
 $-129664x^3y^4 - 2944x^3y^3 + 90816x^3y^2 - 45728x^3y + 1152x^3$   
 $+6144x^2y^7 + 62976x^2y^6 - 12032x^2y^5 - 109184x^2y^4$   
 $+109056x^2y^3 - 12960x^2y^2 - 22592x^2y + 15136x^2 + 13312xy^7 + 5376xy^6$   
 $+18176xy^5 - 87872xy^4 + 138624xy^3 - 99392xy^2 + 26176xy - 9544x - 3072y^8$   
 $-17408y^7 + 90368y^6 - 145920y^5 + 132864y^4 - 78592y^3 + 14640y^2 - 1384y + 2820$   
 $B \quad -2048x^4y^8 - 51200x^4y^7 + 43776x^4y^6 + 51584x^4y^5 - 81984x^4y^4$   
 $+75360x^4y^3 - 61904x^4y^2 + 34120x^4y - 748x^4 - 27648x^3y^8$   
 $-77056x^3y^7 + 173056x^3y^6 - 116992x^3y^5 + 62208x^3y^4$   
 $-53536x^3y^3 + 5056x^3y^2 + 14512x^3y - 4792x^3 - 6144x^2y^9$   
 $-74240x^2y^8 + 52480x^2y^7 + 75776x^2y^6 - 109696x^2y^5$   
 $+89216x^2y^4 - 81856x^2y^3 + 58288x^2y^2 - 19408x^2y - 2144x^2$   
 $-16384xy^9 - 47360xy^8 + 105088xy^7 - 77824xy^6 + 27648xy^5 - 19520xy^4$   
 $-6032xy^3 + 30392xy^2 - 14340xy + 7306x + 3072y^{10} + 10752y^9 - 60416y^8$   
 $+93696y^7 - 102912y^6 + 96384y^5 - 55280y^4 + 44784y^3 - 20236y^2 - 2798y - 2091$

$i = 10, m = 2, \text{ neighbours } 44, 1, 17, 18, 35, 37, 39$   
 $W \quad -8x^4 - 8x^3 - 12x^2 - 8x - 9$   
 $A \quad -64x^5y + 32x^5 - 224x^4y + 112x^4 + 64x^3y^2 - 480x^3y + 192x^3$   
 $+256x^2y^2 - 704x^2y + 224x^2 + 320xy^3 + 288xy^2 - 736xy + 128x + 64y^4$   
 $+480y^3 - 192y^2 - 280y + 80$   
 $B \quad -32x^4y^3 + 576x^4y^2 - 384x^4y - 64x^4 - 64x^3y^4 + 128x^3y^3$   
 $+352x^3y^2 - 256x^3y - 56x^3 - 256x^2y^4 + 32x^2y^3 + 464x^2y^2$

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$$\begin{aligned}
& -312x^2y - 28x^2 - 256xy^5 + 128xy^4 - 176xy^3 + 464xy^2 - 264xy - 14x - 64y^6 \\
& -192y^5 + 160y^4 + 40y^3 + 300y^2 - 278y - 1 \\
i = 11, m = 2, \text{ neighbours } 6, 7, 8, 13, 29, 34, 36, 40 \\
W & -14x^4 + 6x^3 + 12x^2 + 16x - 24 \\
A & -128x^7 + 512x^6y - 384x^5y^2 - 128x^5y + 96x^5 - 512x^4y^3 + 384x^4y^2 \\
& -224x^4 + 512x^3y^4 + 256x^3y^3 + 672x^3y^2 + 1120x^3y + 94x^3 \\
& -1536x^2y^4 - 2304x^2y^3 - 1216x^2y^2 - 672x^2y + 116x^2 + 1024xy^5 \\
& + 2176xy^4 + 960xy^3 + 1056xy^2 + 528xy - 42x - 1024y^5 - 3200y^4 - 1104y^3 \\
& + 1504y^2 + 1020y + 780 \\
B & 768x^4y^5 + 1280x^4y^4 - 224x^4y^3 - 2496x^4y^2 - 728x^4y - 584x^4 \\
& -512x^3y^6 + 256x^3y^5 - 1024x^3y^4 + 736x^3y^3 + 2256x^3y^2 \\
& + 930x^3y - 192x^3 + 1024x^2y^6 + 1216x^2y^5 + 704x^2y^4 - 988x^2y^3 \\
& -1892x^2y^2 - 152x^2y + 190x^2 - 1024xy^7 - 896xy^6 - 128xy^5 - 136xy^4 \\
& + 248xy^3 + 1192xy^2 - 706xy + 248x + 512y^7 + 1920y^6 - 176y^5 - 624y^4 - 2124y^3 \\
& + 280y^2 + 472y + 102 \\
i = 12, m = 1, \text{ neighbours } 43, 1, 2, 18, 25, 26, 34, 39 \\
W & -4x^4 - 10x^3 + 10x^2 - 2x \\
A & -32x^6y + 16x^6 - 32x^5y^2 - 48x^5y + 32x^5 - 64x^4y^2 - 8x^4y + 20x^4 \\
& -104x^3y^2 + 68x^3y - 16x^3 + 64x^2y^3 - 192x^2y^2 + 140x^2y - 70x^2 \\
& + 160xy^4 - 160xy^3 + 36xy^2 + 18xy - 18x + 32y^5 + 80y^4 - 240y^3 + 192y^2 - 112y + 48 \\
B & 128x^4y^4 - 40x^4y^3 - 28x^4y^2 + 40x^4y + 60x^4 + 168x^3y^4 \\
& -140x^3y^3 + 28x^3y^2 - 72x^3y + 6x^3 - 32x^2y^5 + 208x^2y^4 \\
& -136x^2y^3 + 58x^2y^2 - 39x^2y + 22x^2 - 128xy^6 + 160xy^5 - 76xy^4 \\
& + 122xy^3 - 84xy^2 + 28xy - 15x - 32y^7 - 176y^6 + 256y^5 - 144y^4 + 136y^3 - 68y^2 \\
& + 32y - 24 \\
i = 13, m = 1, \text{ neighbours } 6, 9, 11, 14, 38, 40 \\
W & 12x^4 + 12x^3 + 18x^2 \\
A & -64x^3 - 120x^2 - 40xy^2 - 20xy - 270x - 32y^3 + 12y^2 - 62y - 106 \\
B & 32x^4y^3 - 32x^4y^2 + 40x^4y - 100x^4 + 48x^3y^3 - 36x^3y^2 + 64x^3y \\
& -49x^3 - 64x^2y^4 + 128x^2y^3 - 76x^2y^2 + 50x^2y + 18x^2 - 8xy^4 \\
& + 120xy^3 - 10xy^2 + 81xy - 79x + 64y^5 - 140y^4 + 218y^3 - 80y^2 + 40y - 55 \\
i = 14, m = 1, \text{ neighbours } 4, 13, 20, 23, 25, 27, 41 \\
W & -52x^4 - 60x^3 + 20x^2 + 48x + 8 \\
A & 16x^6y - 16x^6 + 16x^5y^2 - 24x^5y + 8x^5 - 72x^4y^2 + 64x^4y + 8x^4 \\
& -48x^3y^3 - 16x^3y^2 + 56x^3y - 1344x^3 - 48x^2y^3 + 24x^2y^2 - 48x^2y \\
& -3048x^2 - 32xy^4 + 216xy^3 - 308xy^2 + 52xy - 688x - 16y^5 + 536y^4 + 132y^3 \\
& -224y^2 - 1572y + 1080 \\
B & 232x^4y^4 + 392x^4y^3 - 104x^4y^2 - 1564x^4y - 2716x^4 + 48x^3y^5 \\
& + 208x^3y^4 - 544x^3y^3 - 1068x^3y^2 - 1968x^3y - 448x^3 + 96x^2y^5 \\
& -928x^2y^4 - 1608x^2y^3 - 660x^2y^2 - 256x^2y + 1600x^2 + 48xy^6 \\
& -936xy^5 - 1036xy^4 - 1734xy^3 + 322xy^2 + 1472xy + 1236x + 16y^7 - 584y^6 \\
& -588y^5 - 372y^4 + 1340y^3 + 390y^2 + 278y - 508 \\
i = 15, m = 1, \text{ neighbours } 45, 17, 20, 21, 26 \\
W & 2x^3 - 2x \\
A & 2x^3 + 8x^2 + 4xy + 16y \\
B & -16x^5 + 4x^4y - 14x^4 - 32x^3y - 12x^3 - 16x^2y^2 - 26x^2y + 4x^2 - 4xy^3 \\
& -32xy^2 - 4xy + 8x - 32y^3 + 4y + 8 \\
i = 16, m = 4, \text{ neighbours } 6, 8, 21, 27, 29, 39 \\
W & -176x^5 - 56x^4 - 40x^3 - 32x^2 - 8x + 56 \\
A & 64x^8 - 13728x^5 - 192x^4y^2 - 1368x^4y - 12971x^4 + 256x^3y^2 - 704x^3y \\
& -7712x^3 - 512x^2y^2 - 1024x^2y - 6592x^2 + 768xy^3 - 5920xy^2 - 18500xy - 7211x \\
& -256y^4 + 13536y^3 + 29916y^2 + 12630y + 3136 \\
B & -128x^{11} + 128x^{10} - 256x^9y - 128x^9 + 512x^8y - 3136x^8 - 128x^7y^2 \\
& + 1136x^7y + 10262x^7 - 128x^6y^3 + 1152x^6y^2 - 7408x^6y - 10198x^6 \\
& -128x^5y^3 + 1120x^5y^2 + 27932x^5y + 10198x^5 - 256x^4y^4 \\
& + 2432x^4y^3 - 6144x^4y^2 + 224x^4y - 1088x^4 + 512x^3y^4 + 2176x^3y^3 \\
& -296x^3y^2 + 6219x^3y - 480x^3 - 128x^2y^5 + 7600x^2y^4 + 13654x^2y^3 \\
& -2712x^2y^2 - 3331x^2y - 6603x^2 + 640xy^5 + 720xy^4 + 2842xy^3 + 13486xy^2 \\
& + 7619xy + 3371x - 256y^6 + 14048y^5 + 30172y^4 + 12374y^3 - 3696y^2 - 14894y - 6315 \\
i = 17, m = 768, \text{ neighbours } 45, 4, 5, 7, 9, 10, 15, 25, 29, 30, 32, 40, 41 \\
W & 13805568x^5 - 13246464x^4 - 12072960x^3 - 7661568x^2 - 3379200x - 13621248
\end{aligned}$$


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$$\begin{aligned}
A \quad & -603979776x^{12}y^2 + 603979776x^{12} + 1509949440x^{11}y^2 - 1509949440x^{11} \\
& + 1811939328x^{10}y^4 - 1207959552x^{10}y^3 - 3925868544x^{10}y^2 \\
& + 1207959552x^{10}y + 2113929216x^{10} + 1207959552x^9y^5 - 2113929216x^9y^4 \\
& + 1811939328x^9y^3 + 4982833152x^9y^2 - 3019898880x^9y - 2868903936x^9 \\
& - 201326592x^8y^6 + 6077546496x^8y^5 + 5503451136x^8y^4 \\
& - 8019050496x^8y^3 + 7909318656x^8y^2 - 233848759296x^8y + 2612939329152x^8 \\
& + 4831838208x^7y^6 - 2415919104x^7y^5 - 13967032320x^7y^4 \\
& + 7474249728x^7y^3 + 10362028032x^7y^2 - 5058330624x^7y - 1226833920x^7 \\
& + 13740539904x^6y^5 + 10116661248x^6y^4 - 18761121792x^6y^3 \\
& - 10078912512x^6y^2 + 5020581888x^6y - 37748736x^6 + 1811939328x^5y^7 \\
& + 12683575296x^5y^6 - 20157825024x^5y^5 - 22196256768x^5y^4 \\
& + 20554186752x^5y^3 + 8248098816x^5y^2 - 2208301056x^5y \\
& - 151337384979264x^5 - 201326592x^4y^8 - 14558429184x^4y^7 \\
& + 8200912896x^4y^6 + 6015418368x^4y^5 - 49989943296x^4y^4 \\
& + 1372898205696x^4y^3 - 16286692310016x^4y^2 + 36072611297328x^4y \\
& + 261710174707650x^4 - 12683575296x^3y^8 + 13136560128x^3y^7 \\
& + 24083693568x^3y^6 - 28009562112x^3y^5 - 11079254016x^3y^4 \\
& + 11862540288x^3y^3 - 1944059904x^3y^2 + 3010461696x^3y + 78941289185280x^3 \\
& - 4831838208x^2y^9 - 3019898880x^2y^8 - 10947133440x^2y^7 \\
& + 905969664x^2y^6 + 22592618496x^2y^5 - 56623104x^2y^4 \\
& - 2982150144x^2y^3 + 2897215488x^2y^2 - 3831496704x^2y - 54152458665984x^2 \\
& - 10871635968xy^9 - 19176357888xy^8 + 28915531776xy^7 + 24234688512xy^6 \\
& - 15576072192xy^5 - 689504256xy^4 - 7408705536xy^3 + 280186891854912xy^2 \\
& - 673619487251544xy + 107682462964098x - 5637144576y^{10} + 4076863488y^9 \\
& - 1447034880y^8 + 64298680320y^7 - 138090774528y^6 + 180038123520y^5 \\
& + 1185301923840y^4 + 87837375590208y^3 - 558650539135656y^2 + 31249240961796y \\
& + 185538397077504 \\
B \quad & 402653184x^{11}y^6 - 1283457024x^{11}y^5 + 1374683136x^{11}y^4 \\
& - 2383282176x^{11}y^3 - 33409548288x^{11}y^2 + 475247265792x^{11}y \\
& - 5220669332736x^{11} - 402653184x^{10}y^6 + 1283457024x^{10}y^5 \\
& - 1374683136x^{10}y^4 + 2383282176x^{10}y^3 + 33409548288x^{10}y^2 \\
& - 475247265792x^{10}y + 5220669332736x^{10} - 805306368x^9y^7 \\
& + 2969567232x^9y^6 - 4032823296x^9y^5 + 6141247488x^9y^4 \\
& + 64435814400x^9y^3 - 983904079872x^9y^2 + 10916585931264x^9y \\
& - 5220669332736x^9 + 1610612736x^8y^7 - 5536481280x^8y^6 \\
& + 6782189568x^8y^5 - 10907811840x^8y^4 - 131254910976x^8y^3 \\
& + 1934398611456x^8y^2 - 21357924596736x^8y + 117304260653952x^8 \\
& - 402653184x^7y^8 + 4253024256x^7y^7 - 14615052288x^7y^6 \\
& + 24553979904x^7y^5 + 2945187840x^7y^4 - 699299414016x^7y^3 \\
& + 9038832433152x^7y^2 - 42957879445728x^7y - 274978616914692x^7 \\
& + 402653184x^6y^9 - 4102029312x^6y^8 + 8194621440x^6y^7 \\
& - 1533935616x^6y^6 - 35506176000x^6y^5 + 734124558336x^6y^4 \\
& - 8388975500544x^6y^3 + 33693721534464x^6y^2 - 191888265495072x^6y \\
& + 277588951581060x^6 + 402653184x^5y^9 - 4404019200x^5y^8 \\
& + 21035483136x^5y^7 - 47574024192x^5y^6 + 29615800320x^5y^5 \\
& + 664474269696x^5y^4 - 9688689365760x^5y^3 + 52222037356992x^5y^2 \\
& + 741845499324456x^5y - 277588951581060x^5 - 805306368x^4y^{10} \\
& + 7801405440x^4y^9 - 11985223680x^4y^8 - 17967611904x^4y^7 \\
& + 118133391360x^4y^6 - 1496421027840x^4y^5 + 16111930212864x^4y^4 \\
& - 57696072510720x^4y^3 - 228825877231104x^4y^2 + 201559978260672x^4y \\
& - 193206307123584x^4 + 7650410496x^3y^{10} - 21441282048x^3y^9 \\
& - 5536481280x^3y^8 - 9338093568x^3y^7 + 114506661888x^3y^6 \\
& - 152654979072x^3y^5 - 511876402176x^3y^4 + 96554431233408x^3y^3 \\
& - 173629350063408x^3y^2 + 29406267245694x^3y - 227630175795648x^3 \\
& + 2013265920x^2y^{11} + 5863636992x^2y^{10} - 7832862720x^2y^9 \\
& + 47509929984x^2y^8 - 89942458368x^2y^7 + 124356943872x^2y^6 \\
& + 489867580416x^2y^5 + 43708270966176x^2y^4 - 160944484544772x^2y^3 \\
& - 245778648285264x^2y^2 + 314137253877426x^2y - 11819909113470x^2 \\
& + 8053063680xy^{11} + 22825402368xy^{10} - 31677480960xy^9 + 7631536128xy^8 \\
& - 97132019712xy^7 + 171629076480xy^6 + 507668726784xy^5 - 236832799158432xy^4 \\
& + 350329382882148xy^3 + 186967208507220xy^2 + 195278793333966xy - 289916307380610x \\
& + 5637144576y^{12} - 1660944384y^{11} + 239075328y^{10} - 73157050368y^9 \\
& + 143149105152y^8 - 175778807808y^7 - 1155644000256y^6 - 87907930926912y^5
\end{aligned}$$


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$$+558741313172136y^4 - 30656439004932y^3 - 141619784779872y^2 - 279325269567828y + 15624620480898$$

$i = 18, m = 1$ , neighbours 45, 1, 3, 5, 10, 12, 19, 21, 26, 37

$$W \quad -11168x^5 - 5540x^4 - 8740x^3 - 7892x^2 - 5592x + 3168$$

$$A \quad -49152x^3y^3 - 1419264x^3y^2 - 74905344x^3y - 3774240672x^3 + 12288x^2y^4 + 419328x^2y^3 + 20589120x^2y^2 + 1041873432x^2y + 4596225729x^2 - 3072xy^5 + 19968xy^4 - 1478400xy^3 - 68394048xy^2 + 8533460184xy + 4626924545x - 7168y^5 - 26624y^4 + 1464704y^3 - 1937718512y^2 - 15865986410y + 8430623265$$

$$B \quad 128x^9y^2 + 64x^9y - 64x^9 + 512x^8y^3 - 64x^8y^2 - 416x^8y - 124724064x^8 + 640x^7y^4 - 384x^7y^3 - 416x^7y^2 + 480x^7y + 982656x^7 + 256x^6y^5 + 256x^6y^4 + 97472x^6y^3 + 2838112x^6y^2 - 99637360x^6y + 7446314528x^6 + 1024x^5y^5 - 28608x^5y^4 - 937152x^5y^3 - 44013424x^5y^2 - 1982144944x^5y - 16912863586x^5 + 512x^4y^6 + 384x^4y^5 + 183744x^4y^4 + 9477024x^4y^3 - 21312624x^4y^2 - 92509232x^4y - 76033872x^4 - 2688x^3y^6 - 42816x^3y^5 - 1970784x^3y^4 - 99497968x^3y^3 - 23528032x^3y^2 - 19099808x^3y - 1500112x^3 - 256x^2y^7 + 3712x^2y^6 + 413888x^2y^5 + 20789280x^2y^4 + 48812512x^2y^3 - 65271728x^2y^2 + 3733782208x^2y - 47396464x^2 - 5120xy^7 + 15360xy^6 - 1487232xy^5 - 68426688xy^4 + 7531474016xy^3 - 992692136xy^2 - 8388813905xy - 3758231584x - 512y^8 - 6912y^7 - 29440y^6 + 1449216y^5 - 1936990416y^4 - 16834468018y^3 + 12656864y^2 + 975187656y + 8420587041$$

$i = 19, m = 48$ , neighbours 4, 18, 24, 33, 37, 38, 39

$$W \quad 96x^5 + 48x^4 + 576x^3 - 336x^2 - 96x + 144$$

$$A \quad -1536x^8y - 6592x^8 + 36864x^7y - 36864x^7 + 18432x^6y - 18432x^6 - 221184x^5y^2 + 129024x^5y + 499040x^5 + 7680x^4y^3 - 155776x^4y^2 + 56120x^4y + 1030465x^4 + 442368x^3y^3 - 239616x^3y^2 - 516096x^3y + 256896x^3 - 36864x^2y^4 + 718848x^2y^3 - 322560x^2y^2 - 576000x^2y + 221184x^2 - 239616xy^4 + 147456xy^3 - 1264864xy^2 - 4602508xy + 1241185x + 67584y^5 - 853504y^4 + 779808y^3 - 6438548y^2 + 3605826y - 211872$$

$$B \quad 3072x^{11}y + 13184x^{11} + 3072x^{10}y + 13184x^{10} + 6144x^9y^2 + 26368x^9y + 12288x^8y^2 + 49664x^8y - 836160x^8 - 3072x^7y^3 - 33536x^7y^2 - 119216x^7y - 2463938x^7 + 3072x^6y^4 + 34688x^6y^3 + 53504x^6y^2 - 1842736x^6y - 1634370x^6 - 3072x^5y^4 - 29312x^5y^3 - 151392x^5y^2 - 6651396x^5y + 829568x^5 + 6144x^4y^5 + 63232x^4y^4 + 43008x^4y^3 + 224512x^4y^2 - 975008x^4y - 4608x^4 + 307200x^3y^5 - 504832x^3y^4 - 199360x^3y^3 - 317880x^3y^2 - 1348961x^3y + 540192x^3 - 3072x^2y^6 + 75520x^2y^5 + 290448x^2y^4 - 3063106x^2y^3 - 1007736x^2y^2 - 924937x^2y + 1568353x^2 - 273408xy^6 + 528128xy^5 - 1907440xy^4 - 1460466xy^3 - 3539842xy^2 + 850072xy + 1036225x + 67584y^7 - 672256y^6 + 75040y^5 - 6139908y^4 + 925760y^3 - 203408y^2 + 3149826y - 232608$$

$i = 20, m = 16$ , neighbours 0, 5, 9, 14, 15, 21, 37

$$W \quad -2656x^4 + 960x^3 + 832x^2 - 128x + 832$$

$$A \quad -256x^8y - 3168x^8 - 287328x^5 + 1536x^4y^3 + 30720x^4y^2 + 181096x^4y - 811293x^4 - 4096x^3y^2 - 4096x^3y - 3527168x^3 - 6144x^2y^2 - 6144x^2y + 2549760x^2 + 4096xy^4 - 14336xy^3 + 1701440xy^2 + 13081256xy + 12572392x - 2048y^5 + 30208y^4 - 677280y^3 + 4847412y^2 + 12785384y + 692224$$

$$B \quad 512x^{11}y + 6336x^{11} + 1024x^{10}y + 12672x^{10} + 1024x^9y^2 + 14720x^9y + 25344x^9 + 4096x^8y^2 + 54784x^8y + 625344x^8 - 1024x^7y^3 - 23552x^7y^2 - 198768x^7y + 2873274x^7 + 512x^6y^4 + 12480x^6y^3 + 62464x^6y^2 + 852896x^6y + 5743380x^6 - 1024x^5y^4 - 22144x^5y^3 - 272608x^5y^2 + 7452340x^5y + 11486760x^5 + 1024x^4y^5 + 22912x^4y^4 + 80640x^4y^3 - 2287360x^4y^2 + 508064x^4y + 1326720x^4 + 4096x^3y^5 - 52224x^3y^4 + 1489984x^3y^3 + 421272x^3y^2 - 769827x^3y - 1295712x^3 - 1024x^2y^6 + 17152x^2y^5 - 309456x^2y^4 - 7494x^2y^3 + 1885296x^2y^2 + 4826146x^2y + 570403x^2 + 6144xy^6 - 53760xy^5 + 2381536xy^4 + 7892132xy^3 + 2232986xy^2 + 4507012xy - 3770810x - 2048y^7 + 46592y^6 - 677280y^5 + 4827956y^4 + 12770280y^3 + 1034960y^2 - 2423706y - 6392692$$

$i = 21, m = 1536$ , neighbours 7, 15, 16, 18, 20, 25, 26, 27, 29, 31

$$W \quad -122880x^5 - 2580480x^4 - 2942976x^3 - 2350080x^2 - 1545216x - 998400$$

$$A \quad -37748736x^{14} + 75497472x^{13}y + 18874368x^{13} + 75497472x^{12} + 75497472x^{11}y^2 - 75497472x^{11}y + 66060288x^{11} + 377487360x^{10}y^2$$


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$$\begin{aligned}
& +207618048x^{10}y + 66060288x^{10} - 717225984x^9y^3 - 1094713344x^9y^2 \\
& +113246208x^9y - 56623104x^9 + 6291456x^8y^5 - 2359296x^8y^4 \\
& -550207488x^8y^3 + 1291137024x^8y^2 - 4933504512x^8y - 19622988096x^8 \\
& -1056964608x^7y^4 - 2642411520x^7y^3 - 839909376x^7y^2 - 419954688x^7y \\
& -37748736x^7 - 75497472x^6y^5 - 1170210816x^6y^4 - 4822401024x^6y^3 \\
& -1604321280x^6y^2 - 391643136x^6y - 14155776x^6 + 1811939328x^5y^5 \\
& +736100352x^5y^4 - 2699034624x^5y^3 - 1736441856x^5y^2 - 264241152x^5y \\
& -2743292863584x^5 - 31457280x^4y^7 + 410517504x^4y^6 + 4133093376x^4y^5 \\
& -24818122752x^4y^4 + 48029614080x^4y^3 + 38121501696x^4y^2 \\
& -153706322520x^4y + 3939098448927x^4 + 75497472x^3y^7 + 4473225216x^3y^6 \\
& +11551113216x^3y^5 - 943718400x^3y^4 - 3638034432x^3y^3 \\
& -828112896x^3y^2 + 94371840x^3y - 3691090280448x^3 + 1056964608x^2y^7 \\
& +3888119808x^2y^6 - 1594884096x^2y^5 + 3038773248x^2y^4 \\
& -169869312x^2y^3 - 639369216x^2y^2 - 184025088x^2y - 7883068538880x^2 \\
& +75497472xy^8 + 868220928xy^7 + 17656971264xy^6 + 4835377152xy^5 \\
& -3416408064xy^4 + 893601792xy^3 + 13669670471904xy^2 - 25625888731620xy \\
& -13511187923391x + 25165824y^9 - 81788928y^8 + 157286400y^7 + 16771645440y^6 \\
& -78855806976y^5 + 295898883072y^4 - 1632560167584y^3 - 35003725159884y^2 \\
& -31507553004798y - 6430691569056 \\
B \quad & -12582912x^{11}y^5 - 146276352x^{11}y^4 + 1364656128x^{11}y^3 \\
& -4620705792x^{11}y^2 + 9451772928x^{11}y + 39132729984x^{11} + 12582912x^{10}y^5 \\
& +146276352x^{10}y^4 - 1364656128x^{10}y^3 + 4620705792x^{10}y^2 \\
& -9451772928x^{10}y + 39132729984x^{10} + 25165824x^9y^6 + 292552704x^9y^5 \\
& -2729312256x^9y^4 + 9241411584x^9y^3 - 18903545856x^9y^2 \\
& -78265459968x^9y - 50331648x^8y^6 - 597688320x^8y^5 + 5312348160x^8y^4 \\
& -17118167040x^8y^3 + 33186385920x^8y^2 + 165982692864x^8y \\
& +5510722771776x^8 + 12582912x^7y^7 + 47185920x^7y^6 - 2424569856x^7y^5 \\
& +16437018624x^7y^4 - 55818878976x^7y^3 + 69088892928x^7y^2 \\
& +239054159088x^7y - 14023173931902x^7 - 12582912x^6y^8 - 58195968x^6y^7 \\
& +2563178496x^6y^6 - 12216852480x^6y^5 + 22011878400x^6y^4 \\
& +45014676096x^6y^3 - 431930311680x^6y^2 - 11431208281968x^6y \\
& +8492884795134x^6 - 12582912x^5y^8 - 48758784x^5y^7 + 2238775296x^5y^6 \\
& -18232614912x^5y^5 + 73188860928x^5y^4 - 127373582976x^5y^3 \\
& -115266899424x^5y^2 + 39238501986684x^5y + 5530289136768x^5 \\
& +25165824x^4y^9 + 141557760x^4y^8 - 5019402240x^4y^7 \\
& +19631751168x^4y^6 - 13524744192x^4y^5 - 184654128384x^4y^4 \\
& +940846196736x^4y^3 - 3930142404864x^4y^2 + 9201672011808x^4y \\
& -8170341138432x^4 - 50331648x^3y^9 - 3460300800x^3y^8 + 2017198080x^3y^7 \\
& -18136596480x^3y^6 + 77648547840x^3y^5 - 270198686208x^3y^4 \\
& +8126644379712x^3y^3 - 4272042348072x^3y^2 - 424895706303x^3y \\
& -9294480551328x^3 + 12582912x^2y^{10} - 1078984704x^2y^9 \\
& -7063732224x^2y^8 + 20043890688x^2y^7 - 67300970496x^2y^6 \\
& +230620875264x^2y^5 - 1142079222864x^2y^4 - 9035789962878x^2y^3 \\
& -10163094392088x^2y^2 + 11101289929287x^2y + 1633274462655x^2 \\
& -62914560xy^{10} - 896532480xy^9 - 16461594624xy^8 + 11682938880xy^7 \\
& -54250426368xy^6 + 244636936704xy^5 - 14842803915312xy^4 + 16832261180046xy^3 \\
& +15284359562430xy^2 + 15795729790392xy - 9169181103135x - 25165824y^{11} \\
& +94371840y^{10} - 1412431872y^9 - 17260806144y^8 + 88049295360y^7 \\
& -339240907776y^6 + 1810490946720y^5 + 34058976285564y^4 + 14626014303552y^3 \\
& -1698595666512y^2 - 16652564746686y - 7427494129056 \\
i = 22, m = 4, \text{ neighbours } 46, 23, 30 \\
W \quad & -4x^5 - 44x^4 - 36x^3 - 52x^2 - 72x - 32 \\
A \quad & 24x^5 - 330x^4 - 1112x^3 - 1792x^2 - 96xy^2 - 200xy - 1506x + 1256y^2 - 316y - 1964 \\
B \quad & -64x^8 + 372x^7 - 128x^6y - 244x^6 + 872x^5y - 500x^5 - 64x^4y^2 - 728x^4y \\
& -2448x^4 - 48x^3y^3 - 704x^3y^2 - 1054x^3y - 2936x^3 + 580x^2y^3 \\
& -624x^2y^2 - 1274x^2y - 3010x^2 - 96xy^4 + 332xy^3 - 76xy^2 - 698xy - 2310x \\
& +1256y^4 + 940y^3 - 1024y^2 - 1460y - 1590 \\
i = 23, m = 1, \text{ neighbours } 45, 14, 22, 37, 41 \\
W \quad & 2x^4 + 10x^3 + 8x^2 - 4x + 8 \\
A \quad & -x^4 - 2x^3 - 28x^2 + 8xy - 57x + 4y^2 + 22y - 90 \\
B \quad & 2x^7 + 2x^6 + 4x^5y - 2x^5 + 8x^4y - 64x^4 + 41x^3y - 104x^3 + 2x^2y^3 \\
& -48x^2y^2 + 75x^2y - 81x^2 + 6xy^3 - 102xy^2 + 21xy - 35x + 4y^4 + 42y^3 - 96y^2
\end{aligned}$$


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$-50y - 77$

$i = 24, m = 4$ , neighbours 45, 7, 8, 19, 31

$W$   $-12x^4 - 8x^3 - 8x^2 + 40x + 20$

$A$   $18x^4 - 72x^3 - 96x^2 - 48x - 72y^2 - 888y - 106$

$B$   $-36x^7 - 72x^6 - 72x^5y - 180x^5 - 288x^4y - 464x^4 + 366x^3y - 384x^3$   
 $-36x^2y^3 - 320x^2y^2 + 164x^2y + 290x^2 + 72xy^3 + 92xy^2 - 810xy + 700x$   
 $-72y^4 - 596y^3 + 320y^2 - 188y + 506$

$i = 25, m = 2$ , neighbours 2, 3, 4, 7, 9, 12, 14, 17, 21, 28, 39

$W$   $72x^5 - 56x^4 + 44x^3 - 64x^2 + 24x - 44$

$A$   $256x^9y - 256x^9 + 512x^8y^2 - 512x^8y - 1024x^7y^3 + 2688x^7y^2$   
 $-2368x^7y + 704x^7 - 2560x^6y^4 + 3712x^6y^3 - 3392x^6y^2 + 1920x^6y$   
 $+320x^6 - 3072x^5y^6 - 1280x^5y^5 + 9856x^5y^4 - 28096x^5y^3$   
 $+20960x^5y^2 + 9008x^5y - 14008x^5 + 3072x^4y^7 + 1536x^4y^6$   
 $-37504x^4y^5 + 61248x^4y^4 - 49440x^4y^3 + 16736x^4y^2 + 12816x^4y$   
 $-2412x^4 + 1024x^3y^7 + 512x^3y^6 - 30336x^3y^5 + 28800x^3y^4$   
 $+896x^3y^3 + 3296x^3y^2 - 4192x^3y - 4736x^3 + 10752x^2y^7$   
 $-35200x^2y^6 + 15552x^2y^5 + 11008x^2y^4 + 15392x^2y^3 - 15696x^2y^2$   
 $-6328x^2y + 8512x^2 + 4096xy^8 + 1792xy^7 - 35584xy^6 + 49216xy^5 - 17248xy^4$   
 $+2416xy^3 - 4360xy^2 + 3632xy - 2112x - 3072y^9 + 9216y^8 + 1792y^7 - 50176y^6$   
 $+65504y^5 + 832y^4 - 37312y^3 + 8052y^2 + 2432y + 1936$

$B$   $3072x^8y^6 + 1024x^8y^5 - 19968x^8y^4 + 39808x^8y^3 - 26272x^8y^2$   
 $-4464x^8y + 10840x^8 - 3072x^7y^7 + 512x^7y^6 + 20480x^7y^5$   
 $-49792x^7y^4 + 46176x^7y^3 - 8672x^7y^2 - 13072x^7y + 5420x^7$   
 $-3072x^6y^7 - 1024x^6y^6 + 19968x^6y^5 - 39808x^6y^4 + 26272x^6y^3$   
 $+4464x^6y^2 - 10840x^6y + 3072x^5y^8 - 512x^5y^7 - 22016x^5y^6$   
 $+49280x^5y^5 - 36192x^5y^4 - 11232x^5y^3 + 26208x^5y^2 - 3188x^5y$   
 $-320x^5 + 3072x^4y^8 + 4096x^4y^7 - 18944x^4y^6 + 19840x^4y^5$   
 $+13536x^4y^4 - 30736x^4y^3 + 6376x^4y^2 + 5740x^4y - 3752x^4$   
 $+3072x^3y^9 + 3840x^3y^8 - 12672x^3y^7 + 30464x^3y^6 - 22080x^3y^5$   
 $-11008x^3y^4 + 11464x^3y^3 - 3352x^3y^2 + 5326x^3y - 1740x^3$   
 $-3072x^2y^{10} - 1536x^2y^9 + 31872x^2y^8 - 49728x^2y^7 + 52064x^2y^6$   
 $-60224x^2y^5 + 47072x^2y^4 - 34492x^2y^3 + 6452x^2y^2 + 13104x^2y$   
 $-6966x^2 - 4096xy^{10} - 10496xy^9 + 75008xy^8 - 116672xy^7 + 80608xy^6$   
 $-39264xy^5 + 4680xy^4 + 13056xy^3 + 1470xy^2 - 4804xy + 1040x + 3072y^{11}$   
 $-13312y^{10} + 16640y^9 + 17920y^8 - 41440y^7 + 3904y^6 + 1088y^5 + 28796y^4$   
 $-1248y^3 - 21360y^2 + 4154y + 1216$

$i = 26, m = 2$ , neighbours 0, 1, 5, 12, 15, 18, 21, 31, 33, 34, 36

$W$   $40x^5 - 80x^4 + 10x^3 + 52x^2 - 28x + 6$

$A$   $512x^{10}y^3 + 256x^{10}y^2 - 256x^{10}y + 512x^9y^4 + 768x^9y^3 - 256x^9y$   
 $+4096x^8y^4 + 1920x^8y^3 - 2112x^8y^2 + 64x^8y + 3072x^7y^5$   
 $+5504x^7y^4 + 448x^7y^3 - 1984x^7y^2 + 11264x^6y^5 + 6784x^6y^4$   
 $-5376x^6y^3 - 736x^6y^2 + 160x^6y + 29696x^5y^6 + 12544x^5y^5$   
 $-15232x^5y^4 + 4256x^5y^3 + 3696x^5y^2 + 4776x^5y + 1280x^5$   
 $-25600x^4y^7 + 2304x^4y^6 + 37952x^4y^5 - 3424x^4y^4 - 15440x^4y^3$   
 $-5952x^4y^2 - 4356x^4y + 2160x^4 - 8192x^3y^7 + 35840x^3y^6$   
 $+13184x^3y^5 - 22656x^3y^4 + 6112x^3y^3 - 1728x^3y^2 + 352x^3y$   
 $-3600x^3 - 3072x^2y^8 - 13184x^2y^7 + 16320x^2y^6 + 19072x^2y^5$   
 $-10976x^2y^4 - 2160x^2y^3 + 1264x^2y^2 - 192x^2y - 1280x^2 - 35328xy^8$   
 $+58240xy^7 + 10880xy^6 - 53408xy^5 + 25712xy^4 - 2568xy^3 + 2916xy^2 - 5006xy$   
 $+1254x + 22528y^9 - 31872y^8 - 11200y^7 + 44416y^6 - 7936y^5 - 3640y^4 - 1728y^3$   
 $-5948y^2 + 2857y + 272$

$B$   $-25088x^8y^6 + 5376x^8y^5 + 20032x^8y^4 - 11968x^8y^3 - 3120x^8y^2$   
 $-5000x^8y - 2080x^8 + 25088x^7y^7 + 7168x^7y^6 - 22720x^7y^5$   
 $+1952x^7y^4 + 9104x^7y^3 + 6560x^7y^2 + 4580x^7y + 1040x^7$   
 $+25088x^6y^7 - 17920x^6y^6 - 17344x^6y^5 + 21984x^6y^4 - 2864x^6y^3$   
 $+3440x^6y^2 - 420x^6y - 1040x^6 - 25088x^5y^8 + 5376x^5y^7$   
 $+26304x^5y^6 - 13312x^5y^5 - 8128x^5y^4 - 2008x^5y^3 - 1300x^5y^2$   
 $+1250x^5y + 520x^5 - 25088x^4y^8 + 5376x^4y^7 + 13760x^4y^6$   
 $-10624x^4y^5 + 1888x^4y^4 - 7992x^4y^3 - 2860x^4y^2 + 5404x^4y$   
 $+3456x^4 - 27136x^3y^9 - 8320x^3y^8 + 5504x^3y^7 + 17312x^3y^6$   
 $+4544x^3y^5 - 23216x^3y^4 + 13080x^3y^3 + 8714x^3y^2 - 1732x^3y$   
 $-2372x^3 + 25600x^2y^{10} - 8960x^2y^9 - 7232x^2y^8 - 18144x^2y^7$

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$$\begin{aligned}
& -26000x^2y^6 + 40704x^2y^5 + 16556x^2y^4 - 12104x^2y^3 - 10658x^2y^2 \\
& -908x^2y + 2216x^2 + 35328xy^{10} - 55936xy^9 - 32128xy^8 + 38592xy^7 \\
& + 14320xy^6 + 6080xy^5 - 29944xy^4 + 8042xy^3 + 5033xy^2 + 2098xy - 1590x \\
& - 22528y^{11} + 16512y^{10} + 21504y^9 - 12160y^8 - 19200y^7 - 20008y^6 + 18156y^5 \\
& + 13618y^4 + 852y^3 - 966y^2 - 3329y - 236
\end{aligned}$$

$i = 27, m = 2$ , neighbours 0, 1, 14, 16, 21

$$\begin{aligned}
W & -12x^4 - 12 \\
A & 12x^4 - 72x^3 + 36x - 60y^2 + 36y + 144 \\
B & -12x^7 + 12x^6 + 12x^5y - 12x^5 - 24x^4y - 24x^4 + 32x^3y^2 + 2x^3y - 120x^3 \\
& + 4x^2y^3 - 64x^2y^2 + 82x^2y - 18x^2 + 68xy^3 - 18xy^2 + 70xy - 18x - 4y^4 \\
& + 28y^3 - 128y^2 - 46y + 18
\end{aligned}$$

$i = 28, m = 1$ , neighbours 6, 8, 9, 25, 30

$$\begin{aligned}
W & 4x^3 - 2x^2 \\
A & -8x^2 - 8x - 8y + 12 \\
B & 16x^5 - 16x^4y - 8x^3y^2 + 28x^3y + 2x^3 + 16x^2y^3 - 28x^2y^2 - 6x^2y \\
& - 2x^2 - 16xy^3 + 16xy^2 + 8xy + 4x + 16y^4 - 2y - 6
\end{aligned}$$

$i = 29, m = 1$ , neighbours 8, 11, 16, 17, 21, 36

$$\begin{aligned}
W & 2x^5 + 2x^4 + 8x^3 + 2x \\
A & -8x^5y + 16x^5 + 8x^4y^2 - 42x^4y + 38x^4 + 4x^3y^2 - 6x^3y - 30x^3 \\
& + 2x^2y^2 - 14x^2y - 2x^2 + 12xy^3 + 10xy^2 - 48xy - 46x - 8y^4 + 28y^3 + 26y^2 \\
& - 46y - 36 \\
B & 8x^8y - 20x^8 - 8x^7y^2 + 40x^7y - 50x^7 - 8x^6y^2 + 12x^6y + 20x^6 \\
& + 8x^5y^3 - 32x^5y^2 + 6x^5y + 24x^5 + 8x^4y^3 + 4x^4y^2 - 16x^4y \\
& - 22x^4 + 8x^3y^4 + 41x^3y^2 - 29x^3y + 4x^3 - 8x^2y^5 + 26x^2y^4 \\
& - 28x^2y^3 + 21x^2y^2 + 10x^2y + 15x^2 - 12xy^5 + 38xy^4 + 68xy^3 + 27xy^2 \\
& + 17xy - 23x + 8y^6 - 68y^5 - 38y^4 + 16y^3 + 18y^2 - 5y - 18
\end{aligned}$$

$i = 30, m = 4$ , neighbours 43, 1, 8, 17, 22, 28, 40

$$\begin{aligned}
W & 24x^4 - 24x^3 + 8x^2 - 8x \\
A & 64x^8 + 392x^5 + 344x^4y + 1096x^4 - 128x^3y^2 + 256x^3y - 384x^3 \\
& - 192x^2y^2 + 288x^2y + 480x^2 - 256xy^2 + 1451xy - 1005x - 64y^4 - 376y^3 \\
& - 774y^2 + 1775y - 643 \\
B & -64x^{11} - 32x^{10} + 64x^9y - 48x^9 + 64x^8y - 432x^8 - 64x^7y^2 - 136x^7y \\
& - 1400x^7 - 64x^6y^3 - 96x^6y^2 + 396x^6y - 884x^6 + 32x^5y^3 \\
& + 56x^5y^2 + 1530x^5y - 1142x^5 + 64x^4y^4 + 80x^4y^3 - 416x^4y^2 \\
& + 328x^4y - 488x^4 + 64x^3y^4 - 592x^3y^3 - 244x^3y^2 - 796x^3y + 536x^3 \\
& - 64x^2y^5 + 280x^2y^4 - 1768x^2y^3 - 306x^2y^2 + 458x^2y + 380x^2 \\
& + 32xy^5 + 508xy^4 - 520xy^3 + 601xy^2 - 1009xy + 362x + 64y^6 - 104y^5 + 2258y^4 \\
& - 2342y^3 + 52y^2 - 489y + 643
\end{aligned}$$

$i = 31, m = 64$ , neighbours 47, 0, 3, 8, 21, 24, 26, 34, 35

$$\begin{aligned}
W & -1152x^5 + 3072x^4 - 2304x^3 + 1152x^2 + 768x \\
A & 491520x^8y^2 + 952320x^8y - 3775104x^8 + 196608x^7y + 196608x^6y^2 \\
& - 98304x^6y - 393216x^5y^3 + 884736x^5y^2 - 786432x^5y - 12211344x^5 \\
& + 577536x^4y^3 - 2117376x^4y^2 - 2376144x^4y + 3235272x^4 - 1179648x^3y^4 \\
& + 589824x^3y^3 - 5455872x^3y^2 + 344064x^3y - 7372800x^3 + 1966080x^2y^4 \\
& - 7667712x^2y^3 - 2310144x^2y^2 + 1671168x^2y + 8404992x^2 + 196608xy^5 \\
& + 1769472xy^4 - 9437184xy^3 + 3526464xy^2 - 3285546xy - 4684221x - 491520y^6 \\
& + 190464y^5 - 1683072y^4 + 803280y^3 + 13286412y^2 + 13442208y + 3618735 \\
B & -491520x^{11}y^2 - 952320x^{11}y + 3775104x^{11} - 245760x^{10}y^2 - 476160x^{10}y \\
& + 1887552x^{10} + 491520x^9y^3 + 1075200x^9y^2 - 3537024x^9y - 943776x^9 \\
& + 491520x^8y^3 + 1136640x^8y^2 - 3417984x^8y + 10132128x^8 - 491520x^7y^4 \\
& - 1505280x^7y^3 + 2849664x^7y^2 + 4529712x^7y + 5841624x^7 - 491520x^6y^5 \\
& - 1689600x^6y^4 + 1640064x^6y^3 + 4520256x^6y^2 - 5682120x^6y \\
& - 4032804x^6 + 245760x^5y^5 + 906240x^5y^4 - 1276992x^5y^3 \\
& - 3694416x^5y^2 - 10947540x^5y - 4937214x^5 + 491520x^4y^6 \\
& + 2058240x^4y^5 - 342144x^4y^4 - 5978784x^4y^3 + 1587264x^4y^2 \\
& + 11528592x^4y - 1076400x^4 + 884736x^3y^6 + 6789120x^3y^5 \\
& - 5030784x^3y^4 + 12139104x^3y^3 + 7191672x^3y^2 + 230700x^3y \\
& - 7499088x^3 - 491520x^2y^7 + 1886208x^2y^6 + 9740160x^2y^5 \\
& - 1558800x^2y^4 + 11529336x^2y^3 - 8826612x^2y^2 - 2958954x^2y \\
& - 487788x^2 + 49152xy^7 + 2952192xy^6 + 6850368xy^5 - 10747176xy^4 \\
& + 4914324xy^3 - 12983010xy^2 + 1628037xy - 1065486x + 491520y^8 + 1136640y^7
\end{aligned}$$


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$$\begin{aligned}
& +1114752y^6 - 3965712y^5 - 11208324y^4 - 7794846y^3 + 680328y^2 + 6204738y \\
& + 3618735 \\
i = 32, m = 1, \text{ neighbours } 1, 4, 5, 17 \\
W & -4x^5 + 16x^4 - 14x^3 + 6x^2 \\
A & -4x^5 + 114x^4 - 184x^3 + 248x^2 + 4xy^2 + 48xy - 182x - 98y^2 + 170y \\
B & -4x^8 - 54x^7 + 4x^6y - 54x^6 + 58x^5y - 54x^5 - 8x^4y^2 + 66x^4y - 66x^4 \\
& + 4x^3y^3 + 32x^3y^2 - 107x^3y + 16x^3 - 102x^2y^3 - 34x^2y^2 + 51x^2y \\
& + 55x^2 - 4xy^4 + 50xy^3 + 73xy^2 - 89xy - 37x + 98y^4 - 162y^3 - 4y^2 - 57y + 89 \\
i = 33, m = 2, \text{ neighbours } 6, 7, 9, 19, 26, 41 \\
W & 16x^5 - 20x^4 - 36x^3 + 4x^2 + 14x + 6 \\
A & 32x^8 - 8x^5 - 96x^4y^2 + 96x^4y + 84x^4 - 48x^3y + 360x^3 + 24x^2y - 780x^2 \\
& + 64xy^3 - 296xy^2 + 352xy - 1082x + 64y^4 - 32y^3 - 164y^2 - 976y + 577 \\
B & -32x^{11} + 32x^{10} + 32x^9y - 48x^9 - 64x^8y - 56x^8 + 64x^7y^2 + 16x^7y \\
& + 284x^7 - 32x^6y^3 + 8x^6y - 296x^6 - 32x^5y^3 + 16x^5y^2 - 284x^5y \\
& + 438x^5 + 32x^4y^4 + 16x^4y^3 + 336x^4y^2 + 164x^4y - 292x^4 \\
& - 64x^3y^4 + 8x^3y^3 + 32x^3y^2 + 278x^3y + 244x^3 + 64x^2y^5 \\
& - 160x^2y^4 - 244x^2y^3 - 60x^2y^2 - 88x^2y + 282x^2 + 136xy^4 - 144xy^3 \\
& - 222xy^2 + 389xy + 164x - 64y^6 + 128y^5 - 92y^4 + 1218y^3 - 16y^2 - 146y - 533 \\
i = 34, m = 1, \text{ neighbours } 3, 4, 8, 9, 11, 12, 26, 31, 39, 40 \\
W & -252x^5 + 192x^4 + 445x^3 - 290x^2 - 148x + 64 \\
A & -4608x^3y^3 + 2048x^3y^2 + 39584x^3y + 23048x^3 + 4608x^2y^4 \\
& - 20480x^2y^3 - 31392x^2y^2 + 135288x^2y + 87320x^2 - 4608xy^5 + 10112xy^4 \\
& + 24192xy^3 - 40976xy^2 + 65972xy + 13816x - 1024y^6 + 14080y^5 - 26496y^4 \\
& - 66624y^3 + 206952y^2 - 56792y - 74192 \\
B & -128x^{11}y + 128x^{11} - 640x^{10}y^2 + 832x^{10}y - 192x^{10} - 896x^9y^3 \\
& + 2048x^9y^2 - 1120x^9y - 32x^9 + 128x^8y^4 + 1856x^8y^3 - 3680x^8y^2 \\
& + 1296x^8y - 31352x^8 + 1024x^7y^5 - 640x^7y^4 - 4832x^7y^3 \\
& + 6464x^7y^2 - 1152x^7y + 63396x^7 + 512x^6y^6 - 2176x^6y^5 + 352x^6y^4 \\
& + 12688x^6y^3 - 10656x^6y^2 - 7280x^6y + 8066x^6 - 1536x^5y^6 \\
& + 5888x^5y^5 - 4896x^5y^4 + 3696x^5y^3 + 44640x^5y^2 - 195752x^5y \\
& - 186657x^5 - 512x^4y^7 + 4096x^4y^6 - 3488x^4y^5 - 17760x^4y^4 \\
& - 2248x^4y^3 + 14440x^4y^2 + 16682x^4y + 6962x^4 - 5632x^3y^6 \\
& - 512x^3y^5 + 23232x^3y^4 + 27872x^3y^3 + 27132x^3y^2 + 38776x^3y \\
& + 57736x^3 - 512x^2y^8 - 1792x^2y^7 + 8192x^2y^6 + 13120x^2y^5 \\
& - 22144x^2y^4 - 31656x^2y^3 - 37096x^2y^2 - 20432x^2y - 10496x^2 \\
& + 512xy^8 + 3584xy^7 + 768xy^6 - 36928xy^5 - 24112xy^4 + 7312xy^3 - 58932xy^2 \\
& - 33088xy + 3184x + 512y^9 - 12928y^7 + 14464y^6 + 78752y^5 - 120296y^4 - 76664y^3 \\
& - 36000y^2 + 67548y + 39144 \\
i = 35, m = 1, \text{ neighbours } 0, 2, 7, 10, 31, 37 \\
W & 12x^5 + 40x^4 + 48x^3 + 36x^2 + 20x + 4 \\
A & -156x^5 - 622x^4 - 1376x^3 - 16x^2y - 2352x^2 + 252xy^2 - 200xy - 1870x + 302y^2 \\
& - 870y - 1164 \\
B & 84x^8 + 58x^7 - 84x^6y - 142x^6 + 26x^5y + 84x^5 - 168x^4y^2 + 494x^4y \\
& - 904x^4 + 204x^3y^3 - 120x^3y^2 + 573x^3y - 670x^3 + 32x^2y^4 \\
& + 594x^2y^3 - 226x^2y^2 + 525x^2y - 505x^2 - 252xy^4 + 722xy^3 - 191xy^2 \\
& + 752xy - 425x - 32y^5 - 318y^4 + 1164y^3 - 4y^2 + 155y - 590 \\
i = 36, m = 1, \text{ neighbours } 2, 11, 26, 29, 38 \\
W & -8x^5 + 28x^4 - 36x^3 + 30x^2 - 18x \\
A & -24x^5 + 392x^4 - 672x^3 + 24x^2y + 1264x^2 + 24xy^2 + 96xy - 856x - 288y^2 + 352y \\
& + 640 \\
B & -8x^8 - 176x^7 + 8x^6y - 168x^6 + 184x^5y + 8x^5 - 16x^4y^2 + 284x^4y \\
& - 592x^4 + 24x^3y^3 + 112x^3y^2 - 416x^3y + 348x^3 - 336x^2y^3 \\
& - 132x^2y^2 + 252x^2y + 36x^2 - 24xy^4 + 176xy^3 + 168xy^2 - 302xy - 112x \\
& + 272y^4 - 640y^3 + 12y^2 - 148y + 320 \\
i = 37, m = 2, \text{ neighbours } 3, 10, 18, 19, 20, 23, 35, 39 \\
W & -20x^5 - 46x^4 - 26x^3 + 8x^2 + 18x + 2 \\
A & 16x^9 + 48x^8y - 32x^8 + 32x^7y^2 - 200x^7y + 44x^7 - 432x^6y^2 + 272x^6y \\
& - 80x^6 - 384x^5y^3 + 2536x^5y^2 - 1796x^5y - 1488x^5 + 128x^4y^4 \\
& - 624x^4y^3 + 1240x^4y^2 + 1168x^4y - 1720x^4 + 688x^3y^4 - 544x^3y^3 \\
& + 1412x^3y^2 - 726x^3y - 1546x^3 - 32x^2y^5 - 728x^2y^4 + 1900x^2y^3 \\
& - 2336x^2y^2 + 1096x^2y - 3288x^2 - 336xy^5 - 1776xy^4 + 492xy^3 + 1250xy^2
\end{aligned}$$


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$$\begin{aligned}
& +1758xy + 40x - 96y^6 + 1736y^5 - 3220y^4 + 816y^3 + 136y^2 + 3548y + 4 \\
B \quad & 128x^8y^3 - 1824x^8y^2 + 1336x^8y + 1336x^8 - 128x^7y^4 + 1888x^7y^3 \\
& - 2248x^7y^2 - 668x^7y + 668x^7 + 128x^6y^4 - 1824x^6y^3 + 1336x^6y^2 \\
& + 1336x^6y - 128x^5y^5 + 1888x^5y^4 - 2312x^5y^3 + 244x^5y^2 + 2194x^5 \\
& + 128x^4y^5 - 1696x^4y^4 - 488x^4y^3 + 2672x^4y^2 - 1526x^4y + 362x^4 \\
& + 352x^3y^6 - 4560x^3y^5 + 3112x^3y^4 + 2248x^3y^3 - 188x^3y^2 \\
& - 1296x^3y + 312x^3 - 128x^2y^7 + 496x^2y^6 + 48x^2y^5 - 4648x^2y^4 \\
& + 2958x^2y^3 + 1161x^2y^2 + 1115x^2y - 284x^2 - 336xy^7 - 32xy^6 - 1840xy^5 \\
& + 576xy^4 + 4782xy^3 - 1434xy^2 + 1365xy - 821x - 96y^8 + 2248y^7 - 3988y^6 \\
& + 608y^5 + 164y^4 + 4806y^3 - 516y^2 + 28y - 1790
\end{aligned}$$

$i = 38, m = 2$ , neighbours 4, 6, 13, 19, 36, 40

$$\begin{aligned}
W \quad & 8x^4 + 6x^2 + 2x - 3 \\
A \quad & -64x^5y + 112x^5 + 64x^4y^2 - 80x^4y - 56x^4 - 32x^3y + 32x^2y^2 - 16x^2y \\
& - 16x^2 + 64xy^3 - 80xy^2 + 16xy - 4x - 64y^4 + 112y^3 - 8y^2 - 52y + 30 \\
B \quad & 64x^8y - 112x^8 - 64x^7y^2 + 80x^7y + 56x^7 + 64x^6y^2 - 144x^6y + 56x^6 \\
& - 64x^5y^3 + 112x^5y^2 + 16x^5y - 28x^5 + 64x^4y^3 - 112x^4y^2 + 24x^4y \\
& - 56x^4 + 64x^3y^4 - 336x^3y^3 + 312x^3y^2 - 132x^3y + 88x^3 - 64x^2y^5 \\
& + 16x^2y^4 + 408x^2y^3 - 240x^2y^2 - 76x^2y - 26x^2 + 64xy^5 - 144xy^4 \\
& + 72xy^2 + 34xy - 24x - 64y^6 - 48y^5 + 32y^4 + 220y^3 - 60y^2 - 106y + 53
\end{aligned}$$

$i = 39, m = 4096$ , neighbours 4, 6, 7, 9, 10, 12, 16, 19, 25, 34, 37, 40

$$\begin{aligned}
W \quad & 120832x^5 + 325632x^4 + 411136x^3 + 193024x^2 + 52736x - 44544 \\
A \quad & 2147483648x^{12}y - 2147483648x^{12} - 2147483648x^{11}y^2 + 2147483648x^{11} \\
& - 10737418240x^{10}y^3 + 6442450944x^{10}y^2 + 4294967296x^{10} \\
& + 2147483648x^9y^4 + 12884901888x^9y^3 - 11811160064x^9y^2 \\
& + 4831838208x^9y - 8053063680x^9 + 57982058496x^8y^6 - 33957085184x^8y^5 \\
& + 90655686656x^8y^4 - 28456779776x^8y^3 + 111790227456x^8y^2 \\
& + 121132849152x^8y + 276624659328x^8 + 8589934592x^7y^6 - 25769803776x^7y^5 \\
& - 10200547328x^7y^4 + 23085449216x^7y^3 - 1744830464x^7y^2 \\
& - 671088640x^7y + 6710886400x^7 - 68719476736x^6y^6 + 68719476736x^6y^5 \\
& + 27380416512x^6y^4 - 22011707392x^6y^3 - 14629732352x^6y^2 \\
& + 17045651456x^6y - 7784628224x^6 - 34359738368x^5y^7 + 60129542144x^5y^6 \\
& - 2147483648x^5y^5 - 536870912x^5y^4 - 57176752128x^5y^3 \\
& + 39862665216x^5y^2 - 8589934592x^5y - 977980509872x^5 - 56103010304x^4y^7 \\
& - 119906762752x^4y^6 + 111577923584x^4y^5 - 358400851968x^4y^4 \\
& + 27051364352x^4y^3 - 414815515392x^4y^2 - 646558330992x^4y \\
& + 1184291529176x^4 + 51539607552x^3y^7 - 73014444032x^3y^6 \\
& - 108179488768x^3y^5 + 300647710720x^3y^4 - 194817032192x^3y^3 \\
& + 27313307648x^3y^2 - 1409286144x^3y - 104776859648x^3 - 8589934592x^2y^9 \\
& + 197568495616x^2y^8 - 329101869056x^2y^7 + 109521666048x^2y^6 \\
& + 253805723648x^2y^5 - 354603237376x^2y^4 + 138982457344x^2y^3 \\
& - 10603200512x^2y^2 - 872415232x^2y - 153310199808x^2 + 137438953472xy^9 \\
& - 326417514496xy^8 + 289373421568xy^7 - 60469280768xy^6 - 195859578880xy^5 \\
& + 269980450816xy^4 - 69949703168xy^3 + 45098265792xy^2 + 1415068244302xy \\
& - 615287755153x - 49392123904y^{10} - 35567697920y^9 + 397678739456y^8 \\
& - 514564030464y^7 + 228933730304y^6 + 137678444544y^5 - 40782823552y^4 \\
& + 547425435248y^3 - 2512212257492y^2 + 1854336476468y - 367644108907 \\
B \quad & -57982058496x^{11}y^6 + 51136954368x^{11}y^5 - 73475817472x^{11}y^4 \\
& - 15029764096x^{11}y^3 - 90315390976x^{11}y^2 - 135762581504x^{11}y \\
& - 274342957952x^{11} + 28991029248x^{10}y^6 - 25568477184x^{10}y^5 \\
& + 36737908736x^{10}y^4 + 7514882048x^{10}y^3 + 45157695488x^{10}y^2 \\
& + 67881290752x^{10}y + 137171478976x^{10} - 57982058496x^9y^7 \\
& + 65632468992x^9y^6 - 86260056064x^9y^5 + 3339190272x^9y^4 \\
& - 86557949952x^9y^3 - 113183733760x^9y^2 - 240402312576x^9y \\
& + 68585739488x^9 + 57982058496x^8y^7 - 72880226304x^8y^6 \\
& + 92652175360x^8y^5 - 12523667456x^8y^4 + 84679229440x^8y^3 \\
& + 101894309888x^8y^2 + 223431989888x^8y + 870620286816x^8 \\
& - 57982058496x^7y^8 + 116366770176x^7y^7 - 12322865152x^7y^6 \\
& - 37040422912x^7y^5 + 76988907520x^7y^4 - 3393718272x^7y^3 \\
& + 63254262144x^7y^2 + 586524861712x^7y - 1693107090424x^7 \\
& - 57982058496x^6y^9 + 138110042112x^6y^8 - 233530458112x^6y^7 \\
& + 149214986240x^6y^6 - 156213411840x^6y^5 - 46578063360x^6y^4 \\
& - 205576521600x^6y^3 + 186015399360x^6y^2 + 533523151768x^6y
\end{aligned}$$


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$$\begin{aligned}
& +548414880780x^6 - 28991029248x^5y^9 + 76302778368x^5y^8 \\
& +46258978816x^5y^7 - 105486483456x^5y^6 + 173615824896x^5y^5 \\
& -18599140352x^5y^4 + 167739382080x^5y^3 + 461890030960x^5y^2 \\
& -2253131097164x^5y + 572346104822x^5 - 57982058496x^4y^{10} \\
& +181596585984x^4y^9 - 340736868352x^4y^8 + 242850725888x^4y^7 \\
& -206728953856x^4y^6 - 30010423296x^4y^5 - 209295306112x^4y^4 \\
& +197507152096x^4y^3 + 164941932864x^4y^2 + 489321512560x^4y \\
& -102057355888x^4 + 83751862272x^3y^{10} - 170590732288x^3y^9 \\
& +191100878848x^3y^8 - 145285971968x^3y^7 + 429131988992x^3y^6 \\
& -154777647104x^3y^5 + 300890683008x^3y^4 + 891459792416x^3y^3 \\
& +289815200616x^3y^2 - 996526311228x^3y - 447908259760x^3 \\
& -57982058496x^2y^{11} + 164685152256x^2y^{10} - 136138719232x^2y^9 \\
& +122338934784x^2y^8 + 286365155328x^2y^7 - 418403207168x^2y^6 \\
& +311290404224x^2y^5 + 709093909712x^2y^4 - 1567319483416x^2y^3 \\
& +54702960284x^2y^2 - 118808821314x^2y + 841322723836x^2 + 112742891520xy^{11} \\
& -445133094912xy^{10} + 636523380736xy^9 - 220401500160xy^8 - 233039347712xy^7 \\
& +241705839616xy^6 + 172503983040xy^5 + 478775738488xy^4 + 644844174948xy^3 \\
& -1155229401150xy^2 + 185709763343xy - 240961333542x - 49392123904y^{12} \\
& -94623498240y^{11} + 487805943808y^{10} - 443533492224y^9 + 119440113664y^8 \\
& +43656288256y^7 + 173902826368y^6 + 536552981552y^5 - 2306869408956y^4 \\
& +717535087382y^3 - 282052411928y^2 + 1115079922782y - 369628276843
\end{aligned}$$

$i = 40, m = 48, \text{ neighbours } 4, 5, 6, 11, 13, 17, 30, 34, 38, 39$

$W \quad -2304x^5 + 4032x^4 - 2928x^3 + 3024x^2 - 48x - 1008$

$A \quad 5505024x^8y^4 - 4489216x^8y^3 - 5545984x^8y^2 + 4886528x^8y - 2869760x^8$   
 $-147456x^7y^3 - 811008x^7y^2 + 147456x^7y + 811008x^7 + 589824x^6y^4$   
 $-663552x^6y^3 - 2174976x^6y^2 + 663552x^6y + 1585152x^6 + 589824x^5y^5$   
 $+1474560x^5y^4 - 2174976x^5y^3 - 2562048x^5y^2 + 1585152x^5y$   
 $-1566720x^5 - 11010048x^4y^6 + 10354688x^4y^5 + 8101888x^4y^4$   
 $-10103296x^4y^3 + 9188224x^4y^2 - 986368x^4y + 17351864x^4$   
 $+1179648x^3y^6 + 2949120x^3y^5 - 1105920x^3y^4 - 5419008x^3y^3$   
 $+1612800x^3y^2 + 2469888x^3y - 16561152x^3 - 147456x^2y^6$   
 $+3465216x^2y^5 - 387072x^2y^4 - 5465088x^2y^3 + 1833984x^2y^2$   
 $+1999872x^2y + 17473536x^2 - 1769472xy^7 - 516096xy^6 + 7004160xy^5$   
 $-3055104xy^4 - 5057856xy^3 + 3569040xy^2 + 1073448xy - 10068471x + 5505024y^8$   
 $-12992512y^7 + 5939200y^6 + 9574912y^5 - 15850240y^4 + 7459424y^3$   
 $-17597384y^2 - 987054y + 5993821$

$B \quad -5505024x^{11}y^4 + 4489216x^{11}y^3 + 5398528x^{11}y^2 - 4886528x^{11}y$   
 $+3017216x^{11} - 2752512x^{10}y^4 + 2244608x^{10}y^3 + 2699264x^{10}y^2$   
 $-2443264x^{10}y + 1508608x^{10} - 5505024x^9y^5 + 5865472x^9y^4$   
 $+4276224x^9y^3 - 6236160x^9y^2 + 4238848x^9y - 754304x^9 - 5505024x^8y^5$   
 $+6553600x^8y^4 + 3715072x^8y^3 - 6910976x^8y^2 + 4849664x^8y$   
 $-1131456x^8 + 5505024x^7y^6 - 5177344x^7y^5 - 278528x^7y^4$   
 $+1843712x^7y^3 - 8098688x^7y^2 + 4423808x^7y - 8840536x^7$   
 $-5505024x^6y^7 + 1736704x^6y^6 + 12115968x^6y^5 - 1835008x^6y^4$   
 $-7074048x^6y^3 + 1556544x^6y^2 + 1098880x^6y - 5363148x^6$   
 $+2752512x^5y^7 - 7061504x^5y^6 + 8368128x^5y^5 + 1065472x^5y^4$   
 $-12557568x^5y^3 + 9251424x^5y^2 - 10503000x^5y + 1738694x^5$   
 $-5505024x^4y^8 + 5865472x^4y^7 + 7716864x^4y^6 - 3708928x^4y^5$   
 $-4559360x^4y^4 - 2053248x^4y^3 + 3902464x^4y^2 - 10219232x^4y$   
 $-2988992x^4 - 6094848x^3y^8 + 8273920x^3y^7 + 10944512x^3y^6$   
 $-8958976x^3y^5 - 4826880x^3y^4 + 975488x^3y^3 + 8885184x^3y^2$   
 $-6600876x^3y - 2286624x^3 + 5505024x^2y^9 - 7143424x^2y^8 - 950272x^2y^7$   
 $+8327680x^2y^6 - 8688512x^2y^5 - 3324416x^2y^4 - 16281656x^2y^3$   
 $-912704x^2y^2 + 4322010x^2y + 8866988x^2 - 4521984xy^9 + 4505600xy^8$   
 $+5300224xy^7 - 2820864xy^6 - 2977472xy^5 - 3263744xy^4 + 15375460xy^3$   
 $-10264876xy^2 - 5958237xy - 5113754x + 5505024y^{10} - 11419648y^9 + 1409024y^8$   
 $+10476032y^7 - 7654912y^6 - 1180832y^5 - 19510040y^4 - 3259322y^3 - 397744y^2$   
 $+9115916y + 4977757$

$i = 41, m = 4, \text{ neighbours } 45, 14, 17, 23, 33$

$W \quad -16x^5 - 32x^4 + 4x^3 + 8x^2$

$A \quad -240x^4 - 448x^3 + 192x^2 + 144xy - 62x + 112y^2 - 388y + 276$

$B \quad -128x^8 - 208x^7 - 128x^6y + 232x^6 - 144x^5y + 92x^5 - 128x^4y^2 + 352x^4y$

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|                                                                                  |
|----------------------------------------------------------------------------------|
| $-96x^4 - 192x^3y^2 + 568x^3y - 448x^3 + 112x^2y^3 + 640x^2y^2 - 284x^2y$        |
| $-56x^2 + 200xy^3 + 440xy^2 - 586xy - 36x + 112y^4 - 20y^3 - 384y^2 + 72y + 138$ |

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Table 2: The data of the identities (3).

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|                                                                                            |
|--------------------------------------------------------------------------------------------|
| $\{i, j\} = \{0, 3\}, k_{ij} = 2$                                                          |
| $\alpha \quad 2$                                                                           |
| $\beta \quad 4x^4 + 4x^3 + 4x^2 + 3x - 2y^2 + 4$                                           |
| $\gamma \quad -4x^4 - 4x^3 - 4x^2 - 4x - 4$                                                |
| $\{i, j\} = \{0, 6\}, k_{ij} = 2 \cdot 3$                                                  |
| $\alpha \quad -2$                                                                          |
| $\beta \quad 4x^4 - 4x^3 + 4x^2 - 3x + 2y^2 + 4$                                           |
| $\gamma \quad -4x^4 + 4x^3 - 4x^2 + 4x - 4$                                                |
| $\{i, j\} = \{0, 7\}, k_{ij} = 2$                                                          |
| $\alpha \quad 2$                                                                           |
| $\beta \quad -4x^4 - 4x^3 - 4x^2 - 5x - 2y^2 - 4$                                          |
| $\gamma \quad 4x^4 + 4x^3 + 4x^2 + 4x + 4$                                                 |
| $\{i, j\} = \{0, 8\}, k_{ij} = 2$                                                          |
| $\alpha \quad -2$                                                                          |
| $\beta \quad 2x^4 + x + 2y^2$                                                              |
| $\gamma \quad -2x^4$                                                                       |
| $\{i, j\} = \{0, 20\}, k_{ij} = 2 \cdot 3^2 \cdot 7$                                       |
| $\alpha \quad 8x^4 - 126$                                                                  |
| $\beta \quad -16x^7 - 32x^6 - 68x^5 - 8x^4y^2 - 128x^4 - 4x^3 + 63x + 126y^2$              |
| $\gamma \quad -16x^7 - 32x^6 - 64x^5 - 128x^4 - 4x^3$                                      |
| $\{i, j\} = \{0, 26\}, k_{ij} = 7$                                                         |
| $\alpha \quad -4x - 8$                                                                     |
| $\beta \quad -4x^4 - 8x^3 + 4xy^2 + 8y^2 - 1$                                              |
| $\gamma \quad 4x^4 + 8x^3 + 2x^2 + 4x + 1$                                                 |
| $\{i, j\} = \{0, 27\}, k_{ij} = 2 \cdot 3$                                                 |
| $\alpha \quad 4x^4 - 6$                                                                    |
| $\beta \quad 4x^7 - 4x^6 + 2x^5 - 4x^4y^2 - 4x^4 - 2x^3 + 3x + 6y^2$                       |
| $\gamma \quad -4x^7 + 4x^6 - 4x^5 + 4x^4 + 2x^3$                                           |
| $\{i, j\} = \{0, 31\}, k_{ij} = 2 \cdot 3 \cdot 7$                                         |
| $\alpha \quad -32x^4 - 62$                                                                 |
| $\beta \quad -32x^7 - 16x^6 + 24x^5 + 32x^4y^2 + 12x^4 - 60x^3 - 20x^2 + 51x + 62y^2 + 20$ |
| $\gamma \quad 32x^7 + 16x^6 - 8x^5 - 12x^4 + 60x^3 + 20x^2 - 20x - 20$                     |
| $\{i, j\} = \{0, 35\}, k_{ij} = 2 \cdot 7$                                                 |
| $\alpha \quad 4x^4 - 2$                                                                    |
| $\beta \quad 4x^7 - 4x^6 - 2x^5 - 4x^4y^2 + 4x^4 - 6x^3 + 7x + 2y^2 - 6$                   |
| $\gamma \quad -4x^7 + 4x^6 - 4x^4 + 6x^3 - 6x + 6$                                         |
| $\{i, j\} = \{1, 5\}, k_{ij} = 3$                                                          |
| $\alpha \quad 0$                                                                           |
| $\beta \quad -1$                                                                           |
| $\gamma \quad 1$                                                                           |
| $\{i, j\} = \{1, 9\}, k_{ij} = 3$                                                          |
| $\alpha \quad -4$                                                                          |
| $\beta \quad -4x^4 + 2x + 4y^2 + 2y$                                                       |
| $\gamma \quad 4x^4 + 1$                                                                    |
| $\{i, j\} = \{1, 10\}, k_{ij} = 13$                                                        |
| $\alpha \quad -4$                                                                          |
| $\beta \quad -4x^4 + 4x^3 - 4x^2 + 6x + 4y^2 + 2y - 4$                                     |
| $\gamma \quad 4x^4 - 4x^3 + 4x^2 - 4x + 5$                                                 |
| $\{i, j\} = \{1, 12\}, k_{ij} = 7$                                                         |
| $\alpha \quad -16$                                                                         |
| $\beta \quad -16x^4 - 8x^3 - 4x^2 + 6x + 16y^2 + 8y - 1$                                   |
| $\gamma \quad 16x^4 + 8x^3 + 4x^2 + 2x + 5$                                                |
| $\{i, j\} = \{1, 18\}, k_{ij} = 3 \cdot 13$                                                |

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|                                                      |                                                                                        |
|------------------------------------------------------|----------------------------------------------------------------------------------------|
| $\alpha$                                             | $40x^4 - 52$                                                                           |
| $\beta$                                              | $-80x^7 + 80x^6 - 100x^5 - 40x^4y^2 - 20x^4y + 70x^4 + 4x^3 + 26x^2 + 52y^2$           |
| $\gamma$                                             | $+26y + 39$                                                                            |
| $\gamma$                                             | $-80x^7 + 80x^6 - 80x^5 + 80x^4 + 4x^3 + 26x^2 - 26x + 26$                             |
| $\{i, j\} = \{1, 26\}, k_{ij} = 3^2$                 |                                                                                        |
| $\alpha$                                             | $-8x - 12$                                                                             |
| $\beta$                                              | $-8x^4 - 12x^3 + 4x^2 + 8xy^2 + 4xy + 8x + 12y^2 + 6y + 1$                             |
| $\gamma$                                             | $8x^4 + 12x^3 + 2$                                                                     |
| $\{i, j\} = \{1, 27\}, k_{ij} = 3$                   |                                                                                        |
| $\alpha$                                             | $-4$                                                                                   |
| $\beta$                                              | $-4x^3 + 4x^2 + 4y^2 + 2y + 1$                                                         |
| $\gamma$                                             | $4x^3 - 4x^2 + 2x$                                                                     |
| $\{i, j\} = \{1, 30\}, k_{ij} = 3 \cdot 7$           |                                                                                        |
| $\alpha$                                             | $-320x^4 - 28$                                                                         |
| $\beta$                                              | $-320x^7 - 160x^6 + 80x^5 + 320x^4y^2 + 160x^4y + 40x^4 - 128x^3 + 56x^2 + 42x$        |
| $\gamma$                                             | $+28y^2 + 14y + 21$                                                                    |
| $\gamma$                                             | $320x^7 + 160x^6 + 80x^5 + 40x^4 + 128x^3 - 56x^2 - 28x - 14$                          |
| $\{i, j\} = \{1, 32\}, k_{ij} = 3$                   |                                                                                        |
| $\alpha$                                             | $-4$                                                                                   |
| $\beta$                                              | $-4x^3 - 4x^2 + 4y^2 + 2y + 1$                                                         |
| $\gamma$                                             | $4x^3 + 4x^2 + 2x$                                                                     |
| $\{i, j\} = \{2, 4\}, k_{ij} = 2^2$                  |                                                                                        |
| $\alpha$                                             | $-2$                                                                                   |
| $\beta$                                              | $2x^4 - 2x^3y + 2x^2y^2 - 2xy^3 + 2y^4 + y$                                            |
| $\gamma$                                             | $-2y^4 - 2y^3 - y - 1$                                                                 |
| $\{i, j\} = \{2, 12\}, k_{ij} = 2$                   |                                                                                        |
| $\alpha$                                             | $0$                                                                                    |
| $\beta$                                              | $1$                                                                                    |
| $\gamma$                                             | $-1$                                                                                   |
| $\{i, j\} = \{2, 25\}, k_{ij} = 2^2 \cdot 3$         |                                                                                        |
| $\alpha$                                             | $4x^5 - 2y - 12$                                                                       |
| $\beta$                                              | $-4x^9 + 4x^8y - 4x^7y^2 + 4x^6y^2 - 2x^6 - 4x^5y^3 + 14x^4 - 14x^3y$                  |
| $\gamma$                                             | $+14x^2y^2 - 16xy^2 + 5xy + x + 16y^3 - 4y^2 + 5y$                                     |
| $\gamma$                                             | $4x^5y^3 - 4x^5y^2 + 2x^5 - 14y^3 + 16y^2 - 5y - 1$                                    |
| $\{i, j\} = \{2, 35\}, k_{ij} = 2^3$                 |                                                                                        |
| $\alpha$                                             | $2x^5 + 4y$                                                                            |
| $\beta$                                              | $-2x^9 + 2x^8y - 2x^7y^2 + 4x^6y - x^6 - 2x^5y^3 - 4x^5y^2 + 4x^5y - x^5$              |
| $\gamma$                                             | $+4xy^2 - 2xy - 4x - 4y^3 + 4y^2 + 2y - 4$                                             |
| $\gamma$                                             | $2x^5y^3 - 4x^5y + x^5 - 4y^2 + 2y + 4$                                                |
| $\{i, j\} = \{2, 36\}, k_{ij} = 2^4$                 |                                                                                        |
| $\alpha$                                             | $-2x^5 - 2y - 8$                                                                       |
| $\beta$                                              | $2x^9 - 2x^8y + 2x^7y^2 - 4x^6y^2 + 4x^6y - 3x^6 + 2x^5y^3 + 3x^5 + 4x^4$              |
| $\gamma$                                             | $-4x^3y + 4x^2y^2 - 10xy^2 + 9xy - 4x + 6y^3 + 2y^2 - y + 4$                           |
| $\gamma$                                             | $-2x^5y^3 + 4x^5y^2 - 4x^5y + 3x^5 - 4y^3 + 10y^2 - 9y + 4$                            |
| $\{i, j\} = \{3, 5\}, k_{ij} = 2^6 \cdot 3 \cdot 5$  |                                                                                        |
| $\alpha$                                             | $2$                                                                                    |
| $\beta$                                              | $-4x^4 + 8x^3y - 4x^3 - 16x^2y^2 + 16x^2y - 4x^2 + 32xy^3 - 48xy^2 + 24xy - 4x$        |
| $\gamma$                                             | $-64y^4 + 128y^3 - 96y^2 + 30y - 4$                                                    |
| $\gamma$                                             | $64y^4 - 224y^3 + 382y^2 - 460y + 479$                                                 |
| $\{i, j\} = \{3, 18\}, k_{ij} = 3 \cdot 5 \cdot 7^3$ |                                                                                        |
| $\alpha$                                             | $-3872x^5 - 117128y + 51704$                                                           |
| $\beta$                                              | $7744x^9 - 15488x^8y + 7744x^8 + 30976x^7y^2 - 30976x^7y + 7744x^7$                    |
| $\gamma$                                             | $+61952x^6y^2 - 1936x^6y + 44528x^6 - 61952x^5y^3 + 34848x^5y^2$                       |
| $\gamma$                                             | $-42592x^5y + 81312x^5 + 10816x^4 - 21632x^3y + 10816x^3 + 43264x^2y^2$                |
| $\gamma$                                             | $-43264x^2y + 10816x^2 + 27964xy^2 - 32712xy - 11739x + 30600y^3 + 108688y^2 - 28754y$ |
| $\gamma$                                             | $-34294$                                                                               |
| $\gamma$                                             | $-61952x^5y^3 + 30976x^5y^2 - 44528x^5y - 36784x^5 - 86528y^3 + 101828y^2$             |
| $\gamma$                                             | $-32184y + 22555$                                                                      |
| $\{i, j\} = \{3, 25\}, k_{ij} = 2^4$                 |                                                                                        |

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|----------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $\alpha$                                                 | $384x^5 - 72y - 70$                                                                                                                                                                                                                                                                        |
| $\beta$                                                  | $-768x^9 + 1536x^8y - 768x^8 - 3072x^7y^2 + 3072x^7y - 768x^7 + 1536x^6y^2$<br>$-960x^6y - 144x^6 - 3072x^5y^3 + 3456x^5y^2 - 1056x^5y - 144x^5 + 380x^4$<br>$-760x^3y + 380x^3 + 1520x^2y^2 - 1520x^2y + 380x^2 - 796xy^2 + 544xy + 54x$<br>$+1592y^3 - 1812y^2 + 506y + 54$              |
| $\gamma$                                                 | $3072x^5y^3 - 5376x^5y^2 + 2784x^5y - 312x^5 - 1520y^3 + 2678y^2 - 1412y + 163$                                                                                                                                                                                                            |
| $\{i, j\} = \{3, 31\}, k_{ij} = 2 \cdot 7^2$             |                                                                                                                                                                                                                                                                                            |
| $\alpha$                                                 | $64x - 24$                                                                                                                                                                                                                                                                                 |
| $\beta$                                                  | $-128x^5 + 256x^4y - 80x^4 - 512x^3y^2 + 416x^3y - 80x^3 + 704x^2y^2$<br>$-672x^2y + 32x^2 - 512xy^3 + 256xy^2 + 292xy - 194x + 192y^3 - 168y^2 - 42y + 11$                                                                                                                                |
| $\gamma$                                                 | $512xy^3 - 1024xy^2 + 624xy - 56x - 192y^3 + 384y^2 - 234y + 85$                                                                                                                                                                                                                           |
| $\{i, j\} = \{3, 34\}, k_{ij} = 3 \cdot 7^2$             |                                                                                                                                                                                                                                                                                            |
| $\alpha$                                                 | $-256x^5 - 32y - 224$                                                                                                                                                                                                                                                                      |
| $\beta$                                                  | $512x^9 - 1024x^8y + 512x^8 + 2048x^7y^2 - 2048x^7y + 512x^7 + 384x^6y$<br>$-256x^6 + 2048x^5y^3 - 3840x^5y^2 + 2880x^5y - 640x^5 + 320x^4 - 640x^3y$<br>$+320x^3 + 1280x^2y^2 - 1280x^2y + 320x^2 - 16xy^2 + 208xy - 166x + 1312y^3$<br>$-2328y^2 + 1828y - 409$                          |
| $\gamma$                                                 | $-2048x^5y^3 + 3072x^5y^2 - 1728x^5y + 384x^5 - 1280y^3 + 1928y^2 - 1064y + 243$                                                                                                                                                                                                           |
| $\{i, j\} = \{3, 37\}, k_{ij} = 2^2 \cdot 3^2 \cdot 5^2$ |                                                                                                                                                                                                                                                                                            |
| $\alpha$                                                 | $-1664x^5 - 1352y - 762$                                                                                                                                                                                                                                                                   |
| $\beta$                                                  | $3328x^9 - 6656x^8y + 3328x^8 + 13312x^7y^2 - 13312x^7y + 3328x^7$<br>$+6656x^6y^2 - 2496x^6y + 1040x^6 - 13312x^5y^3 + 11648x^5y^2 - 2912x^5y$<br>$+1040x^5 + 900x^4 - 1800x^3y + 900x^3 + 3600x^2y^2 - 3600x^2y + 900x^2$<br>$+1124xy^2 - 1000xy - 138x - 2248y^3 + 4476y^2 + 38y - 138$ |
| $\gamma$                                                 | $-13312x^5y^3 + 16640x^5y^2 - 8736x^5y + 1144x^5 - 3600y^3 + 4838y^2 - 2200y$<br>$+519$                                                                                                                                                                                                    |
| $\{i, j\} = \{4, 8\}, k_{ij} = 2^3$                      |                                                                                                                                                                                                                                                                                            |
| $\alpha$                                                 | $2$                                                                                                                                                                                                                                                                                        |
| $\beta$                                                  | $-2x^4 - 2x^3 - 2x^2 - 3x - 2y^2 - 2y - 5$                                                                                                                                                                                                                                                 |
| $\gamma$                                                 | $2x^4 + 2x^3 + 2x^2 + 2x + 3$                                                                                                                                                                                                                                                              |
| $\{i, j\} = \{4, 14\}, k_{ij} = 2$                       |                                                                                                                                                                                                                                                                                            |
| $\alpha$                                                 | $2$                                                                                                                                                                                                                                                                                        |
| $\beta$                                                  | $-2x^4 - x - 2y^2 - 2y - 3$                                                                                                                                                                                                                                                                |
| $\gamma$                                                 | $2x^4 + 1$                                                                                                                                                                                                                                                                                 |
| $\{i, j\} = \{4, 17\}, k_{ij} = 2^5 \cdot 3 \cdot 7$     |                                                                                                                                                                                                                                                                                            |
| $\alpha$                                                 | $2x^4 - 64$                                                                                                                                                                                                                                                                                |
| $\beta$                                                  | $4x^7 - 4x^6 - 5x^5 - 2x^4y^2 - 2x^4y + 10x^4 - 130x^3 + 108x^2 + 184x + 64y^2$<br>$+64y - 304$                                                                                                                                                                                            |
| $\gamma$                                                 | $-4x^7 + 4x^6 + 4x^5 - 12x^4 + 130x^3 - 108x^2 - 152x + 368$                                                                                                                                                                                                                               |
| $\{i, j\} = \{4, 19\}, k_{ij} = 2 \cdot 3 \cdot 5$       |                                                                                                                                                                                                                                                                                            |
| $\alpha$                                                 | $-22x^4 + 152$                                                                                                                                                                                                                                                                             |
| $\beta$                                                  | $44x^7 + 44x^6 + 99x^5 + 22x^4y^2 + 22x^4y + 154x^4 - 62x^3 + 92x^2 - 46x$<br>$-152y^2 - 152y - 30$                                                                                                                                                                                        |
| $\gamma$                                                 | $44x^7 + 44x^6 + 88x^5 + 132x^4 - 62x^3 + 92x^2 + 30x + 122$                                                                                                                                                                                                                               |
| $\{i, j\} = \{4, 25\}, k_{ij} = 2^2 \cdot 5$             |                                                                                                                                                                                                                                                                                            |
| $\alpha$                                                 | $-6x + 2$                                                                                                                                                                                                                                                                                  |
| $\beta$                                                  | $-6x^4 + 2x^3 + 9x^2 + 6xy^2 + 6xy + 3x - 2y^2 - 2y - 11$                                                                                                                                                                                                                                  |
| $\gamma$                                                 | $6x^4 - 2x^3 - 6x^2 + 2x + 9$                                                                                                                                                                                                                                                              |
| $\{i, j\} = \{4, 32\}, k_{ij} = 2 \cdot 7$               |                                                                                                                                                                                                                                                                                            |
| $\alpha$                                                 | $-2x^4 + 4$                                                                                                                                                                                                                                                                                |
| $\beta$                                                  | $-2x^7 - 2x^6 + x^5 + 2x^4y^2 + 2x^4y + 4x^4 + 5x^3 + 2x^2 - 5x - 4y^2 - 4y - 9$                                                                                                                                                                                                           |
| $\gamma$                                                 | $2x^7 + 2x^6 - 2x^4 - 5x^3 - 2x^2 + 3x + 5$                                                                                                                                                                                                                                                |
| $\{i, j\} = \{4, 34\}, k_{ij} = 2 \cdot 7$               |                                                                                                                                                                                                                                                                                            |
| $\alpha$                                                 | $288x^4 + 14$                                                                                                                                                                                                                                                                              |
| $\beta$                                                  | $288x^7 - 144x^6 - 72x^5 - 288x^4y^2 - 288x^4y - 324x^4 + 176x^3 + 56x^2 - 35x$<br>$-14y^2 - 14y$                                                                                                                                                                                          |
| $\gamma$                                                 | $-288x^7 + 144x^6 - 72x^5 + 36x^4 - 176x^3 - 56x^2 + 28x - 14$                                                                                                                                                                                                                             |
| $\{i, j\} = \{4, 38\}, k_{ij} = 1$                       |                                                                                                                                                                                                                                                                                            |
| $\alpha$                                                 | $12x - 8$                                                                                                                                                                                                                                                                                  |
| $\beta$                                                  | $-12x^4 + 8x^3 - 12x^2 - 12xy^2 - 12xy - 4x + 8y^2 + 8y - 1$                                                                                                                                                                                                                               |

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| $\gamma$                                                | $-12x^4 + 8x^3 - 6x^2 + 4x - 9$                                                                                                                                                                                                            |
| $\{i, j\} = \{4, 39\}, k_{ij} = 2^2 \cdot 5^3$          |                                                                                                                                                                                                                                            |
| $\alpha$                                                | $-608x^4 + 358$                                                                                                                                                                                                                            |
| $\beta$                                                 | $608x^7 - 304x^6 + 760x^5 + 608x^4y^2 + 608x^4y + 228x^4 + 364x^3 - 68x^2 + 37x$<br>$-358y^2 - 358y - 500$                                                                                                                                 |
| $\gamma$                                                | $608x^7 - 304x^6 + 456x^5 - 380x^4 + 364x^3 - 68x^2 + 216x - 142$                                                                                                                                                                          |
| $\{i, j\} = \{4, 40\}, k_{ij} = 2^3 \cdot 5 \cdot 7$    |                                                                                                                                                                                                                                            |
| $\alpha$                                                | $-608x^4 + 678$                                                                                                                                                                                                                            |
| $\beta$                                                 | $608x^7 + 304x^6 + 760x^5 + 608x^4y^2 + 608x^4y + 988x^4 + 44x^3 + 516x^2 - 59x$<br>$-678y^2 - 678y - 280$                                                                                                                                 |
| $\gamma$                                                | $608x^7 + 304x^6 + 456x^5 + 380x^4 + 44x^3 + 516x^2 + 280x + 398$                                                                                                                                                                          |
| $\{i, j\} = \{5, 17\}, k_{ij} = 2^2 \cdot 5^2 \cdot 13$ |                                                                                                                                                                                                                                            |
| $\alpha$                                                | $42x^4 - 692$                                                                                                                                                                                                                              |
| $\beta$                                                 | $84x^7 - 84x^6 + 231x^5 - 42x^4y^2 + 42x^4y - 462x^4 - 502x^3 - 464x^2 - 194x$<br>$+692y^2 - 692y + 304$                                                                                                                                   |
| $\gamma$                                                | $-84x^7 + 84x^6 - 252x^5 + 420x^4 + 502x^3 + 464x^2 + 540x + 388$                                                                                                                                                                          |
| $\{i, j\} = \{5, 18\}, k_{ij} = 2 \cdot 3^2 \cdot 5^4$  |                                                                                                                                                                                                                                            |
| $\alpha$                                                | $-2x^4 - 328$                                                                                                                                                                                                                              |
| $\beta$                                                 | $4x^7 - 4x^6 - 7x^5 + 2x^4y^2 - 2x^4y + 22x^4 + 658x^3 - 724x^2 - 1086x$<br>$+328y^2 - 328y + 3750$                                                                                                                                        |
| $\gamma$                                                | $4x^7 - 4x^6 - 8x^5 + 20x^4 + 658x^3 - 724x^2 - 1250x + 3422$                                                                                                                                                                              |
| $\{i, j\} = \{5, 20\}, k_{ij} = 2 \cdot 3 \cdot 5^2$    |                                                                                                                                                                                                                                            |
| $\alpha$                                                | $6x - 14$                                                                                                                                                                                                                                  |
| $\beta$                                                 | $-12x^4 + 4x^3 + 29x^2 - 6xy^2 + 6xy + 57x + 14y^2 - 14y + 68$                                                                                                                                                                             |
| $\gamma$                                                | $-12x^4 + 4x^3 + 32x^2 + 56x + 54$                                                                                                                                                                                                         |
| $\{i, j\} = \{5, 26\}, k_{ij} = 3 \cdot 5^2$            |                                                                                                                                                                                                                                            |
| $\alpha$                                                | $28x + 24$                                                                                                                                                                                                                                 |
| $\beta$                                                 | $28x^4 + 24x^3 + 28x^2 - 28xy^2 + 28xy - 4x - 24y^2 + 24y + 25$                                                                                                                                                                            |
| $\gamma$                                                | $-28x^4 - 24x^3 - 42x^2 - 36x - 49$                                                                                                                                                                                                        |
| $\{i, j\} = \{5, 32\}, k_{ij} = 2 \cdot 3 \cdot 5$      |                                                                                                                                                                                                                                            |
| $\alpha$                                                | $2x^4 - 12$                                                                                                                                                                                                                                |
| $\beta$                                                 | $2x^7 + 2x^6 + 3x^5 - 2x^4y^2 + 2x^4y + 4x^4 - 3x^3 + 3x + 12y^2 - 12y + 9$                                                                                                                                                                |
| $\gamma$                                                | $-2x^7 - 2x^6 - 4x^5 - 6x^4 + 3x^3 + 3x + 3$                                                                                                                                                                                               |
| $\{i, j\} = \{5, 40\}, k_{ij} = 2^2 \cdot 3 \cdot 13$   |                                                                                                                                                                                                                                            |
| $\alpha$                                                | $32x^4 - 2$                                                                                                                                                                                                                                |
| $\beta$                                                 | $-32x^7 - 16x^6 + 24x^5 - 32x^4y^2 + 32x^4y + 12x^4 - 20x^3 - 28x^2 + 17x$<br>$+2y^2 - 2y + 52$                                                                                                                                            |
| $\gamma$                                                | $-32x^7 - 16x^6 + 40x^5 + 44x^4 - 20x^3 - 28x^2 + 16x + 50$                                                                                                                                                                                |
| $\{i, j\} = \{6, 9\}, k_{ij} = 3^2$                     |                                                                                                                                                                                                                                            |
| $\alpha$                                                | $-12$                                                                                                                                                                                                                                      |
| $\beta$                                                 | $-24x^4 - 48x^3y + 24x^3 - 96x^2y^2 + 96x^2y - 24x^2 - 192xy^3 + 288xy^2$<br>$-144xy + 24x - 128y^4 + 256y^3 - 184y^2 + 64y - 6$                                                                                                           |
| $\gamma$                                                | $128y^4 - 256y^3 + 196y^2 - 58y + 9$                                                                                                                                                                                                       |
| $\{i, j\} = \{6, 11\}, k_{ij} = 3 \cdot 13^2$           |                                                                                                                                                                                                                                            |
| $\alpha$                                                | $-32$                                                                                                                                                                                                                                      |
| $\beta$                                                 | $-64x^4 - 128x^3y + 64x^3 - 256x^2y^2 + 256x^2y - 64x^2 - 512xy^3 + 768xy^2$<br>$-384xy + 64x - 512y^4 + 640y^3 + 112y^2 - 556y + 347$                                                                                                     |
| $\gamma$                                                | $512y^4 - 1408y^3 + 1648y^2 - 1036y + 411$                                                                                                                                                                                                 |
| $\{i, j\} = \{6, 13\}, k_{ij} = 2 \cdot 3^2$            |                                                                                                                                                                                                                                            |
| $\alpha$                                                | $-6$                                                                                                                                                                                                                                       |
| $\beta$                                                 | $-12x^4 - 24x^3y + 12x^3 - 48x^2y^2 + 48x^2y - 12x^2 - 96xy^3 + 144xy^2 - 72xy$<br>$+12x - 64y^4 + 64y^3 + 36y^2 - 66y + 26$                                                                                                               |
| $\gamma$                                                | $64y^4 - 160y^3 + 162y^2 - 78y + 19$                                                                                                                                                                                                       |
| $\{i, j\} = \{6, 16\}, k_{ij} = 2^4 \cdot 3^2$          |                                                                                                                                                                                                                                            |
| $\alpha$                                                | $-56x^5 + 196y - 104$                                                                                                                                                                                                                      |
| $\beta$                                                 | $-112x^9 - 224x^8y + 112x^8 - 448x^7y^2 + 448x^7y - 112x^7 + 252x^6y - 56x^6$<br>$+896x^5y^3 - 840x^5y^2 + 504x^5y - 112x^5 - 40x^4 - 80x^3y + 40x^3$<br>$-160x^2y^2 + 160x^2y - 40x^2 - 98xy^2 - 64xy - 83x + 124y^3 - 412y^2 - 30y - 40$ |
| $\gamma$                                                | $896x^5y^3 - 1344x^5y^2 + 924x^5y - 168x^5 + 320y^3 - 578y^2 + 176y - 123$                                                                                                                                                                 |

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$$\begin{aligned}
&\{i, j\} = \{6, 28\}, k_{ij} = 2 \cdot 3^3 \cdot 13 \\
&\alpha \quad -4480x^5 + 9800y - 4714 \\
&\beta \quad -8960x^9 - 17920x^8y + 8960x^8 - 35840x^7y^2 + 35840x^7y - 8960x^7 \\
&\quad -17920x^6y^2 + 33600x^6y - 12880x^6 - 35840x^5y^3 + 85120x^5y^2 \\
&\quad -63840x^5y + 12880x^5 + 4012x^4 + 8024x^3y - 4012x^3 + 16048x^2y^2 \\
&\quad -16048x^2y + 4012x^2 + 3124xy^2 - 20260xy + 7666x + 6248y^3 - 33844y^2 + 30878y - 7666 \\
&\gamma \quad 35840x^5y^3 - 62720x^5y^2 + 43680x^5y - 10920x^5 - 16048y^3 + 25634y^2 \\
&\quad -22166y + 5839 \\
&\{i, j\} = \{6, 33\}, k_{ij} = 3^2 \cdot 13^2 \\
&\alpha \quad -41216x^5 + 414736y - 209396 \\
&\beta \quad -82432x^9 - 164864x^8y + 82432x^8 - 329728x^7y^2 + 329728x^7y - 82432x^7 \\
&\quad -494592x^6y^2 + 886144x^6y - 561568x^6 - 329728x^5y^3 + 783104x^5y^2 \\
&\quad -669760x^5y - 82432x^5 + 26856x^4 + 53712x^3y - 26856x^3 + 107424x^2y^2 \\
&\quad -107424x^2y + 26856x^2 - 46232xy^2 - 436500xy + 335182x - 307312y^3 - 135992y^2 \\
&\quad +299832y + 26856 \\
&\gamma \quad 329728x^5y^3 - 741888x^5y^2 + 690368x^5y - 322000x^5 - 107424y^3 \\
&\quad +138020y^2 - 298818y + 181019 \\
&\{i, j\} = \{6, 38\}, k_{ij} = 3^2 \\
&\alpha \quad -2816x^5 + 6336y^2 - 7392y + 2100 \\
&\beta \quad -5632x^9 - 11264x^8y + 5632x^8 - 22528x^7y^2 + 22528x^7y - 5632x^7 \\
&\quad +11264x^6y^2 - 7040x^6y + 528x^6 + 22528x^5y^3 - 25344x^5y^2 + 5280x^5y \\
&\quad -528x^5 + 1368x^4 + 2736x^3y + 2712x^3 + 5472x^2y^2 + 2688x^2y - 2712x^2 \\
&\quad -5640xy^2 - 2844xy + 3594x - 6336y^4 - 1248y^3 - 2760y^2 + 7848y - 2688 \\
&\gamma \quad 22528x^5y^3 - 28160x^5y^2 + 13376x^5y - 2552x^5 + 3168x^3y^2 + 1320x^3y \\
&\quad -1812x^3 + 2040x^2 + 3168xy^3 - 264xy^2 - 2472xy + 906x - 5472y^3 - 2772y^2 + 4674y \\
&\quad -579 \\
&\{i, j\} = \{6, 39\}, k_{ij} = 2 \cdot 3^2 \\
&\alpha \quad 64x^5 - 80y + 16 \\
&\beta \quad 128x^9 + 256x^8y - 128x^8 + 512x^7y^2 - 512x^7y + 128x^7 - 32x^6y - 64x^6 \\
&\quad -512x^5y^3 + 704x^5y^2 - 432x^5y + 96x^5 - 96x^4 - 192x^3y + 96x^3 \\
&\quad -384x^2y^2 + 384x^2y - 96x^2 + 40xy^2 + 72xy + 34x + 464y^3 - 532y^2 + 336y - 65 \\
&\gamma \quad -512x^5y^3 + 768x^5y^2 - 400x^5y + 32x^5 + 384y^3 - 556y^2 + 324y - 31 \\
&\{i, j\} = \{6, 40\}, k_{ij} = 2^4 \cdot 3 \\
&\alpha \quad -608x^5 - 760x + 2888y - 1508 \\
&\beta \quad -1216x^9 - 2432x^8y + 1216x^8 - 4864x^7y^2 + 4864x^7y - 1216x^7 \\
&\quad +4864x^6y^2 - 2128x^6y + 1976x^6 + 4864x^5y^3 - 4256x^5y^2 + 2888x^5y \\
&\quad -3876x^5 - 3040x^4y + 1848x^4 - 6080x^3y^2 + 6736x^3y - 1848x^3 \\
&\quad +7392x^2y^2 - 3972x^2y + 2798x^2 + 6080xy^3 - 8076xy^2 + 1410xy - 1749x \\
&\quad -4200y^3 + 654y^2 - 2052y + 2222 \\
&\gamma \quad 4864x^5y^3 - 4864x^5y^2 + 2584x^5y + 380x^5 + 6080xy^3 - 6080xy^2 + 3230xy \\
&\quad +475x - 1312y^3 + 590y^2 - 2084y + 762 \\
&\{i, j\} = \{7, 11\}, k_{ij} = 7^2 \\
&\alpha \quad -32 \\
&\beta \quad -64x^4 - 128x^3y - 64x^3 - 256x^2y^2 - 256x^2y - 64x^2 - 512xy^3 - 768xy^2 \\
&\quad -384xy - 64x - 512y^4 - 1152y^3 - 912y^2 - 356y - 37 \\
&\gamma \quad 512y^4 + 896y^3 + 624y^2 + 188y + 27 \\
&\{i, j\} = \{7, 17\}, k_{ij} = 2^2 \cdot 7^2 \\
&\alpha \quad 60x^5 - 3600y - 1226 \\
&\beta \quad 120x^9 + 240x^8y + 120x^8 + 480x^7y^2 + 480x^7y + 120x^7 + 960x^6y^2 \\
&\quad -270x^6y + 1110x^6 + 960x^5y^3 - 60x^5y^2 + 1020x^5y + 2100x^5 - 412x^4 \\
&\quad -824x^3y - 412x^3 - 1648x^2y^2 - 1648x^2y - 412x^2 - 1496xy^2 - 363xy - 721x \\
&\quad +304y^3 - 2374y^2 - 922y - 1030 \\
&\gamma \quad -960x^5y^3 - 480x^5y^2 - 990x^5y + 990x^5 + 3296y^3 + 3448y^2 + 2109y - 309 \\
&\{i, j\} = \{7, 21\}, k_{ij} = 5^2 \cdot 11 \\
&\alpha \quad 98064x^5 - 2542752y^2 - 8223768y - 3476416 \\
&\beta \quad 524178x^9 + 812160x^8y + 406080x^8 + 1624320x^7y^2 + 1624320x^7y + 406080x^7 \\
&\quad +3248640x^6y^2 - 101520x^6y + 2538000x^6 + 3248640x^5y^3 + 1053864x^5y^2 \\
&\quad +3899232x^5y + 5116068x^5 - 233237x^4 - 1955200x^3y - 977600x^3 \\
&\quad -3910400x^2y^2 - 3910400x^2y - 977600x^2 - 4394500xy^2 - 1532200xy - 1552982x \\
&\quad +2542752y^4 + 1674344y^3 - 3221572y^2 - 2445928y - 2142380
\end{aligned}$$


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$$\begin{aligned}
\gamma & 328050x^8 - 236196x^7y - 446148x^7 - 419904x^6y + 118098x^6 - 3248640x^5y^3 \\
& -1151928x^5y^2 - 989604x^5y + 2459970x^5 - 635688x^3y^2 - 1614168x^3y \\
& -758379x^3 + 635688x^2y^2 + 125442x^2y + 14016x^2 + 1271376xy^3 \\
& +3228336xy^2 + 3005484xy + 744363x + 5278048y^3 + 3221792y^2 + 1174442y - 1333761 \\
\{i, j\} &= \{7, 24\}, k_{ij} = 2^2 \cdot 5 \cdot 7^3 \\
\alpha & -2096x^5 + 343220y + 166122 \\
\beta & -4192x^9 - 8384x^8y - 4192x^8 - 16768x^7y^2 - 16768x^7y - 4192x^7 \\
& +50304x^6y^2 + 126808x^6y + 165060x^6 + 33536x^5y^3 + 102704x^5y^2 \\
& +150912x^5y - 173444x^5 - 8356x^4 - 16712x^3y - 8356x^3 - 33424x^2y^2 \\
& -33424x^2y - 8356x^2 - 71338xy^2 - 89796xy - 99025x - 276372y^3 + 34422y^2 - 42404y \\
& +82313 \\
\gamma & 33536x^5y^3 + 100608x^5y^2 + 151960x^5y + 169252x^5 + 66848y^3 + 28934y^2 \\
& -39660y - 90669 \\
\{i, j\} &= \{7, 25\}, k_{ij} = 2^3 \cdot 3^2 \cdot 5 \cdot 7 \\
\alpha & 3968x^5 - 7688y - 4350 \\
\beta & 7936x^9 + 15872x^8y + 7936x^8 + 31744x^7y^2 + 31744x^7y + 7936x^7 \\
& +15872x^6y^2 + 1984x^6y + 3472x^6 + 31744x^5y^3 + 19840x^5y^2 \\
& +12896x^5y + 3472x^5 - 9196x^4 - 18392x^3y - 9196x^3 - 36784x^2y^2 \\
& -36784x^2y - 9196x^2 - 14548xy^2 - 3384xy - 1830x - 29096y^3 - 29004y^2 - 11394y \\
& -1830 \\
\gamma & -31744x^5y^3 - 39680x^5y^2 - 22816x^5y - 2232x^5 + 36784y^3 + 47902y^2 \\
& +25896y + 3683 \\
\{i, j\} &= \{7, 33\}, k_{ij} = 5^3 \cdot 7 \cdot 11 \\
\alpha & 2438656x^5 - 6817152y^2 - 74427504y - 18265812 \\
\beta & 7620800x^9 + 13745152x^8y + 6872576x^8 + 27490304x^7y^2 + 27490304x^7y \\
& +6872576x^7 + 41235456x^6y^2 - 5154432x^6y + 33074272x^6 + 27490304x^5y^3 \\
& -10059456x^5y^2 + 26520384x^5y + 50234928x^5 - 10191832x^4 - 33044016x^3y \\
& -16522008x^3 - 66088032x^2y^2 - 66088032x^2y - 16522008x^2 - 52634776xy^2 \\
& -11796740xy - 26006650x + 6817152y^4 + 11748048y^3 - 25007464y^2 - 1991984y - 16085852 \\
\gamma & 1371744x^8 - 748224x^7y - 1745856x^7 - 623520x^6y + 2431728x^6 \\
& -27490304x^5y^3 - 19869504x^5y^2 - 21199680x^5y + 9796192x^5 \\
& -3408576x^3y^2 - 9463648x^3y - 5830144x^3 + 3408576x^2y^2 + 3133472x^2y \\
& +2665056x^2 + 3408576xy^3 + 4350784xy^2 - 2035152xy - 5580128x + 59270880y^3 \\
& +67034692y^2 + 44604486y + 2170335 \\
\{i, j\} &= \{7, 35\}, k_{ij} = 2^3 \cdot 7^2 \\
\alpha & -68256x^5 + 132720y^2 + 496120y + 295046 \\
\beta & -199080x^9 - 364032x^8y - 182016x^8 - 728064x^7y^2 - 728064x^7y - 182016x^7 \\
& -1092096x^6y^2 - 409536x^6y - 329904x^6 - 728064x^5y^3 - 824760x^5y^2 \\
& -585864x^5y - 557424x^5 + 42176x^4 + 424008x^3y + 212004x^3 + 848016x^2y^2 \\
& +848016x^2y + 212004x^2 + 1047348xy^2 + 511848xy + 195286x - 132720y^4 + 285536y^3 \\
& +750732y^2 + 288254y + 160448 \\
\gamma & -31284x^8 + 17064x^7y + 39816x^7 + 14220x^6y - 8532x^6 + 728064x^5y^3 \\
& +528984x^5y^2 + 270180x^5y - 105228x^5 + 66360x^3y^2 + 189284x^3y \\
& +75854x^3 - 66360x^2y^2 - 19456x^2y + 9060x^2 - 66360xy^3 - 189284xy^2 \\
& -245682xy - 84914x - 715296y^3 - 473250y^2 - 124010y + 67495 \\
\{i, j\} &= \{7, 39\}, k_{ij} = 2^2 \cdot 5 \\
\alpha & -8x - 4 \\
\beta & -16x^5 - 32x^4y - 24x^4 - 64x^3y^2 - 80x^3y - 24x^3 - 32x^2y^2 - 60x^2y \\
& -24x^2 + 64xy^3 + 40xy^2 - 48xy - 8x + 32y^3 + 20y^2 - 17y + 8 \\
\gamma & 64xy^3 + 96xy^2 + 34xy + 32y^3 + 48y^2 + 17y + 8 \\
\{i, j\} &= \{8, 11\}, k_{ij} = 7^3 \\
\alpha & -243 \\
\beta & -243x^4 - 243x^3y - 243x^2y^2 - 243xy^3 - 162y^4 - 54y^3 + 117y^2 - 159y + 25 \\
\gamma & 162y^4 - 108y^3 + 234y^2 - 75y + 50 \\
\{i, j\} &= \{8, 16\}, k_{ij} = 2^3 \cdot 3 \\
\alpha & -20x^5 + 50y - 24 \\
\beta & -20x^9 - 20x^8y - 20x^7y^2 + 20x^6y^2 + 40x^6y + 50x^6 + 40x^5y^3 \\
& +60x^5y^2 + 80x^5y + 50x^5 - 14x^4 - 14x^3y - 14x^2y^2 - 36xy^2 - 37xy - 25x \\
& -22y^3 - 48y^2 - 74y - 25 \\
\gamma & 40x^5y^3 + 40x^5y^2 + 80x^5y + 100x^5 + 28y^3 - 72y^2 - 74y - 50
\end{aligned}$$


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$$\begin{aligned}
&\{i, j\} = \{8, 24\}, k_{ij} = 2^6 \cdot 11 \\
&\alpha \quad -26406x^5 - 738720y^2 + 11008872y + 2639266 \\
&\beta \quad -52731x^9 - 42606x^8y - 42606x^7y^2 + 170424x^6y^2 + 766908x^6y \\
&\quad + 3259359x^6 + 85212x^5y^3 + 450360x^5y^2 + 2327940x^5y - 5755293x^5 \\
&\quad - 38973x^4 - 3598366x^3y - 3598366x^2y^2 + 589120xy^2 - 587016xy - 4344189x \\
&\quad + 738720y^4 - 6767020y^3 + 1834360y^2 - 2162172y + 1379885 \\
&\gamma \quad -52650x^8 + 20250x^7y - 105300x^7 - 64800x^6y - 263250x^6 + 85212x^5y^3 \\
&\quad + 381348x^5y^2 + 1213866x^5y + 5886918x^5 - 369360x^3y^2 + 2480544x^3y \\
&\quad + 504034x^3 - 738720x^2y^2 - 2157698x^2y + 1008068x^2 - 738720xy^3 \\
&\quad + 3114288xy^2 - 826784xy + 2520170x + 4241852y^3 + 2352700y^2 - 2969162y - 2639970 \\
&\{i, j\} = \{8, 28\}, k_{ij} = 2 \cdot 3 \\
&\alpha \quad -52x^5 + 338y - 30 \\
&\beta \quad -52x^9 - 52x^8y - 52x^7y^2 - 52x^6y^2 + 156x^6y - 182x^6 - 52x^5y^3 \\
&\quad + 156x^5y^2 - 208x^5y + 178x^4 + 178x^3y + 178x^2y^2 - 160xy^2 - 157xy + 12x \\
&\quad - 160y^3 + 12y^2 - 3y \\
&\gamma \quad 52x^5y^3 - 52x^5y^2 + 156x^5y - 182x^5 - 178y^3 - 160y^2 - 157y + 12 \\
&\{i, j\} = \{8, 29\}, k_{ij} = 2 \cdot 7 \\
&\alpha \quad 12x^5 + 18y + 14 \\
&\beta \quad 12x^9 + 12x^8y + 12x^7y^2 + 12x^6y^2 - 12x^6y - 6x^6 + 12x^5y^3 \\
&\quad - 12x^5y^2 + 2x^4 + 2x^3y + 2x^2y^2 - 16xy^2 - 5xy + 14x - 16y^3 + 4y^2 + 21y \\
&\gamma \quad -12x^5y^3 + 12x^5y^2 - 12x^5y - 6x^5 - 2y^3 - 16y^2 - 5y + 14 \\
&\{i, j\} = \{8, 30\}, k_{ij} = 2^5 \cdot 7 \\
&\alpha \quad 608x^5 - 722y - 170 \\
&\beta \quad 608x^9 + 608x^8y + 608x^7y^2 + 304x^6y^2 - 1064x^6y + 380x^6 + 608x^5y^3 \\
&\quad - 1216x^5y^2 + 1216x^5y - 190x^5 - 664x^4 - 664x^3y - 664x^2y^2 + 390xy^2 \\
&\quad + 668xy - 54x + 58y^3 + 112y^2 - 473y + 27 \\
&\gamma \quad -608x^5y^3 + 304x^5y^2 - 1064x^5y + 380x^5 + 664y^3 + 390y^2 + 668y - 54 \\
&\{i, j\} = \{8, 31\}, k_{ij} = 2^5 \cdot 7 \\
&\alpha \quad 288x^5 + 162y - 50 \\
&\beta \quad 288x^9 + 288x^8y + 288x^7y^2 + 144x^6y^2 - 216x^6y - 108x^6 + 288x^5y^3 \\
&\quad - 288x^5y^2 + 144x^5y + 54x^5 - 248x^4 - 248x^3y - 248x^2y^2 - 286xy^2 - 12xy \\
&\quad + 174x - 410y^3 + 212y^2 + 155y - 87 \\
&\gamma \quad -288x^5y^3 + 144x^5y^2 - 216x^5y - 108x^5 + 248y^3 - 286y^2 - 12y + 174 \\
&\{i, j\} = \{8, 34\}, k_{ij} = 2^3 \cdot 7 \cdot 11 \\
&\alpha \quad 224x^5 - 3430y + 300 \\
&\beta \quad 224x^9 + 224x^8y + 224x^7y^2 + 336x^6y^2 - 952x^6y + 1652x^6 + 224x^5y^3 \\
&\quad - 784x^5y^2 + 1288x^5y + 826x^5 - 1464x^4 - 1464x^3y - 1464x^2y^2 \\
&\quad + 1234xy^2 + 1028xy - 458x + 1966y^3 - 70y^2 + 206y - 229 \\
&\gamma \quad -224x^5y^3 + 336x^5y^2 - 952x^5y + 1652x^5 + 1464y^3 + 1234y^2 + 1028y - 458 \\
&\{i, j\} = \{9, 13\}, k_{ij} = 3 \\
&\alpha \quad 0 \\
&\beta \quad -1 \\
&\gamma \quad 1 \\
&\{i, j\} = \{9, 17\}, k_{ij} = 3 \cdot 7 \\
&\alpha \quad -32x^5 + 16y + 28 \\
&\beta \quad 32x^9 - 32x^8y + 16x^8 + 32x^7y^2 - 32x^7y + 8x^7 - 32x^6y^2 + 40x^6 \\
&\quad + 64x^5y^3 - 96x^5y^2 + 56x^5 - 16x^4 + 16x^3y - 8x^3 - 16x^2y^2 + 16x^2y \\
&\quad - 4x^2 + 32xy^2 + 4xy - 34x - 48y^3 + 60y^2 + 14y - 49 \\
&\gamma \quad -64x^5y^3 + 160x^5y^2 - 48x^5y - 72x^5 + 32y^3 - 112y^2 + 16y + 64 \\
&\{i, j\} = \{9, 20\}, k_{ij} = 3^3 \\
&\alpha \quad 32x^5 - 16y + 68 \\
&\beta \quad -32x^9 + 32x^8y - 16x^8 - 32x^7y^2 + 32x^7y - 8x^7 + 32x^6 + 64x^5y^3 \\
&\quad - 96x^5y^2 - 56x^5 - 80x^4 + 80x^3y - 40x^3 - 80x^2y^2 + 80x^2y - 20x^2 \\
&\quad - 16xy^2 - 20xy + 50x + 176y^3 - 188y^2 + 66y - 95 \\
&\gamma \quad 64x^5y^3 - 96x^5y^2 + 48x^5y - 72x^5 + 160y^3 - 208y^2 + 160y - 120 \\
&\{i, j\} = \{9, 25\}, k_{ij} = 17 \\
&\alpha \quad -8y - 16 \\
&\beta \quad -2x + 8y^3 + 20y^2 + 10y - 1 \\
&\gamma \quad 8x^3y + 16x^3 - 8xy^2 - 16xy + 2 \\
&\{i, j\} = \{9, 28\}, k_{ij} = 2^3 \cdot 3
\end{aligned}$$


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|----------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $\alpha$                                                       | $448x^5 + 784y - 1448$                                                                                                                                                                                                                                                                      |
| $\beta$                                                        | $-448x^9 + 448x^8y - 224x^8 - 448x^7y^2 + 448x^7y - 112x^7 + 448x^6y^2$<br>$-672x^6y + 1120x^6 - 448x^5y^3 + 896x^5y^2 - 1680x^5y + 560x^5 + 776x^4$<br>$-776x^3y + 388x^3 + 776x^2y^2 - 776x^2y + 194x^2 + 8xy^2 + 100xy - 820x - 8y^3$<br>$-488y^2 + 1594y - 410$                         |
| $\gamma$                                                       | $448x^5y^3 - 1120x^5y^2 + 1008x^5y - 1176x^5 - 776y^3 + 1156y^2 - 682y + 917$                                                                                                                                                                                                               |
| $\{i, j\} = \{9, 33\}, k_{ij} = 3^2$                           |                                                                                                                                                                                                                                                                                             |
| $\alpha$                                                       | $32x^5 - 16y - 4$                                                                                                                                                                                                                                                                           |
| $\beta$                                                        | $-32x^9 + 32x^8y - 16x^8 - 32x^7y^2 + 32x^7y - 8x^7 + 32x^6y + 16x^6$<br>$-32x^5y^3 + 16x^5y^2 - 8x^5y + 28x^5 + 24x^4 - 24x^3y + 12x^3 + 24x^2y^2$<br>$-24x^2y + 6x^2 - 16xy^2 - 12xy - 10x + 40y^3 - 40y^2 + 12y - 18$                                                                    |
| $\gamma$                                                       | $32x^5y^3 - 48x^5y^2 - 8x^5y - 20x^5 - 24y^3 + 52y^2 - 6y + 13$                                                                                                                                                                                                                             |
| $\{i, j\} = \{9, 34\}, k_{ij} = 2 \cdot 3 \cdot 7$             |                                                                                                                                                                                                                                                                                             |
| $\alpha$                                                       | $64x^5 - 4y + 102$                                                                                                                                                                                                                                                                          |
| $\beta$                                                        | $-64x^9 + 64x^8y - 32x^8 - 64x^7y^2 + 64x^7y - 16x^7 + 32x^6y^2 - 16x^6y$<br>$+72x^6 - 64x^5y^3 + 96x^5y^2 - 144x^5y + 76x^5 - 112x^4 + 112x^3y - 56x^3$<br>$-112x^2y^2 + 112x^2y - 28x^2 + 52xy^2 - 38xy + 123x - 108y^3 + 176y^2 - 254y + 130$                                            |
| $\gamma$                                                       | $64x^5y^3 - 128x^5y^2 + 64x^5y - 80x^5 + 112y^3 - 220y^2 + 122y - 137$                                                                                                                                                                                                                      |
| $\{i, j\} = \{9, 39\}, k_{ij} = 2 \cdot 3^2 \cdot 17^2$        |                                                                                                                                                                                                                                                                                             |
| $\alpha$                                                       | $-832x^5 - 676y - 6598$                                                                                                                                                                                                                                                                     |
| $\beta$                                                        | $832x^9 - 832x^8y + 416x^8 + 832x^7y^2 - 832x^7y + 208x^7 + 1248x^6y^2$<br>$-208x^6y + 520x^6 - 832x^5y^3 + 832x^5y^2 + 468x^5 + 6624x^4 - 6624x^3y$<br>$+3312x^3 + 6624x^2y^2 - 6624x^2y + 1656x^2 + 9260xy^2 - 2982xy + 1319x - 5948y^3$<br>$+7612y^2 + 1482y + 905$                      |
| $\gamma$                                                       | $-832x^5y^3 - 416x^5y - 416x^5 - 6624y^3 + 676y^2 - 1986y - 491$                                                                                                                                                                                                                            |
| $\{i, j\} = \{10, 17\}, k_{ij} = 5$                            |                                                                                                                                                                                                                                                                                             |
| $\alpha$                                                       | $-4$                                                                                                                                                                                                                                                                                        |
| $\beta$                                                        | $4x^4 - 4x^3y - 2x^3 + 4x^2y^2 + 4x^2y + x^2 - 4xy^2 + x + 8y^3 + 4y^2 + 4y + 1$                                                                                                                                                                                                            |
| $\gamma$                                                       | $-8y^3 - 4y^2 - 6y - 3$                                                                                                                                                                                                                                                                     |
| $\{i, j\} = \{10, 18\}, k_{ij} = 2 \cdot 5 \cdot 7^2 \cdot 13$ |                                                                                                                                                                                                                                                                                             |
| $\alpha$                                                       | $-928x^5 - 26912y + 10568$                                                                                                                                                                                                                                                                  |
| $\beta$                                                        | $928x^9 - 928x^8y - 464x^8 + 928x^7y^2 + 928x^7y + 232x^7 + 2784x^6y^2$<br>$+8352x^6y + 16472x^6 - 1856x^5y^3 - 5568x^5y^2 - 11136x^5y + 8352x^5$<br>$+1380x^4 - 1380x^3y - 690x^3 + 1380x^2y^2 + 1380x^2y + 345x^2 - 22772xy^2$<br>$-16000xy - 11407x + 24152y^3 + 20140y^2 - 842y - 5531$ |
| $\gamma$                                                       | $-1856x^5y^3 - 8352x^5y^2 - 18096x^5y - 33176x^5 - 2760y^3 + 41404y^2$<br>$+29930y + 22469$                                                                                                                                                                                                 |
| $\{i, j\} = \{10, 35\}, k_{ij} = 2^2 \cdot 11$                 |                                                                                                                                                                                                                                                                                             |
| $\alpha$                                                       | $64x^5 - 16y + 40$                                                                                                                                                                                                                                                                          |
| $\beta$                                                        | $-64x^9 + 64x^8y + 32x^8 - 64x^7y^2 - 64x^7y - 16x^7 + 96x^6y - 80x^6$<br>$-64x^5y^3 - 192x^5y^2 + 48x^5y - 48x^5 - 72x^4 + 72x^3y + 36x^3$<br>$-72x^2y^2 - 72x^2y - 18x^2 - 16xy^2 + 100xy - 50x - 56y^3 - 208y^2 + 26y - 34$                                                              |
| $\gamma$                                                       | $64x^5y^3 + 96x^5y^2 - 48x^5y + 88x^5 + 72y^3 + 124y^2 - 46y + 59$                                                                                                                                                                                                                          |
| $\{i, j\} = \{10, 37\}, k_{ij} = 2^2 \cdot 5$                  |                                                                                                                                                                                                                                                                                             |
| $\alpha$                                                       | $64x^5 + 16y - 40$                                                                                                                                                                                                                                                                          |
| $\beta$                                                        | $-64x^9 + 64x^8y + 32x^8 - 64x^7y^2 - 64x^7y - 16x^7 - 64x^6y^2 - 32x^6y$<br>$-64x^6 + 64x^5y^3 + 64x^5y^2 + 48x^5y + 32x^5 + 56x^4 - 56x^3y - 28x^3$<br>$+56x^2y^2 + 56x^2y + 14x^2 + 72xy^2 + 36xy + 40x - 72y^3 - 80y^2 - 38y - 20$                                                      |
| $\gamma$                                                       | $64x^5y^3 + 160x^5y^2 + 80x^5y + 72x^5 - 56y^3 - 156y^2 - 78y - 47$                                                                                                                                                                                                                         |
| $\{i, j\} = \{10, 39\}, k_{ij} = 2 \cdot 3 \cdot 11$           |                                                                                                                                                                                                                                                                                             |
| $\alpha$                                                       | $64x^5 + 4y + 98$                                                                                                                                                                                                                                                                           |
| $\beta$                                                        | $-64x^9 + 64x^8y + 32x^8 - 64x^7y^2 - 64x^7y - 16x^7 - 96x^6y^2 - 112x^6y$<br>$-88x^6 + 64x^5y^3 + 64x^5y^2 + 32x^5y - 4x^5 - 112x^4 + 112x^3y + 56x^3$<br>$-112x^2y^2 - 112x^2y - 28x^2 - 164xy^2 - 210xy - 157x + 108y^3 + 124y^2 + 66y - 7$                                              |
| $\gamma$                                                       | $64x^5y^3 + 192x^5y^2 + 160x^5y + 96x^5 + 112y^3 + 332y^2 + 294y + 171$                                                                                                                                                                                                                     |
| $\{i, j\} = \{11, 13\}, k_{ij} = 3$                            |                                                                                                                                                                                                                                                                                             |
| $\alpha$                                                       | $-1$                                                                                                                                                                                                                                                                                        |
| $\beta$                                                        | $2x^4 - 4x^3y - 4x^3 + 8x^2y^2 + 16x^2y + 8x^2 - 16xy^3 - 48xy^2 - 48xy - 16x$<br>$-32y^4 - 128y^3 - 190y^2 - 131y - 30$                                                                                                                                                                    |
| $\gamma$                                                       | $32y^4 + 128y^3 + 191y^2 + 130y + 31$                                                                                                                                                                                                                                                       |

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$$\begin{aligned}
&\{i, j\} = \{11, 29\}, k_{ij} = 3^5 \cdot 5^2 \cdot 7 \\
&\alpha \quad 5056x^5 - 24964y - 39139 \\
&\beta \quad -10112x^9 + 20224x^8y + 20224x^8 - 40448x^7y^2 - 80896x^7y - 40448x^7 \\
&\quad + 20224x^6y^2 + 37920x^6y + 8216x^6 - 40448x^5y^3 - 116288x^5y^2 \\
&\quad - 97328x^5y - 16432x^5 + 43518x^4 - 87036x^3y - 87036x^3 + 174072x^2y^2 \\
&\quad + 348144x^2y + 174072x^2 - 99518xy^2 - 131316xy - 70306x + 199036y^3 + 486632y^2 \\
&\quad + 442383y + 140612 \\
&\gamma \quad 40448x^5y^3 + 111232x^5y^2 + 102384x^5y + 36340x^5 - 174072y^3 - 472457y^2 \\
&\quad - 456558y - 138919 \\
&\{i, j\} = \{11, 34\}, k_{ij} = 2^2 \cdot 3^2 \cdot 7 \\
&\alpha \quad -16x^5 - 56y - 128 \\
&\beta \quad 32x^9 - 64x^8y - 64x^8 + 128x^7y^2 + 256x^7y + 128x^7 + 120x^6y + 152x^6 \\
&\quad + 128x^5y^3 + 144x^5y^2 - 84x^5y - 100x^5 + 32x^4 - 64x^3y - 64x^3 \\
&\quad + 128x^2y^2 + 256x^2y + 128x^2 - 28xy^2 + 148xy + 194x + 184y^3 + 186y^2 - 98y - 163 \\
&\gamma \quad -128x^5y^3 - 384x^5y^2 - 444x^5y - 204x^5 - 128y^3 - 370y^2 - 458y - 225 \\
&\{i, j\} = \{11, 36\}, k_{ij} = 3^4 \cdot 11^2 \\
&\alpha \quad 2624x^5 + 6724y + 3457 \\
&\beta \quad -5248x^9 + 10496x^8y + 10496x^8 - 20992x^7y^2 - 41984x^7y - 20992x^7 \\
&\quad + 31488x^6y^2 + 40672x^6y + 20664x^6 - 20992x^5y^3 - 49856x^5y^2 \\
&\quad - 40016x^5y - 20008x^5 + 14406x^4 - 28812x^3y - 28812x^3 + 57624x^2y^2 \\
&\quad + 115248x^2y + 57624x^2 - 83074xy^2 - 118924xy - 51286x + 50900y^3 + 134602y^2 \\
&\quad + 110143y + 38610 \\
&\gamma \quad 20992x^5y^3 + 47232x^5y^2 + 42640x^5y + 10660x^5 - 57624y^3 - 131335y^2 \\
&\quad - 113410y - 31981 \\
&\{i, j\} = \{11, 40\}, k_{ij} = 3 \cdot 7 \\
&\alpha \quad -512x^5 + 384y^2 - 1504y - 1744 \\
&\beta \quad -1280x^9 + 4096x^8y + 4096x^8 - 8192x^7y^2 - 16384x^7y - 8192x^7 + 6656x^6y \\
&\quad + 4608x^6 + 8192x^5y^3 + 12032x^5y^2 - 2304x^5y - 2896x^5 + 1584x^4 \\
&\quad + 448xy^2 - 896xy + 730x - 384y^4 + 1696y^3 - 152y^2 - 20y - 733 \\
&\gamma \quad 1152x^8 - 768x^7y - 192x^7 + 768x^6y - 672x^6 + 8192x^5y^3 + 23808x^5y^2 \\
&\quad + 21824x^5y + 5648x^5 - 192x^3y^2 + 1440x^3y + 768x^3 - 96x^2y^2 - 864x^2y \\
&\quad - 1200x^2 - 192xy^3 + 1488xy^2 - 384xy - 984x - 192y^3 - 296y^2 - 512y - 257 \\
&\{i, j\} = \{12, 18\}, k_{ij} = 2^4 \cdot 5 \cdot 17 \\
&\alpha \quad -178x^5 - 15842y - 1792 \\
&\beta \quad 178x^9 - 178x^8y + 178x^8 + 178x^7y^2 - 356x^7y + 178x^7 + 534x^6y^2 \\
&\quad + 534x^6y + 3827x^6 - 356x^5y^3 - 2136x^5y + 7476x^5 + 12828x^4 - 12828x^3y \\
&\quad + 12828x^3 + 12828x^2y^2 - 25656x^2y + 12828x^2 + 22642xy^2 - 21769xy + 8624x \\
&\quad - 9814y^3 + 36490y^2 - 12782y + 4420 \\
&\gamma \quad -356x^5y^3 - 2136x^5y - 7298x^5 - 25656y^3 + 31684y^2 - 33430y + 8408 \\
&\{i, j\} = \{12, 25\}, k_{ij} = 2^2 \\
&\alpha \quad 2x - 2 \\
&\beta \quad -2x^5 + 2x^4y - 2x^3y^2 + 2x^3y + 4x^2y^2 - 6x^2y + 3x^2 - 2xy^3 + 2xy^2 \\
&\quad - 2xy - 2x + 2y^3 - 4y^2 + 4y - 1 \\
&\gamma \quad 2xy^3 - 8xy^2 + 8xy - 3x - 2y^3 + 8y^2 - 8y + 5 \\
&\{i, j\} = \{12, 26\}, k_{ij} = 17 \\
&\alpha \quad 24x^5 - 18y + 2 \\
&\beta \quad -24x^9 + 24x^8y - 24x^8 - 24x^7y^2 + 48x^7y - 24x^7 + 24x^6y^2 - 36x^6y \\
&\quad + 48x^6 - 24x^5y^3 + 60x^5y^2 - 96x^5y + 48x^5 + 22x^4 - 22x^3y + 22x^3 \\
&\quad + 22x^2y^2 - 44x^2y + 22x^2 - 40xy^2 + 48xy - 41x + 40y^3 - 79y^2 + 88y - 41 \\
&\gamma \quad 24x^5y^3 - 96x^5y^2 + 108x^5y - 72x^5 - 22y^3 + 106y^2 - 114y + 63 \\
&\{i, j\} = \{12, 34\}, k_{ij} = 2^3 \cdot 3 \cdot 7 \\
&\alpha \quad -736x^5 + 1058y - 890 \\
&\beta \quad 736x^9 - 736x^8y + 736x^8 + 736x^7y^2 - 1472x^7y + 736x^7 - 368x^6y^2 \\
&\quad + 184x^6y - 1380x^6 + 736x^5y^3 - 1840x^5y^2 + 3128x^5y - 2438x^5 + 200x^4 \\
&\quad - 200x^3y + 200x^3 + 200x^2y^2 - 400x^2y + 200x^2 + 958xy^2 + 418xy + 522x \\
&\quad - 858y^3 + 190y^2 + 850y + 683 \\
&\gamma \quad -736x^5y^3 + 2576x^5y^2 - 2392x^5y + 2116x^5 - 200y^3 - 358y^2 - 1018y - 322 \\
&\{i, j\} = \{12, 39\}, k_{ij} = 2^4 \cdot 3^2 \\
&\alpha \quad -160x^5 + 50y - 2888 \\
&\beta \quad 160x^9 - 160x^8y + 160x^8 + 160x^7y^2 - 320x^7y + 160x^7 + 240x^6y^2
\end{aligned}$$


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|------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
|                                                                        | $-200x^6y + 100x^6 - 160x^5y^3 + 320x^5y^2 - 80x^5y + 70x^5 + 2848x^4$                |
|                                                                        | $-2848x^3y + 2848x^3 + 2848x^2y^2 - 5696x^2y + 2848x^2 + 4322xy^2 - 3450xy$           |
|                                                                        | $+2110x - 2898y^3 + 5636y^2 - 1569y + 1741$                                           |
| $\gamma$                                                               | $-160x^5y^3 + 240x^5y^2 - 280x^5y + 60x^5 - 2848y^3 + 4222y^2 - 5094y + 738$          |
| $\{i, j\} = \{13, 14\}, k_{ij} = 2^2 \cdot 3$                          |                                                                                       |
| $\alpha$                                                               | $-4$                                                                                  |
| $\beta$                                                                | $4x^4 - 4x^3y - 4x^3 + 4x^2y^2 + 8x^2y + 4x^2 - 4xy^3 - 12xy^2 - 12xy - 4x + 2y^4$    |
|                                                                        | $+6y^3 + 6y^2 - 3y - 7$                                                               |
| $\gamma$                                                               | $-2y^4 - 10y^3 - 18y^2 - 21y - 11$                                                    |
| $\{i, j\} = \{13, 38\}, k_{ij} = 3^2$                                  |                                                                                       |
| $\alpha$                                                               | $-8x^5 + 6y - 18$                                                                     |
| $\beta$                                                                | $8x^9 - 8x^8y - 8x^8 + 8x^7y^2 + 16x^7y + 8x^7 + 8x^6y^2 + 4x^6y$                     |
|                                                                        | $-8x^5y^3 - 12x^5y^2 + 18x^4 - 18x^3y - 18x^3 + 18x^2y^2 + 36x^2y + 18x^2$            |
|                                                                        | $+24xy^2 - 3x - 24y^3 - 27y^2 + 12y + 3$                                              |
| $\gamma$                                                               | $-8x^5y^3 - 32x^5y^2 - 28x^5y - 8x^5 - 18y^3 - 78y^2 - 54y - 15$                      |
| $\{i, j\} = \{13, 40\}, k_{ij} = 2^3 \cdot 3^4$                        |                                                                                       |
| $\alpha$                                                               | $-352x^5 - 242y - 816$                                                                |
| $\beta$                                                                | $352x^9 - 352x^8y - 352x^8 + 352x^7y^2 + 704x^7y + 352x^7 + 176x^6y^2$                |
|                                                                        | $-440x^6y - 44x^6 - 352x^5y^3 - 352x^5y^2 + 352x^5y - 110x^5 + 904x^4$                |
|                                                                        | $-904x^3y - 904x^3 + 904x^2y^2 + 1808x^2y + 904x^2 + 210xy^2 - 558xy - 410x$          |
|                                                                        | $-662y^3 - 992y^2 + 299y + 163$                                                       |
| $\gamma$                                                               | $-352x^5y^3 - 1232x^5y^2 - 616x^5y - 308x^5 - 904y^3 - 2922y^2 - 2154y - 494$         |
| $\{i, j\} = \{14, 20\}, k_{ij} = 2 \cdot 3^3 \cdot 5$                  |                                                                                       |
| $\alpha$                                                               | $41876x^5 + 13697280y^2 - 97091478y + 49299220$                                       |
| $\beta$                                                                | $494276x^9 + 320276x^8y - 320276x^8 + 320276x^7y^2 - 640552x^7y + 320276x^7$          |
|                                                                        | $-1281104x^6y^2 - 3523036x^6y - 23220010x^6 - 640552x^5y^3$                           |
|                                                                        | $-2018980x^5y^2 - 21565908x^5y + 60389078x^5 + 1658668x^4 + 55540276x^3y$             |
|                                                                        | $-55540276x^3 + 55540276x^2y^2 - 111080552x^2y + 55540276x^2 - 96773050xy^2$          |
|                                                                        | $+134899699xy + 46770211x - 13697280y^4 + 27102766y^3 + 35293438y^2 + 46016460y$      |
|                                                                        | $-24649745$                                                                           |
| $\gamma$                                                               | $904800x^8 - 348000x^7y + 2157600x^7 + 1113600x^6y + 4315200x^6$                      |
|                                                                        | $-640552x^5y^3 - 1336552x^5y^2 - 2425328x^5y - 37169068x^5$                           |
|                                                                        | $+6848640x^3y^2 - 44065152x^3y + 25538928x^3 + 13697280x^2y^2$                        |
|                                                                        | $+19632912x^2y - 56685360x^2 + 13697280xy^3 - 60735744xy^2 + 90343680xy$              |
|                                                                        | $-113370720x - 56291432y^3 + 57489892y^2 + 3874382y - 22120466$                       |
| $\{i, j\} = \{14, 23\}, k_{ij} = 2 \cdot 5^4$                          |                                                                                       |
| $\alpha$                                                               | $-3744x^5 + 104000y^2 - 217152y + 114402$                                             |
| $\beta$                                                                | $5616x^9 + 416x^8y - 416x^8 + 416x^7y^2 - 832x^7y + 416x^7 - 1248x^6y^2$              |
|                                                                        | $-1664x^6y - 11024x^6 - 832x^5y^3 - 17056x^5y^2 - 26000x^5y + 46592x^5$               |
|                                                                        | $-6087x^4 + 28958x^3y - 28958x^3 + 28958x^2y^2 - 57916x^2y + 28958x^2$                |
|                                                                        | $-43610xy^2 + 79064xy - 30056x - 104000y^4 + 263236y^3 - 278318y^2 + 181748y - 61974$ |
| $\gamma$                                                               | $18720x^8 - 10400x^7y + 29120x^7 + 27040x^6y - 8320x^6 - 832x^5y^3$                   |
|                                                                        | $-20800x^5y^2 + 100256x^5y - 87776x^5 + 52000x^3y^2 - 86944x^3y + 24014x^3$           |
|                                                                        | $+52000x^2y^2 - 16854x^2y - 46076x^2 + 104000xy^3 - 225888xy^2 + 205062xy$            |
|                                                                        | $-94104x + 150084y^3 - 277068y^2 + 130998y - 4148$                                    |
| $\{i, j\} = \{14, 25\}, k_{ij} = 2^2 \cdot 7$                          |                                                                                       |
| $\alpha$                                                               | $2x - 4$                                                                              |
| $\beta$                                                                | $2x^5 + 2x^4y - 6x^4 + 2x^3y^2 - 8x^3y + 6x^3 - 2x^2y^2 + 2x^2y - 7x^2$               |
|                                                                        | $+2xy^3 - 12xy^2 + 16xy + 11x - 4y^3 + 16y^2 - 8y - 12$                               |
| $\gamma$                                                               | $-2xy^3 + 8xy^2 - 12xy - x + 4y^3 - 16y^2 + 24y + 4$                                  |
| $\{i, j\} = \{14, 27\}, k_{ij} = 2 \cdot 3$                            |                                                                                       |
| $\alpha$                                                               | $-12x^5 + 54y - 14$                                                                   |
| $\beta$                                                                | $-12x^9 - 12x^8y + 12x^8 - 12x^7y^2 + 24x^7y - 12x^7 - 24x^6y^2 + 96x^6y$             |
|                                                                        | $-102x^6 - 12x^5y^3 + 60x^5y^2 - 72x^5y - 12x^5 + 4x^4 + 4x^3y - 4x^3$                |
|                                                                        | $+4x^2y^2 - 8x^2y + 4x^2 - 46xy^2 - 41xy + 70x - 50y^3 - 2y^2 + 51y + 4$              |
| $\gamma$                                                               | $12x^5y^3 - 60x^5y^2 + 132x^5y - 114x^5 - 4y^3 - 34y^2 - 53y + 74$                    |
| $\{i, j\} = \{14, 41\}, k_{ij} = 2^3 \cdot 3 \cdot 5 \cdot 7 \cdot 11$ |                                                                                       |
| $\alpha$                                                               | $2848x^5 - 15842y + 25082$                                                            |
| $\beta$                                                                | $2848x^9 + 2848x^8y - 2848x^8 + 2848x^7y^2 - 5696x^7y + 2848x^7$                      |
|                                                                        | $-1424x^6y^2 - 712x^6y - 9612x^6 - 2848x^5y^3 + 4272x^5y^2 - 12104x^5y$               |

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|------------------------------------------------------------|-------------------------------------------------------------------------------------|
|                                                            | $+6230x^5 + 10664x^4 + 10664x^3y - 10664x^3 + 10664x^2y^2 - 21328x^2y$              |
|                                                            | $+10664x^2 + 10510xy^2 + 30442xy + 7886x + 5178y^3 + 33262y^2 - 10790y + 1389$      |
| $\gamma$                                                   | $-2848x^5y^3 + 7120x^5y^2 - 9256x^5y - 6764x^5 - 10664y^3 + 42502y^2 - 1550y$       |
|                                                            | $+18550$                                                                            |
| $\{i, j\} = \{15, 17\}, k_{ij} = 1$                        |                                                                                     |
| $\alpha$                                                   | $-1$                                                                                |
| $\beta$                                                    | $2x^3 + xy^2 - 4xy - 7y^2 - y$                                                      |
| $\gamma$                                                   | $-xy^2 + 8y^2 + y$                                                                  |
| $\{i, j\} = \{15, 20\}, k_{ij} = 2^4$                      |                                                                                     |
| $\alpha$                                                   | $1120y^2 + 400y - 16$                                                               |
| $\beta$                                                    | $245x^6 - 490x^5y - 490x^5 - 280x^3y^2 + 180x^3y + 68x^3 - 72x^2y - 72x^2$          |
|                                                            | $+560xy^3 - 360xy^2 - 64xy - 100x - 560y^4 - 200y^3 + 152y^2 + 352y + 200$          |
| $\gamma$                                                   | $-245x^6 + 490x^5y - 36x^3 + 72x^2y + 100x + 560y^4 + 200y^3 + 136y^2 - 208y$       |
| $\{i, j\} = \{15, 21\}, k_{ij} = 11$                       |                                                                                     |
| $\alpha$                                                   | $-4$                                                                                |
| $\beta$                                                    | $-x^6 + 2x^3y + 9x^3 - 4xy^2 - 36xy + 7x - 8y^3 - 36y^2 - 4y$                       |
| $\gamma$                                                   | $x^6 - x^5 + 18xy - 7x + 8y^3 + 36y^2 - 14y + 7$                                    |
| $\{i, j\} = \{15, 26\}, k_{ij} = 11$                       |                                                                                     |
| $\alpha$                                                   | $24x + 20$                                                                          |
| $\beta$                                                    | $-48x^4 - 40x^3 - 96x^2y + 72x^2 - 48xy^2 - 56xy + 48x - 40y^2 + 20y + 26$          |
| $\gamma$                                                   | $96x^2y - 36x^2 + 24xy^2 + 68xy - 24x + 20y^2 - 10y - 31$                           |
| $\{i, j\} = \{16, 21\}, k_{ij} = 3 \cdot 5 \cdot 13$       |                                                                                     |
| $\alpha$                                                   | $256x - 96$                                                                         |
| $\beta$                                                    | $-512x^4 + 704x^3 - 512x^2y + 128xy^2 + 1184xy + 448x - 48y^2 - 372y - 99$          |
| $\gamma$                                                   | $-512x^2y - 704x^2 - 128xy^2 + 1248xy + 256x + 48y^2 - 396y + 99$                   |
| $\{i, j\} = \{16, 27\}, k_{ij} = 2 \cdot 3^2$              |                                                                                     |
| $\alpha$                                                   | $12x^4 - 18$                                                                        |
| $\beta$                                                    | $-24x^7 + 24x^6 - 16x^5y + 28x^5 + 8x^4y^2 + 32x^4y + 8x^4 + 12x^3 + 8xy - 14x$     |
|                                                            | $-12y^2 - 8y$                                                                       |
| $\gamma$                                                   | $-16x^5y - 26x^5 - 4x^4y^2 + 32x^4y + 8x^4 + 8xy + 7x + 6y^2 - 8y$                  |
| $\{i, j\} = \{16, 29\}, k_{ij} = 2 \cdot 13^3$             |                                                                                     |
| $\alpha$                                                   | $413712x^5 - 1551216y^2 - 1498856y - 332378$                                        |
| $\beta$                                                    | $-298792x^9 - 528632x^8 - 896376x^7y + 229840x^7 - 160888x^6y - 447440x^6$          |
|                                                            | $-321776x^5y^2 - 1526192x^5y - 217600x^5 - 93178x^4 - 136136x^3y^2$                 |
|                                                            | $-174488x^3y + 193506x^3 + 136136x^2y^2 + 868754x^2y + 362516x^2 - 272272xy^3$      |
|                                                            | $-485112xy^2 - 157142xy + 150784x - 1034144y^4 - 473008y^3 + 1183072y^2 + 2023246y$ |
|                                                            | $+826400$                                                                           |
| $\gamma$                                                   | $149396x^9 + 108800x^6 - 735488x^5y^2 + 1073380x^5y + 332520x^5 + 46589x^4$         |
|                                                            | $-244800x^3y^2 - 37400x^3y + 156550x^3 - 486900x^2y - 324600x^2 + 244800xy^3$       |
|                                                            | $+282200xy^2 - 281450xy - 413200x + 517072y^4 + 508776y^3 + 1006674y^2 + 240964y$   |
|                                                            | $-163992$                                                                           |
| $\{i, j\} = \{16, 39\}, k_{ij} = 2^2 \cdot 5$              |                                                                                     |
| $\alpha$                                                   | $-72x^5 - 144y^2 - 16y - 12$                                                        |
| $\beta$                                                    | $162x^9 + 63x^8 + 162x^7y + 18x^7 - 36x^6y - 18x^6 - 468x^5y^2 - 18x^5$             |
|                                                            | $-72x^4 - 36x^3y^2 - 4x^3y - 30x^3 + 36x^2y^2 - 68x^2y - 6x^2 - 72xy^3$             |
|                                                            | $-44xy^2 + 8xy - 8x - 288y^4 - 32y^3 + 156y^2 + 40y - 15$                           |
| $\gamma$                                                   | $-81x^9 + 396x^5y^2 + 36x^5y + 18x^5 + 36x^4 + 7x + 144y^4 + 88y^3 - 124y^2 - 30y$  |
|                                                            | $+8$                                                                                |
| $\{i, j\} = \{17, 25\}, k_{ij} = 2^9$                      |                                                                                     |
| $\alpha$                                                   | $144x^5 + 48y^2 - 64y - 512$                                                        |
| $\beta$                                                    | $-216x^9 - 72x^8 + 216x^7y + 72x^7 - 72x^6y - 72x^6 + 144x^5y^2 + 962x^4$           |
|                                                            | $-24x^3y^2 + 56x^3y + 206x^3 + 24x^2y^2 - 1018x^2y - 206x^2 + 48xy^3$               |
|                                                            | $-136xy^2 + 606xy + 64x - 96y^4 + 224y^3 - 1148y^2 + 242y$                          |
| $\gamma$                                                   | $108x^9 - 288x^5y^2 - 36x^5y + 36x^5 - 481x^4 + 71x + 48y^4 - 160y^3 + 1660y^2$     |
|                                                            | $-218y - 32$                                                                        |
| $\{i, j\} = \{17, 29\}, k_{ij} = 2^2 \cdot 7$              |                                                                                     |
| $\alpha$                                                   | $4$                                                                                 |
| $\beta$                                                    | $-6x^4 - 2x^3 + 6x^2y - 4x^2 - 2xy - 14x + 4y^2 + 6y - 18$                          |
| $\gamma$                                                   | $3x^4 + 9x - 8y^2 + 2y + 16$                                                        |
| $\{i, j\} = \{17, 30\}, k_{ij} = 2^{11} \cdot 3^3 \cdot 7$ |                                                                                     |

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|-----------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $\alpha$                                                        | $6000x^5 - 5600y^2 + 1440y - 361408$                                                                                                                                                                                                                                                                                                                                                                            |
| $\beta$                                                         | $-7500x^9 - 750x^8 + 7500x^7y + 4500x^7 - 6000x^6y - 4500x^6 + 3000x^5y^2$<br>$+1000x^5y + 1500x^5 + 432590x^4 + 4200x^3y^2 - 10880x^3y + 81431x^3$<br>$-4200x^2y^2 - 421710x^2y - 297726x^2 - 8400xy^3 + 25960xy^2 + 258848xy + 76992x$<br>$+11200y^4 - 25280y^3 - 73604y^2 + 103328y + 84703$                                                                                                                 |
| $\gamma$                                                        | $3750x^9 - 9000x^5y^2 - 2000x^5y + 1500x^5 - 216295x^4 + 110367x - 5600y^4$<br>$+21040y^3 + 445532y^2 + 116638y - 25664$                                                                                                                                                                                                                                                                                        |
| $\{i, j\} = \{17, 32\}, k_{ij} = 2 \cdot 5 \cdot 7$             |                                                                                                                                                                                                                                                                                                                                                                                                                 |
| $\alpha$                                                        | $-848x^5 + 3032y^2 - 16164y - 70$                                                                                                                                                                                                                                                                                                                                                                               |
| $\beta$                                                         | $848x^9 - 2304x^7y + 2304x^6y - 1456x^5y^2 + 100280x^4 - 2304x^3y^2$<br>$+12672x^3y - 67680x^3 + 2304x^2y^2 + 381360x^2y + 67680x^2 + 4608xy^3$<br>$-27648xy^2 - 246000xy - 95246x - 6064y^4 + 35720y^3 + 283124y^2 + 99872y - 5388$                                                                                                                                                                            |
| $\gamma$                                                        | $-424x^9 + 728x^7y + 728x^6y + 2304x^5y^2 - 424x^5y - 50140x^4 - 1456x^3y^2$<br>$+9828x^3y - 66794x^3 - 1456x^2y^2 - 237328x^2y - 66794x^2 + 1456xy^3$<br>$-11284xy^2 - 170534xy - 53011x + 3032y^4 - 19556y^3 - 283054y^2 - 98356y - 2694$                                                                                                                                                                     |
| $\{i, j\} = \{17, 40\}, k_{ij} = 2^7 \cdot 7 \cdot 13^2$        |                                                                                                                                                                                                                                                                                                                                                                                                                 |
| $\alpha$                                                        | $571536x^5 - 163296y^2 + 316224y - 219024$                                                                                                                                                                                                                                                                                                                                                                      |
| $\beta$                                                         | $-333396x^9 - 642978x^8 + 1000188x^7y + 476280x^7 + 285768x^6y - 476280x^6$<br>$-1047816x^5y^2 - 158760x^5y + 158760x^5 + 260730x^4 + 40824x^3y^2$<br>$-92664x^3y + 459729x^3 - 40824x^2y^2 - 689526x^2y - 329364x^2 - 81648xy^3$<br>$+226152xy^2 - 229932xy + 266082x + 108864y^4 - 138240y^3 + 480060y^2 + 129648y - 99241$                                                                                   |
| $\gamma$                                                        | $166698x^9 - 476280x^5y^2 - 539784x^5y + 158760x^5 - 130365x^4 + 31641x$<br>$-54432y^4 + 150768y^3 + 336420y^2 + 379506y - 67600$                                                                                                                                                                                                                                                                               |
| $\{i, j\} = \{17, 41\}, k_{ij} = 2^2 \cdot 3 \cdot 5 \cdot 7^2$ |                                                                                                                                                                                                                                                                                                                                                                                                                 |
| $\alpha$                                                        | $-439397496x^5 - 576709584y^2 + 1099509784y - 486695128$                                                                                                                                                                                                                                                                                                                                                        |
| $\beta$                                                         | $-163721822x^9 + 113103903x^8 - 128994534x^7y - 70105725x^7 - 97213272x^6y$<br>$+70105725x^6 + 585648020x^5y^2 - 259212630x^5y + 238065725x^5 + 220054072x^4$<br>$-54784068x^3y^2 + 124926900x^3y + 147401094x^3 + 54784068x^2y^2$<br>$-248006076x^2y - 106374702x^2 + 109568136xy^3 - 304637868xy^2 - 46796112xy$<br>$+22506570x + 384473056y^4 - 815018528y^3 + 867081900y^2 - 469645356y + 15057862$         |
| $\gamma$                                                        | $81860911x^9 + 423776000x^8 + 310080000x^7y - 315248000x^7 + 268736000x^6y$<br>$-266152000x^6 + 146250524x^5y^2 - 511377324x^5y + 294238275x^5$<br>$-110027036x^4 + 205686400x^3y^2 - 594423360x^3y - 155955960x^3$<br>$-102843200x^2y^2 + 28670160x^2y + 167491820x^2 + 205686400xy^3$<br>$-748688160xy^2 + 424132320xy + 114144116x - 192236528y^4 + 92254728y^3$<br>$+472641500y^2 + 291350936y - 243346094$ |
| $\{i, j\} = \{18, 19\}, k_{ij} = 2^5 \cdot 3 \cdot 7$           |                                                                                                                                                                                                                                                                                                                                                                                                                 |
| $\alpha$                                                        | $-128y - 832$                                                                                                                                                                                                                                                                                                                                                                                                   |
| $\beta$                                                         | $2x^9 - 4x^8 + 4x^7 - 8x^6y - 8x^5y^2 + 12x^5y - 2x^5 + 749x^4 + 32x^3y^2$<br>$+232x^3y + 170x^3 - 32x^2y^2 - 232x^2y - 170x^2 + 64xy^3 + 464xy^2 + 340xy$<br>$+165x + 64y^4 + 368y^3 - 276y^2 + 22y - 80$                                                                                                                                                                                                      |
| $\gamma$                                                        | $-2x^9 + 8x^5y^2 + 4x^5y - 2x^5 - 749x^4 - 165x - 64y^4 - 496y^3 - 620y^2 - 470y$<br>$-80$                                                                                                                                                                                                                                                                                                                      |
| $\{i, j\} = \{18, 21\}, k_{ij} = 2^2 \cdot 7$                   |                                                                                                                                                                                                                                                                                                                                                                                                                 |
| $\alpha$                                                        | $8x^4 - 4$                                                                                                                                                                                                                                                                                                                                                                                                      |
| $\beta$                                                         | $-16x^7 + 16x^6 - 16x^5y + 10x^5 + 4x^4y^2 + 32x^4y + 8x^4 + 24x^3 + 24xy - 13x$<br>$-2y^2 - 24y$                                                                                                                                                                                                                                                                                                               |
| $\gamma$                                                        | $-16x^5y - 10x^5 - 4x^4y^2 + 32x^4y - 24x^4 + 24xy - 11x + 2y^2 - 24y + 24$                                                                                                                                                                                                                                                                                                                                     |
| $\{i, j\} = \{18, 26\}, k_{ij} = 3 \cdot 5^3 \cdot 7 \cdot 17$  |                                                                                                                                                                                                                                                                                                                                                                                                                 |
| $\alpha$                                                        | $-737568x^5 + 5654688y^2 - 5140912y + 1097284$                                                                                                                                                                                                                                                                                                                                                                  |
| $\beta$                                                         | $532688x^9 + 942448x^8 + 1598064x^7y - 1741480x^7 + 286832x^6y + 799032x^6$<br>$+573664x^5y^2 + 2048800x^5y - 799032x^5 - 81094x^4 + 942448x^3y^2$<br>$+1185152x^3y - 1181270x^3 - 942448x^2y^2 - 1428434x^2y + 1302911x^2$<br>$+1884896xy^3 + 2370304xy^2 - 2119258xy - 332391x + 3769792y^4 - 2798976y^3$<br>$-1467908y^2 + 146442y + 437766$                                                                 |
| $\gamma$                                                        | $-266344x^9 + 1311232x^5y^2 - 3708328x^5y + 870740x^5 + 40547x^4 + 105375x$<br>$-1884896y^4 - 485408y^3 - 1408408y^2 + 3003536y - 704143$                                                                                                                                                                                                                                                                       |
| $\{i, j\} = \{18, 37\}, k_{ij} = 2^3 \cdot 5$                   |                                                                                                                                                                                                                                                                                                                                                                                                                 |
| $\alpha$                                                        | $-2800x^5 + 26152y^2 - 64442y + 20516$                                                                                                                                                                                                                                                                                                                                                                          |
| $\beta$                                                         | $7000x^9 - 1400x^8 + 7000x^7y - 5600x^7 - 9800x^6y + 51128x^6 - 19600x^5y^2$<br>$+70728x^5y - 51128x^5 + 21038x^4 + 7560x^3y^2 + 11144x^3y - 10196x^3$                                                                                                                                                                                                                                                          |

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$$\begin{aligned}
& -7560x^2y^2 - 7048x^2y - 10842x^2 + 15120xy^3 + 22288xy^2 - 7546xy - 20556x \\
& + 52304y^4 - 46340y^3 + 6574y^2 - 12488y + 20556 \\
\gamma & -3500x^9 - 22064x^6 + 16800x^5y^2 - 39564x^5y + 2800x^5 - 10519x^4 \\
& + 11032x^3y^2 - 24822x^3y - 1773x^3 + 8471x^2y + 11032xy^3 - 24822xy^2 \\
& - 10244xy + 20797x - 26152y^4 + 8050y^3 - 9228y^2 + 2699y + 5421 \\
\{i, j\} &= \{19, 24\}, k_{ij} = 2^2 \cdot 5^2 \cdot 11 \\
\alpha & 80 \\
\beta & 2x^9 - 2x^8 - 6x^7 - 4x^6y - 4x^6 - 8x^5y^2 - 36x^5y + 8x^5 - 81x^4 \\
& - 16x^3y^2 + 60x^3y - 93x^3 - 16x^2y^2 + 60x^2y + 69x^2 - 32xy^3 + 120xy^2 \\
& - 186xy - 13x - 64y^4 + 144y^3 + 20y^2 + 214y + 724 \\
\gamma & -2x^9 + 8x^5y^2 + 20x^5y - 6x^5 + 81x^4 + 175x + 64y^4 - 208y^3 + 236y^2 - 322y \\
& - 456 \\
\{i, j\} &= \{19, 33\}, k_{ij} = 2 \cdot 3 \cdot 5 \cdot 17^2 \\
\alpha & 226090368x^5 + 62114304y^2 - 71757336y + 20338532 \\
\beta & 127220352x^9 - 204198400x^8 + 382872000x^7y - 395634400x^7 - 25524800x^6y \\
& - 191436000x^6 + 406378560x^5y^2 - 297013696x^5y - 1893584x^5 - 188044808x^4 \\
& - 10209920x^3y^2 + 57941296x^3y + 110408176x^3 - 10209920x^2y^2 \\
& - 201048908x^2y + 239903278x^2 - 20419840xy^3 + 115882592xy^2 - 38173852xy \\
& - 103356998x + 41409536y^4 - 68732944y^3 - 136549860y^2 + 29564372y + 32579515 \\
\gamma & -63610176x^9 + 3229184x^8 - 605472x^7y - 2926448x^7 - 2623712x^6y + 4541040x^6 \\
& + 180288192x^5y^2 - 446078528x^5y + 97041728x^5 + 94022404x^4 \\
& - 20989696x^3y^2 - 130681040x^3y - 101074588x^3 + 20989696x^2y^2 \\
& - 21891070x^2y + 177360643x^2 + 20989696xy^3 + 99196496xy^2 + 57625138xy \\
& - 111471887x - 20704768y^4 - 28032760y^3 - 136538300y^2 + 60627304y + 12232313 \\
\{i, j\} &= \{19, 37\}, k_{ij} = 2^2 \\
\alpha & 212x^5 + 1376y^2 - 3064y + 942 \\
\beta & -216x^9 + 72x^8 - 216x^7y + 288x^7 - 72x^6y - 140x^6 + 856x^5y^2 + 64x^5y \\
& - 140x^5 - 3682x^4 + 24x^3y^2 - 200x^3y + 838x^3 + 24x^2y^2 - 3882x^2y \\
& + 4520x^2 + 48xy^3 - 400xy^2 - 2006xy + 946x + 2752y^4 - 3472y^3 + 6600y^2 - 5692y \\
& + 946 \\
\gamma & 108x^9 - 356x^8 - 356x^6y + 178x^6 - 644x^5y^2 + 328x^5y - 144x^5 + 1841x^4 \\
& - 1424x^3y^2 + 2848x^3y - 3026x^3 - 1424xy^3 + 2848xy^2 - 3026xy + 1368x \\
& - 1376y^4 + 1784y^3 - 7394y^2 + 8304y - 2260 \\
\{i, j\} &= \{19, 38\}, k_{ij} = 3^2 \cdot 7 \\
\alpha & -32x^5 + 32y^2 + 16y - 100 \\
\beta & -48x^9 + 16x^8 - 48x^7y + 40x^7 - 16x^6y + 24x^6 + 32x^5y^2 - 96x^5y + 24x^5 \\
& - 162x^4 - 16x^3y^2 - 32x^3y + 2x^3 - 16x^2y^2 - 194x^2y + 83x^2 - 32xy^3 \\
& - 64xy^2 - 158xy - 21x + 64y^4 + 128y^3 + 268y^2 - 318y + 30 \\
\gamma & 24x^9 - 64x^5y^2 + 104x^5y - 20x^5 + 81x^4 + 51x - 32y^4 - 96y^3 - 352y^2 + 260y \\
& - 67 \\
\{i, j\} &= \{19, 39\}, k_{ij} = 2 \cdot 3^2 \cdot 5 \cdot 11 \\
\alpha & -1961328x^5 - 2983520y^2 - 8986288y + 5237704 \\
\beta & -1900428x^9 - 30450x^8 + 304500x^7y - 182700x^7 + 243600x^6y - 152250x^6 \\
& - 2326728x^5y^2 + 385700x^5y - 321030x^5 + 17017490x^4 + 170520x^3y^2 \\
& + 751912x^3y + 1272817x^3 + 170520x^2y^2 + 16566354x^2y - 6634404x^2 \\
& + 341040xy^3 + 1503824xy^2 + 18360076xy - 2683042x - 5967040y^4 - 13369696y^3 \\
& - 21959804y^2 + 20947924y - 696345 \\
\gamma & 950214x^9 + 551232x^8 - 1102464x^7y + 275616x^7 + 1102464x^6y - 413424x^6 \\
& + 365400x^5y^2 - 130384x^5y + 244644x^5 - 8508745x^4 + 4134240x^3y^2 \\
& + 8406288x^3y + 5946534x^3 - 2067120x^2y^2 - 3601620x^2y - 3274029x^2 \\
& + 4134240xy^3 + 7372728xy^2 + 3544200xy - 3948342x + 2983520y^4 + 2891648y^3 \\
& + 21958484y^2 - 19456824y + 5935039 \\
\{i, j\} &= \{20, 21\}, k_{ij} = 2 \cdot 3 \cdot 7 \cdot 11 \\
\alpha & 1818915904x^5 + 1589308032y^2 + 1239067072y + 316380984 \\
\beta & 826032454x^9 + 596285404x^8 - 348391472x^7y + 1105472940x^7 + 495787864x^6y \\
& + 2210945880x^6 - 3091454808x^5y^2 - 3405845508x^5y + 584247196x^5 \\
& + 1528551025x^4 + 1954290624x^3y^2 - 7874567008x^3y + 5699978416x^3 \\
& + 3908581248x^2y^2 - 3862648064x^2y + 14371578320x^2 + 3908581248xy^3 \\
& - 7931971520xy^2 + 3674660704xy + 13234468137x + 794654016y^4 + 3137470256y^3 \\
& + 2721946252y^2 + 17708640662y + 8634070206 \\
\gamma & -826032454x^9 - 1756019850x^8 + 3652521288x^7y + 2669150172x^7 - 140481588x^6y
\end{aligned}$$


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$$\begin{aligned}
& -913130322x^6 - 4910370712x^5y^2 - 2241972928x^5y + 2081624714x^5 \\
& -1528551025x^4 - 2550281136x^3y^2 + 4643650756x^3y - 2009796119x^3 \\
& + 2550281136x^2y^2 - 10415932608x^2y + 566725656x^2 + 5100562272xy^3 \\
& - 9287301512xy^2 + 9791874090xy + 16951758966x - 794654016y^4 + 2295730192y^3 \\
& + 1257700172y^2 + 17874444592y + 316381446 \\
\{i, j\} = \{20, 37\}, k_{ij} = 2 \cdot 5 \cdot 11 \\
\alpha & 56y^2 - 196y - 110 \\
\beta & 224x^6 - 112x^5y + 24x^3 - 12x^2y + 98x + 112y^4 - 392y^3 - 208y^2 - 104y \\
\gamma & -112x^6 + 56x^5y + 224x^5 - 112x^3y^2 + 420x^3y + 98x^3 + 6x^2y + 24x^2 \\
& - 112xy^3 + 420xy^2 + 110xy - 49x - 56y^4 + 196y^3 + 98y^2 + 76y + 98 \\
\{i, j\} = \{21, 25\}, k_{ij} = 2^4 \cdot 5 \cdot 17 \\
\alpha & 48x^5 + 48y^2 - 160y - 1352 \\
\beta & -72x^9 - 24x^8 + 72x^7y + 96x^7 - 24x^6y - 72x^6 + 48x^5y^2 + 120x^5y \\
& + 72x^5 + 2394x^4 - 24x^3y^2 + 152x^3y + 286x^3 + 24x^2y^2 - 2546x^2y \\
& - 2680x^2 + 48xy^3 - 304xy^2 + 1822xy - 8x - 96y^4 + 512y^3 - 3180y^2 - 2856y + 8 \\
\gamma & 36x^9 - 96x^5y^2 - 180x^5y - 48x^5 - 1197x^4 + 1201x + 48y^4 - 304y^3 \\
& + 4276y^2 + 4470y + 1340 \\
\{i, j\} = \{21, 26\}, k_{ij} = 7 \cdot 11 \cdot 13 \cdot 17 \\
\alpha & -1152x^5 + 5184y^2 + 6192y - 5540 \\
\beta & 2304x^6 + 2304x^5y^2 - 3456x^5y - 2880x^5 - 3752x - 10368y^4 + 3168y^3 \\
& + 6328y^2 - 4308y + 8902 \\
\gamma & 1152x^8 - 1152x^6y - 576x^6 - 1152x^5y^2 + 1728x^5y + 288x^5 - 5184x^3y^2 \\
& - 1008x^3y + 6548x^3 + 5184xy^3 - 1584xy^2 - 7052xy + 5150x + 5184y^4 - 1584y^3 \\
& - 3164y^2 + 2154y - 2575 \\
\{i, j\} = \{21, 27\}, k_{ij} = 2 \cdot 3^3 \\
\alpha & 2x^4 - 16 \\
\beta & -4x^7 + 4x^6 - 8x^5y + 22x^5 - 4x^4y^2 + 20x^4y - 16x^4 + 14x^3 + 4x^2 + 28xy \\
& - 14x + 32y^2 - 52y + 38 \\
\gamma & 8x^5y - 11x^5 + 2x^4y^2 - 18x^4y + 6x^4 - 28xy - 2x - 16y^2 + 36y - 12 \\
\{i, j\} = \{21, 29\}, k_{ij} = 2^3 \cdot 13 \\
\alpha & 150x^5 - 90y^2 - 264y - 232 \\
\beta & -750x^6 - 300x^5y^2 - 300x^5 + 128x + 180y^4 + 528y^3 + 464y^2 - 90y - 136 \\
\gamma & -150x^8 + 150x^6y + 525x^6 + 150x^5y^2 + 525x^5 + 90x^3y^2 + 39x^3y \\
& - 68x^3 - 90xy^3 - 129xy^2 + 29xy + 4x - 90y^4 - 264y^3 - 232y^2 + 45y + 4 \\
\{i, j\} = \{21, 31\}, k_{ij} = 2 \cdot 5 \cdot 7 \cdot 17 \\
\alpha & -430464x^5 + 944832y^2 - 812592y - 643456 \\
\beta & 439584x^9 + 4560x^8 - 45600x^7y - 27360x^7 + 36480x^6y + 22800x^6 \\
& + 375744x^5y^2 + 516192x^5y + 27360x^5 - 3495928x^4 + 54720x^3y^2 \\
& - 280896x^3y + 843464x^3 - 54720x^2y^2 + 4368016x^2y + 1200096x^2 \\
& - 109440xy^3 + 561792xy^2 - 5774048xy - 1301927x - 1889664y^4 - 702240y^3 \\
& + 9068320y^2 + 7332694y + 880515 \\
\gamma & -219792x^9 + 98496x^8 - 196992x^7y - 49248x^7 - 196992x^6y - 73872x^6 \\
& + 54720x^5y^2 - 176928x^5y - 23712x^5 + 1747964x^4 - 1526688x^3y^2 \\
& + 779760x^3y + 2626614x^3 - 763344x^2y^2 + 94284x^2y + 1165509x^2 \\
& + 1526688xy^3 - 398088xy^2 - 2969352xy - 1101369x + 944832y^4 + 1987248y^3 \\
& - 9067368y^2 - 6860754y - 1522781 \\
\{i, j\} = \{22, 23\}, k_{ij} = 2^6 \cdot 11 \\
\alpha & 128y + 576 \\
\beta & -2x^9 + 4x^8 - 4x^7 + 8x^6y - 4x^6 + 8x^5y^2 - 20x^5y + 10x^5 - 425x^4 \\
& - 32x^3y^2 - 200x^3y - 326x^3 + 32x^2y^2 + 200x^2y + 326x^2 - 64xy^3 \\
& - 368xy^2 - 452xy - 483x - 64y^4 - 240y^3 + 204y^2 + 842y - 6 \\
\gamma & 2x^9 - 8x^5y^2 + 4x^5y + 2x^5 + 425x^4 + 809x + 64y^4 + 368y^3 + 500y^2 - 138y \\
& + 646 \\
\{i, j\} = \{22, 30\}, k_{ij} = 2^4 \cdot 7^4 \\
\alpha & -26980406640x^5 + 78790203360y^2 - 125321430960y + 56284952112 \\
\beta & 24369259860x^9 + 9139013730x^8 + 23500321020x^7y - 28722614580x^7 \\
& - 5222293560x^6y + 10444587120x^6 - 18855876600x^5y^2 + 44385165000x^5y \\
& - 11694588840x^5 - 25734565746x^4 + 9477495720x^3y^2 + 19377646200x^3y \\
& - 8264468961x^3 - 9477495720x^2y^2 - 58972831974x^2y + 41260457106x^2 \\
& + 18954991440xy^3 + 29277796680xy^2 + 3688601652xy - 28218150987x + 52526802240y^4
\end{aligned}$$


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$$\begin{aligned}
& -66038686560y^3 - 42607823580y^2 + 31040602122y + 10094944534 \\
\gamma \quad & -12184629930x^9 + 4133954880x^8 + 24803729280x^7y - 18602796960x^7 \\
& + 8267909760x^6y - 7234421040x^6 + 8124530040x^5y^2 - 36665754840x^5y \\
& + 16429006440x^5 + 12867282873x^4 + 20440110240x^3y^2 + 24790970160x^3y \\
& - 26432904582x^3 + 10220055120x^2y^2 - 6408770652x^2y + 2453760819x^2 \\
& - 20440110240xy^3 - 9460887480xy^2 + 35624004336xy - 10246255980x - 26263401120y^4 \\
& - 6375758400y^3 - 36041968332y^2 + 95022543216y - 36095101460 \\
\{i, j\} = \{23, 37\}, k_{ij} = 2^3 \cdot 5 \cdot 11 \\
\alpha \quad & 71910x^5 + 451554y^2 - 1466794y + 943440 \\
\beta \quad & -11016x^9 + 3672x^8 - 11016x^7y + 25704x^7 - 3672x^6y - 219402x^6 \\
& + 165852x^5y^2 + 265608x^5y - 438804x^5 - 1001622x^4 - 1224x^3y^2 \\
& - 42024x^3y + 138714x^3 - 1224x^2y^2 - 1043646x^2y + 2141958x^2 - 2448xy^3 \\
& - 82824xy^2 - 682170xy + 235970x + 903108y^4 - 1122476y^3 + 1809596y^2 - 2521400y \\
& + 471940 \\
\gamma \quad & 5508x^9 - 79254x^8 - 79254x^6y + 118881x^6 - 93942x^5y^2 - 97920x^5y \\
& + 83997x^5 + 500811x^4 - 449106x^3y^2 + 867391x^3y - 1038849x^3 - 449106xy^3 \\
& + 867391xy^2 - 1038849xy + 814280x - 451554y^4 + 558790y^3 - 1988632y^2 + 4105219y \\
& - 2259943 \\
\{i, j\} = \{23, 41\}, k_{ij} = 2 \cdot 5 \cdot 11 \\
\alpha \quad & 555264x^5 + 2119632y^2 - 4675380y + 2555858 \\
\beta \quad & 563184x^9 - 3960x^8 + 39600x^7y - 43560x^7 + 31680x^6y - 35640x^6 \\
& + 507744x^5y^2 - 457584x^5y + 329604x^5 + 12156448x^4 + 79200x^3y^2 \\
& + 474144x^3y + 2067956x^3 + 79200x^2y^2 + 6553624x^2y - 4011524x^2 \\
& + 158400xy^3 + 869088xy^2 + 9741248xy - 8857910x + 4239264y^4 - 12103992y^3 \\
& - 3385426y^2 + 21635369y - 10494024 \\
\gamma \quad & -281592x^9 - 130896x^8 + 261792x^7y - 196344x^7 - 261792x^6y + 229068x^6 \\
& + 47520x^5y^2 - 165768x^5y - 57252x^5 - 6078224x^4 - 3337848x^3y^2 \\
& + 5334012x^3y + 5532780x^3 + 1668924x^2y^2 + 371478x^2y - 5804874x^2 \\
& - 3337848xy^3 + 7837398xy^2 + 13029xy - 2275106x - 2119632y^4 + 9548244y^3 \\
& + 3385536y^2 - 22695075y + 11772008 \\
\{i, j\} = \{24, 31\}, k_{ij} = 2^2 \cdot 3 \cdot 5 \\
\alpha \quad & 851328x^5 - 3024432y^2 - 2834028y - 660954 \\
\beta \quad & -336696x^9 + 132300x^8 - 74088x^7y + 227556x^7 + 116424x^6y + 587412x^6 \\
& + 830160x^5y^2 - 653904x^5y - 1073568x^5 + 5606992x^4 + 580608x^3y^2 \\
& - 3460320x^3y - 2852568x^3 + 1161216x^2y^2 + 4128336x^2y - 180648x^2 \\
& + 1161216xy^3 - 4017600xy^2 - 908784xy + 1206543x - 2016288y^4 + 433080y^3 \\
& - 1784886y^2 - 3415266y - 1120930 \\
\gamma \quad & 168348x^9 - 1170000x^8 - 468000x^7y - 819000x^7 + 936000x^6y + 175500x^6 \\
& - 21168x^5y^2 + 2154888x^5y + 1735224x^5 - 2803496x^4 + 351000x^3y^2 \\
& + 351000x^3y + 5313750x^3 + 175500x^2y^2 + 3061500x^2y + 4099875x^2 \\
& - 351000xy^3 - 438750xy^2 - 3958500xy - 2817141x + 1008144y^4 + 1754892y^3 \\
& - 1784838y^2 - 4927506y - 1781824 \\
\{i, j\} = \{25, 28\}, k_{ij} = 2 \\
\alpha \quad & 0 \\
\beta \quad & 1 \\
\gamma \quad & -1 \\
\{i, j\} = \{25, 39\}, k_{ij} = 2^6 \cdot 3^2 \cdot 5 \cdot 17 \\
\alpha \quad & 95360y^2 - 84632y - 26818 \\
\beta \quad & -14900x^6 - 59600x^5y + 7450x^5 + 23840x^3y^2 + 32482x^3y - 9276x^3 \\
& - 174176x^2y + 21772x^2 - 23840xy^3 - 32482xy^2 - 34268xy + 22142x - 47680y^4 \\
& + 30396y^3 + 195184y^2 - 90654y + 10701 \\
\gamma \quad & 14900x^6 + 59600x^5y - 14900x^5 + 43544x^3 + 174176x^2y - 43544x^2 - 22142x \\
& + 47680y^4 - 30396y^3 + 153168y^2 - 61750y + 22142 \\
\{i, j\} = \{26, 31\}, k_{ij} = 2 \cdot 7 \cdot 17 \\
\alpha \quad & -8 \\
\beta \quad & -8x^6 - 12x^5 - 4x^3y - 6x^3 + 4xy^2 + 12xy - 67x - 8y^3 - 4y^2 + 30y - 99 \\
\gamma \quad & 8x^6 + 16x^5 - 8xy + 64x + 8y^3 + 4y^2 - 34y + 131 \\
\{i, j\} = \{26, 33\}, k_{ij} = 3 \\
\alpha \quad & -4 \\
\beta \quad & 4x^3 + 4xy^2 - 2xy + 3x + 4y \\
\gamma \quad & -4xy^2 - 2xy - x + 4y^2 - 2y + 1
\end{aligned}$$


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$$\begin{aligned}
&\{i, j\} = \{26, 34\}, k_{ij} = 1 \\
&\alpha \quad 4 \\
&\beta \quad -8x^6 - 12x^5 + 4x^3y - 2x^3 - 4xy^2 - 8xy + 7x - 8y^3 + 12y^2 - 20y + 13 \\
&\gamma \quad 8x^6 + 8x^5 + 12xy - 8x + 8y^3 - 12y^2 + 14y - 9 \\
&\{i, j\} = \{26, 36\}, k_{ij} = 3^4 \\
&\alpha \quad 96 \\
&\beta \quad 64x^6 + 32x^5 + 64x^3y + 16x^3 - 64xy^2 + 88xy - 46x + 64y^3 - 48y^2 - 12y - 23 \\
&\gamma \quad -64x^6 - 96x^5 - 72xy + 54x - 64y^3 + 48y^2 - 60y + 77 \\
&\{i, j\} = \{28, 30\}, k_{ij} = 2 \\
&\alpha \quad 2 \\
&\beta \quad 4x^6 + 2x^5 + 2x^3y - 2x^3 - 2xy^2 + 16xy - 12x + 4y^3 - 12y^2 + 16y - 9 \\
&\gamma \quad -4x^6 - 4x^5 - 12xy + 10x - 4y^3 + 12y^2 - 22y + 14 \\
&\{i, j\} = \{29, 36\}, k_{ij} = 2 \cdot 3 \\
&\alpha \quad -2 \\
&\beta \quad 2x^3 - 2xy^2 - 4x + 4y^2 + 3y + 4 \\
&\gamma \quad 2xy^2 - 2xy + 2x - 2y^2 - 5y - 2 \\
&\{i, j\} = \{30, 40\}, k_{ij} = 2 \cdot 3 \cdot 7 \\
&\alpha \quad -640x^4 - 56 \\
&\beta \quad 640x^7 + 320x^6 - 320x^5y + 240x^5 + 320x^4y^2 - 160x^4y + 280x^4 + 256x^3 \\
&\quad - 112x^2 - 128xy + 46x + 28y^2 + 6y - 3 \\
&\gamma \quad -320x^5y + 240x^5 - 320x^4y^2 - 480x^4y + 120x^4 - 128xy + 26x - 28y^2 - 22y - 17 \\
&\{i, j\} = \{31, 34\}, k_{ij} = 5 \cdot 7 \cdot 11 \cdot 13 \\
&\alpha \quad 640y + 240 \\
&\beta \quad 256x^9 + 256x^8 - 256x^7 - 256x^6y - 256x^6 - 256x^5y^2 - 320x^5y - 128x^5 \\
&\quad + 616x^4 + 256x^3y^2 - 32x^3y + 568x^3 + 128x^2y^2 - 16x^2y - 640x^2 \\
&\quad - 256xy^3 - 32xy^2 - 560xy + 936x - 256y^4 - 288y^3 - 288y^2 - 580y + 2787 \\
&\gamma \quad -256x^9 + 256x^5y^2 + 448x^5y + 128x^5 - 616x^4 - 1540x + 256y^4 + 32y^3 \\
&\quad + 224y^2 + 948y - 1229 \\
&\{i, j\} = \{31, 35\}, k_{ij} = 2^5 \cdot 7 \cdot 11 \\
&\alpha \quad 21686292y + 16865658 \\
&\beta \quad 27092556x^9 + 41990400x^8 + 34992000x^7 - 41990400x^6y - 3499200x^6 \\
&\quad - 27092556x^5y^2 - 53954748x^5y - 14597496x^5 - 38103102x^4 \\
&\quad + 41990400x^3y^2 + 20412000x^3y - 41107500x^3 + 20995200x^2y^2 \\
&\quad + 10206000x^2y - 22721850x^2 - 41990400xy^3 - 30909600xy^2 + 36004500xy + 24648870x \\
&\quad - 29795688y^4 - 67302252y^3 + 18208854y^2 + 63441882y + 22185902 \\
&\gamma \quad -27092556x^9 - 1351566x^8 + 901044x^7 + 1351566x^6y + 450522x^6 \\
&\quad + 27092556x^5y^2 - 4285062x^5y - 5999670x^5 + 38103102x^4 + 2703132x^3y^2 \\
&\quad + 9761310x^3y - 19614393x^3 - 2703132x^2y^2 - 9761310x^2y + 2730942x^2 \\
&\quad - 2703132xy^3 - 9761310xy^2 + 19614393xy + 1427406x + 29795688y^4 + 30718116y^3 \\
&\quad - 32141502y^2 - 40404807y - 2658890 \\
&\{i, j\} = \{33, 41\}, k_{ij} = 2^3 \cdot 7 \cdot 11^2 \\
&\alpha \quad 83712x^5 - 774336y^2 + 678416y - 76824 \\
&\beta \quad -15696x^9 - 75864x^8 + 31392x^7y + 52320x^7 + 44472x^6y - 90252x^6 \\
&\quad - 57552x^5y^2 + 222360x^5y - 206664x^5 - 279732x^4 + 193584x^3y^2 \\
&\quad + 265960x^3y - 167862x^3 - 193584x^2y^2 + 293504x^2y - 251736x^2 - 193584xy^3 \\
&\quad + 24416xy^2 + 7338xy + 2437x + 387168y^4 + 144752y^3 - 253964y^2 + 18874y - 117310 \\
&\gamma \quad 15696x^9 + 26160x^5y^2 - 303456x^5y + 323076x^5 + 279732x^4 + 165368x \\
&\quad - 387168y^4 + 242416y^3 - 160748y^2 + 372276y - 93679 \\
&\{i, j\} = \{34, 39\}, k_{ij} = 3 \cdot 11 \\
&\alpha \quad -1024x^4 + 416 \\
&\beta \quad 1024x^7 - 512x^6 - 512x^5y + 512x^5 + 512x^4y^2 + 896x^4y - 160x^4 + 288x^3 \\
&\quad - 320x^2 - 144xy - 32x - 208y^2 + 76y - 331 \\
&\gamma \quad -512x^5y + 768x^5 - 512x^4y^2 + 128x^4y - 992x^4 - 144xy + 480x + 208y^2 + 388y \\
&\quad - 213 \\
&\{i, j\} = \{34, 40\}, k_{ij} = 2^2 \cdot 7^2 \cdot 13 \\
&\alpha \quad -2048x^5 + 9728y^2 + 8448y - 12416 \\
&\beta \quad 512x^9 + 1536x^8 - 1024x^7y - 1024x^6y + 1536x^6 + 1536x^5y^2 - 1792x^5y \\
&\quad + 1152x^5 + 17448x^4 - 4864x^3y^2 - 13952x^3y - 3944x^3 + 2432x^2y^2 \\
&\quad - 27920x^2y + 28144x^2 + 4864xy^3 + 7872xy^2 + 3952xy - 6681x - 4864y^4 - 10304y^3 \\
&\quad + 12904y^2 - 23514y + 11895
\end{aligned}$$


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|-----------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $\gamma$                                      | $-512x^9 - 512x^5y^2 + 1280x^5y - 1920x^5 - 17448x^4 - 11335x + 4864y^4$<br>$+5440y^3 + 13512y^2 - 34526y + 13922$                                                                                                                         |
| $\{i, j\} = \{35, 37\}, k_{ij} = 2^6 \cdot 7$ |                                                                                                                                                                                                                                            |
| $\alpha$                                      | $176x^5 + 264y^2 - 440y - 542$                                                                                                                                                                                                             |
| $\beta$                                       | $-22x^9 - 154x^8 + 44x^7y + 176x^7 + 110x^6y - 22x^6 - 110x^5y^2 - 176x^5y$<br>$+22x^5 + 175x^4 - 132x^3y^2 + 418x^3y + 433x^3 + 132x^2y^2 - 768x^2y$<br>$-608x^2 + 132xy^3 - 418xy^2 - 83xy + 47x - 132y^4 + 154y^3 + 721y^2 + 435y - 47$ |
| $\gamma$                                      | $22x^9 + 66x^5y^2 - 154x^5y - 176x^5 - 175x^4 + 128x + 132y^4 - 418y^3 + 333y^2$<br>$+1024y + 608$                                                                                                                                         |
| $\{i, j\} = \{36, 38\}, k_{ij} = 3^2$         |                                                                                                                                                                                                                                            |
| $\alpha$                                      | $-32$                                                                                                                                                                                                                                      |
| $\beta$                                       | $-12x^4 + 20x^3 + 24x^2y + 26x^2 + 4xy + 4y^2 - 18y + 3$                                                                                                                                                                                   |
| $\gamma$                                      | $12x^4 + 6x - 28y^2 - 28y - 29$                                                                                                                                                                                                            |
| $\{i, j\} = \{37, 39\}, k_{ij} = 2^2$         |                                                                                                                                                                                                                                            |
| $\alpha$                                      | $-2$                                                                                                                                                                                                                                       |
| $\beta$                                       | $4x^6 - 2x^5 + 2x^3y + 2xy^2 + 2x - 4y^3 - 2y - 1$                                                                                                                                                                                         |
| $\gamma$                                      | $-4x^6 + 4x^5 - 2x + 4y^3 + 2y + 2$                                                                                                                                                                                                        |
| $\{i, j\} = \{38, 40\}, k_{ij} = 3$           |                                                                                                                                                                                                                                            |
| $\alpha$                                      | $-4$                                                                                                                                                                                                                                       |
| $\beta$                                       | $4x^3 + 8xy^2 + 8xy + 4y^2 - 6y + 1$                                                                                                                                                                                                       |
| $\gamma$                                      | $-8xy^2 - 4xy - 2x - 8y^2 + 4y - 2$                                                                                                                                                                                                        |
| $\{i, j\} = \{39, 40\}, k_{ij} = 2^5 \cdot 3$ |                                                                                                                                                                                                                                            |
| $\alpha$                                      | $512y - 384$                                                                                                                                                                                                                               |
| $\beta$                                       | $-512x^9 + 512x^8 - 256x^7 + 512x^6y - 128x^6 + 512x^5y^2 - 640x^5y + 160x^5$<br>$+248x^4 - 512x^3y^2 - 64x^3y + 40x^3 + 256x^2y^2 + 32x^2y - 20x^2$<br>$-512xy^3 + 64xy^2 + 56xy + 293x - 512y^4 + 576y^3 - 296y^2 + 122y - 139$          |
| $\gamma$                                      | $512x^9 - 512x^5y^2 + 128x^5y + 32x^5 - 248x^4 - 303x + 512y^4 - 64y^3 + 168y^2$<br>$-186y - 149$                                                                                                                                          |

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Table 3: The polynomials  $\varphi_i(x) = k_i^3 F(x, -r_i(x)/k_i)$  of Step 2 and the signs of  $H_i$  for  $x \geq 8$  and for  $x \leq -11$ .

| $i$ | $k_i$ | $\varphi_i(x)$                                     | $x \geq 8$ | $x \leq -11$ |
|-----|-------|----------------------------------------------------|------------|--------------|
| 0   | 2     | $16x^5 - 8$                                        | -          | +            |
| 1   | 2     | $16x^5 + 4x - 6$                                   | -          | +            |
| 2   | 2     | $16x^5 - 16x^3 - 8x^2 - 8$                         | -          | +            |
| 3   | 2     | $16x^5 - 2x^3 + 2x^2 - 2x - 6$                     | -          | +            |
| 4   | 2     | $16x^5 + 8x + 8$                                   | -          | +            |
| 5   | 2     | $16x^5 - 8x - 24$                                  | -          | +            |
| 6   | 2     | $16x^5 + 2x^3 + 10x^2 + 10x - 6$                   | -          | +            |
| 7   | 2     | $16x^5 + 2x^3 - 2x^2 + 2x - 10$                    | -          | +            |
| 8   | 2     | $16x^5 + 16x^3 + 8x^2 - 8$                         | -          | +            |
| 9   | 2     | $16x^5 - 16x^3 + 16x^2 - 8x - 6$                   | -          | +            |
| 10  | 2     | $16x^5 - 16x^3 - 32x^2 - 16x - 10$                 | -          | +            |
| 11  | 2     | $16x^5 - 2x^3 - 16x^2 - 32x - 24$                  | -          | +            |
| 12  | 2     | $16x^5 - 16x^3 + 40x^2 - 40x + 8$                  | -          | +            |
| 13  | 2     | $16x^5 - 16x^3 - 56x^2 - 56x - 24$                 | -          | +            |
| 14  | 2     | $16x^5 + 16x^3 + 56x^2 + 56x + 8$                  | -          | +            |
| 15  | 2     | $-2x^6 + 16x^5 - 4x^3 - 8$                         | +          | +            |
| 16  | -2    | $-2x^6 - 22x^5 - 6x^4 - 6x^3 - 4x^2 + 8$           | +          | +            |
| 17  | 2     | $-2x^6 + 10x^5 - 6x^4 - 6x^3 - 4x^2 - 8$           | +          | +            |
| 18  | -2    | $-2x^6 - 22x^5 - 12x^4 - 18x^3 - 16x^2 - 10x + 6$  | +          | +            |
| 19  | -2    | $-2x^6 - 10x^5 - 12x^4 + 10x^3 - 8x^2 + 2x + 6$    | +          | +            |
| 20  | -2    | $-2x^6 - 4x^5 - 24x^4 + 12x^3 + 8x^2 + 8$          | +          | +            |
| 21  | 2     | $-2x^6 + 10x^5 - 12x^4 - 18x^3 - 16x^2 - 10x - 10$ | +          | +            |
| 22  | -2    | $-2x^6 - 22x^5 - 18x^4 - 30x^3 - 40x^2 - 32x - 8$  | +          | +            |
| 23  | -2    | $-2x^6 - 10x^5 - 18x^4 + 22x^3 - 32x^2 + 16x - 8$  | +          | +            |
| 24  | -2    | $-2x^6 - 4x^5 - 18x^4 - 12x^3 + 26x^2 + 16x + 10$  | +          | +            |
| 25  | 2     | $-16x^6 + 16x^5 - 8x^3 - 8$                        | +          | +            |
| 26  | 2     | $-16x^6 + 16x^5 + 24x^4 - 8x^3 - 12x^2 + 4x - 6$   | +          | +            |
| 27  | 2     | $-16x^6 - 32x^5 - 48x^4 - 24x^3 - 8x^2 - 8$        | +          | +            |

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| $i$ | $k_i$ | $\varphi_i(x)$                                        | $x \geq 8$ | $x \leq -11$ |
|-----|-------|-------------------------------------------------------|------------|--------------|
| 28  | 2     | $-16x^6 + 16x^5 + 48x^4 - 8x^3 - 48x^2 + 8x + 8$      | +          | +            |
| 29  | 2     | $-16x^6 + 16x^5 - 48x^4 - 8x^3 - 48x^2 - 8x - 24$     | +          | +            |
| 30  | 2     | $-16x^6 + 40x^5 + 12x^4 - 30x^3 - 2x^2 + 10x - 6$     | +          | +            |
| 31  | 2     | $-16x^6 + 40x^5 - 36x^4 + 18x^3 - 14x^2 + 2x - 10$    | +          | +            |
| 32  | 2     | $-16x^6 + 64x^5 - 48x^4 + 8x^3 + 8x^2 - 8$            | +          | +            |
| 33  | 2     | $-16x^6 - 32x^5 - 24x^4 + 24x^3 + 4x^2 - 8x - 6$      | +          | +            |
| 34  | 2     | $-16x^6 - 8x^5 + 36x^4 + 38x^3 - 40x^2 - 16x + 8$     | +          | +            |
| 35  | 2     | $-16x^6 - 32x^5 - 96x^4 - 120x^3 - 104x^2 - 56x - 24$ | +          | +            |
| 36  | 2     | $-16x^6 + 64x^5 - 96x^4 + 104x^3 - 88x^2 + 40x - 24$  | +          | +            |
| 37  | -2    | $-16x^6 - 16x^5 - 8x^3 + 8$                           | +          | +            |
| 38  | -2    | $-16x^6 - 16x^5 - 24x^4 - 8x^3 - 12x^2 - 4x + 6$      | +          | +            |
| 39  | -2    | $-16x^6 - 40x^5 - 36x^4 - 34x^3 - 22x^2 - 10x + 6$    | +          | +            |
| 40  | -2    | $-16x^6 + 8x^5 - 36x^4 + 18x^3 - 14x^2 + 2x + 6$      | +          | +            |
| 41  | -2    | $-16x^6 - 40x^5 - 60x^4 - 58x^3 - 64x^2 - 32x - 8$    | +          | +            |

Table 4: The terminal residue classes  $x \equiv a$ ,  $y \equiv b \pmod{p^k}$  of Step 4, grouped by  $p$ ,  $k$  and the value of  $B_p$ .

| $p$ | $k$ | $B_p$ | classes $(a, b)$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|-----|-----|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2   | 6   | 1     | (7, 11), (15, 3), (23, 59), (31, 51), (39, 43), (47, 35), (55, 27), (63, 19)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 2   | 7   | 1     | (1, 5), (9, 125), (17, 117), (25, 109), (33, 101), (41, 93), (49, 85),<br>(53, 1), (57, 77), (65, 69), (69, 113), (73, 61), (81, 53), (89, 45),<br>(97, 37), (105, 29), (113, 21), (121, 13)                                                                                                                                                                                                                                                                                                                                                                                                   |
| 2   | 8   | 1     | (5, 177), (11, 215), (13, 105), (19, 15), (21, 161), (35, 255), (37, 145),<br>(43, 183), (51, 239), (61, 57), (75, 151), (83, 207), (85, 97), (101, 81),<br>(107, 119), (115, 175), (117, 65), (133, 49), (139, 87), (141, 233),<br>(147, 143), (149, 33), (165, 17), (171, 55), (179, 111), (189, 185),<br>(195, 95), (203, 23), (211, 79), (213, 225), (219, 7), (229, 209),<br>(235, 247), (243, 47), (245, 193), (251, 231)                                                                                                                                                                |
| 2   | 9   | 1     | (3, 287), (27, 455), (29, 345), (45, 73), (59, 423), (67, 223), (77, 41),<br>(91, 391), (93, 281), (99, 191), (109, 9), (123, 359), (125, 249),<br>(131, 159), (155, 327), (157, 217), (163, 127), (173, 457), (187, 295),<br>(205, 425), (221, 153), (227, 63), (237, 393), (253, 121), (259, 31),<br>(283, 199), (285, 89), (301, 329), (315, 167), (323, 479), (333, 297),<br>(347, 135), (349, 25), (355, 447), (365, 265), (379, 103), (381, 505),<br>(387, 415), (411, 71), (413, 473), (419, 383), (429, 201), (443, 39),<br>(461, 169), (477, 409), (483, 319), (493, 137), (509, 377) |
| 3   | 3   | -1    | (2, 9), (6, 2), (6, 11), (8, 6), (11, 18), (14, 12), (15, 14), (15, 23),<br>(17, 15), (20, 0), (23, 21), (24, 8), (24, 17), (24, 26), (26, 24)                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 3   | 4   | -1    | (5, 3), (6, 20), (6, 47), (15, 5), (15, 59), (32, 30), (59, 57)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 3   | 5   | -1    | (6, 74), (6, 155), (6, 236), (177, 32), (177, 113), (177, 194)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 5   | 2   | 1     | (0, 17), (3, 5), (3, 16), (5, 2), (8, 5), (8, 21), (10, 12), (15, 22), (18, 5),<br>(18, 6), (20, 7), (23, 5)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 5   | 3   | 1     | (8, 24), (13, 69), (13, 80), (13, 101), (18, 14), (23, 109), (23, 111),<br>(28, 104), (38, 1), (38, 44), (38, 80), (43, 114), (48, 11), (48, 84),<br>(53, 79), (58, 99), (63, 19), (63, 26), (63, 80), (68, 89), (73, 59),<br>(78, 54), (83, 74), (88, 51), (88, 80), (88, 119), (93, 64), (98, 34),<br>(98, 61), (103, 29), (108, 49), (113, 76), (113, 80), (113, 94), (118, 39),<br>(123, 9), (123, 86)                                                                                                                                                                                     |
| 5   | 4   | 1     | (33, 124), (73, 411), (128, 504), (158, 624), (198, 536), (253, 379),<br>(283, 499), (323, 36), (378, 254), (408, 374), (448, 161), (503, 129),<br>(533, 249), (573, 286)                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 5   | 5   | 1     | (3, 629), (628, 4), (1253, 2504), (1878, 1879), (2503, 1254)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 7   | 1   | 1     | (1, 4)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 7   | 2   | 1     | (2, 42), (9, 21), (10, 29), (11, 4), (16, 0), (18, 46), (23, 28), (25, 39),<br>(30, 7), (31, 15), (32, 5), (32, 12), (32, 19), (32, 26), (32, 33), (32, 40),<br>(32, 47), (37, 35), (39, 25), (44, 14), (45, 22), (46, 18)                                                                                                                                                                                                                                                                                                                                                                     |

| $p$ | $k$ | $B_p$ | classes $(a, b)$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|-----|-----|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7   | 3   | 1     | (3, 99), (4, 207), (5, 292), (17, 106), (19, 285), (24, 232), (26, 306),<br>(32, 81), (33, 229), (38, 43), (40, 54), (47, 124), (52, 295), (53, 158),<br>(54, 96), (66, 302), (68, 89), (73, 85), (75, 110), (81, 32), (82, 33),<br>(87, 239), (89, 201), (96, 271), (101, 148), (103, 243), (115, 155),<br>(117, 236), (122, 281), (124, 257), (130, 326), (131, 180), (136, 92),<br>(138, 5), (145, 75), (150, 1), (151, 60), (152, 47), (164, 8), (166, 40),<br>(171, 134), (173, 61), (179, 277), (180, 327), (185, 288), (187, 152),<br>(194, 222), (199, 197), (200, 11), (201, 194), (213, 204), (215, 187),<br>(220, 330), (222, 208), (229, 131), (234, 141), (236, 299), (243, 26),<br>(248, 50), (249, 305), (250, 341), (262, 57), (264, 334), (269, 183),<br>(271, 12), (277, 179), (278, 278), (283, 337), (285, 103), (292, 173),<br>(297, 246), (298, 256), (299, 145), (311, 253), (313, 138), (318, 36),<br>(320, 159), (326, 130), (327, 82), (332, 190), (334, 250), (341, 320) |
| 7   | 4   | 1     | (12, 1538), (61, 313), (110, 1489), (159, 264), (208, 1440), (228, 2286),<br>(257, 215), (306, 1391), (355, 166), (404, 1342), (445, 109), (453, 117),<br>(502, 1293), (551, 68), (571, 1943), (600, 1244), (649, 19), (698, 1195),<br>(747, 2371), (788, 2167), (796, 1146), (845, 2322), (894, 1097),<br>(914, 1600), (943, 2273), (992, 1048), (1041, 2224), (1090, 999),<br>(1131, 1824), (1139, 2175), (1188, 950), (1237, 2126), (1257, 1257),<br>(1286, 901), (1335, 2077), (1384, 852), (1433, 2028), (1474, 1481),<br>(1482, 803), (1531, 1979), (1580, 754), (1600, 914), (1629, 1930),<br>(1678, 705), (1727, 1881), (1776, 656), (1817, 1138), (1825, 1832),<br>(1874, 607), (1923, 1783), (1943, 571), (1972, 558), (2021, 1734),<br>(2070, 509), (2119, 1685), (2160, 795), (2168, 460), (2217, 1636),<br>(2266, 411), (2286, 228), (2315, 1587), (2364, 362)                                                                                                                         |
| 7   | 5   | 1     | (102, 7655), (2503, 5254), (4904, 2853), (7305, 452), (9706, 14858),<br>(12107, 12457), (14508, 10056)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 11  | 1   | 1     | (0, 8), (3, 6), (4, 7), (6, 10), (10, 2), (10, 4)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 11  | 2   | 1     | (7, 31), (9, 37), (10, 38), (18, 64), (20, 59), (21, 71), (29, 97), (31, 81),<br>(32, 104), (40, 9), (42, 103), (43, 16), (51, 42), (53, 4), (54, 49),<br>(62, 75), (64, 26), (65, 82), (73, 108), (75, 48), (76, 115), (84, 20),<br>(86, 70), (87, 27), (95, 53), (97, 92), (98, 60), (106, 86), (108, 9),<br>(108, 20), (108, 31), (108, 42), (108, 53), (108, 64), (108, 75), (108, 86),<br>(108, 97), (108, 108), (108, 114), (108, 119), (109, 93), (117, 119),<br>(119, 15), (120, 5)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 13  | 1   | 1     | (2, 6), (3, 2), (4, 12), (9, 2), (9, 4), (9, 7), (11, 0), (11, 1)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 13  | 2   | 1     | (5, 42), (10, 159), (11, 90), (12, 72), (18, 42), (23, 81), (24, 38),<br>(25, 137), (31, 42), (36, 3), (37, 155), (38, 33), (44, 42), (49, 94),<br>(50, 103), (51, 98), (57, 42), (62, 16), (63, 51), (64, 163), (70, 42),<br>(75, 107), (76, 168), (77, 59), (89, 116), (90, 124), (96, 42), (101, 120),<br>(102, 64), (103, 20), (109, 42), (114, 42), (115, 12), (116, 85), (122, 42),<br>(127, 133), (128, 129), (129, 150), (135, 42), (140, 55), (141, 77),<br>(142, 46), (148, 42), (153, 146), (154, 25), (155, 111), (161, 42),<br>(166, 68), (167, 142), (168, 7)                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 13  | 3   | 1     | (83, 718), (88, 705), (252, 718), (257, 1888), (421, 718), (590, 718),<br>(595, 2057), (759, 718), (764, 1043), (928, 718), (933, 29), (1097, 718),<br>(1102, 1212), (1266, 718), (1271, 198), (1435, 718), (1440, 1381),<br>(1604, 718), (1609, 367), (1773, 718), (1778, 1550), (1942, 718),<br>(1947, 536), (2111, 718), (2116, 1719)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 13  | 4   | 1     | (426, 22844), (2623, 9662), (4820, 25041), (7017, 11859),<br>(9214, 27238), (11411, 14056), (13608, 874), (15805, 16253),<br>(18002, 3071), (20199, 18450), (22396, 5268), (24593, 20647),<br>(26790, 7465)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 17  | 1   | 1     | (0, 15), (1, 15), (3, 6), (4, 15), (5, 1), (5, 7), (5, 9), (6, 3), (7, 5), (7, 10),<br>(8, 0), (8, 8), (8, 9), (10, 12), (11, 14), (12, 10), (12, 14), (13, 5),<br>(13, 14), (13, 15), (14, 3), (14, 11), (15, 3), (16, 15)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |

| $p$ | $k$ | $B_p$ | classes $(a, b)$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|-----|-----|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 17  | 2   | 1     | $(2, 39), (7, 138), (11, 58), (11, 115), (19, 243), (24, 87), (28, 166),$<br>$(28, 279), (36, 158), (41, 36), (45, 211), (45, 217), (53, 73), (58, 274),$<br>$(62, 143), (62, 268), (70, 277), (75, 223), (79, 30), (79, 75), (87, 192),$<br>$(92, 172), (96, 7), (104, 107), (109, 121), (113, 132), (113, 228),$<br>$(121, 22), (126, 70), (130, 160), (130, 183), (138, 226), (143, 19),$<br>$(147, 92), (147, 234), (155, 141), (160, 257), (164, 24), (164, 285),$<br>$(172, 56), (177, 206), (181, 47), (181, 245), (189, 260), (194, 155),$<br>$(198, 98), (198, 177), (206, 175), (211, 104), (215, 109), (215, 149),$<br>$(223, 90), (228, 53), (232, 41), (232, 200), (240, 5), (245, 2), (249, 251),$<br>$(249, 262), (257, 209), (262, 240), (266, 13), (266, 194), (274, 124),$<br>$(279, 189), (283, 64), (283, 126)$ |
| 17  | 3   | 1     | $(96, 4416), (385, 370), (674, 1237), (963, 2104), (1252, 2971),$<br>$(1541, 3838), (1830, 4705), (2119, 659), (2408, 1526), (2697, 2393),$<br>$(2986, 3260), (3275, 4127), (3564, 81), (3853, 948), (4142, 1815),$<br>$(4431, 2682), (4720, 3549)$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |